Introduction to IT Project Management

By

Casmir, R. (Ph.D)

Introduction

What is a Project?

A **project** is a **temporary** endeavor undertaken to create a **unique** product or service.

Project Definition

- A **temporary** endeavor – means that every project has a definite beginning and end.
- **Unique** means that the *product* or *service* or an improvement or *result* is different from other product or service or result.
- **Progressively elaborated** - proceeding in steps

Project Definition

A project can create:

- A product that can be either a component of another item, an enhancement of an item, or an end item in itself;

- A service or a capability to perform a service (e.g., a business function that supports production or distribution);

- An improvement in the existing product or service lines (e.g., A Six Sigma project undertaken to reduce defects); or

- A result, such as an outcome or document (e.g., a research project that develops knowledge that can be used to determine whether a trend exists or a new process will benefit society).

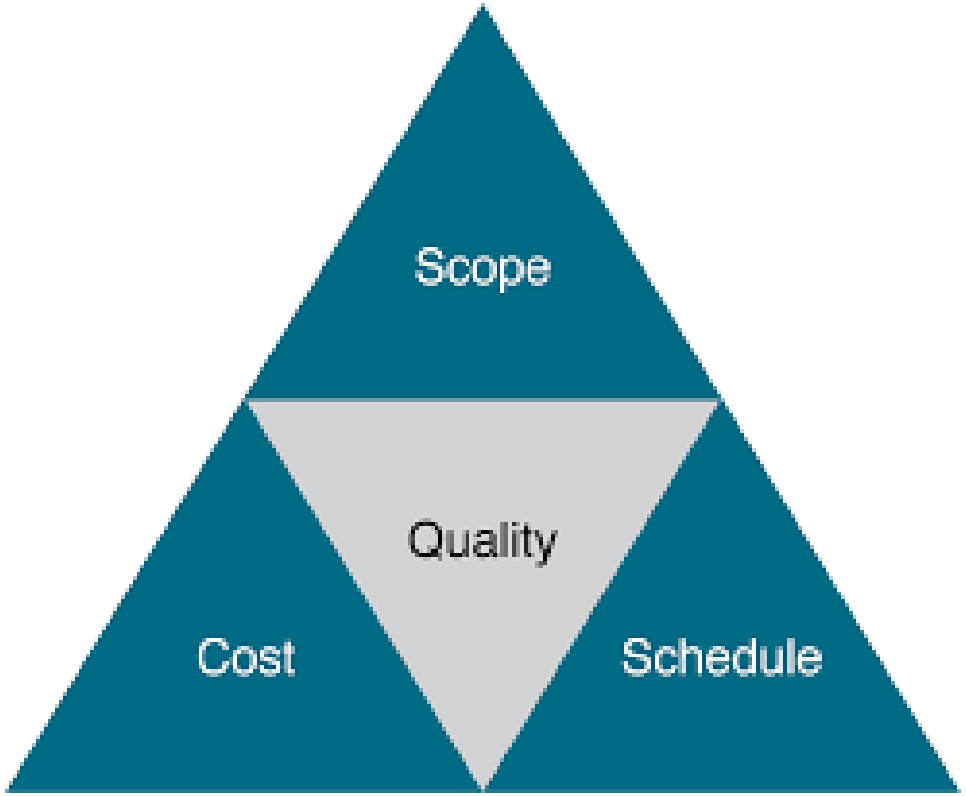
Examples of Project

- ✓ Software development
- ✓ System implementation
- ✓ Network upgrades
- ✓ Cloud migration
- ✓ Security enhancements

Project Stakeholders

- Software development is the process of creating new applications, while system implementation is the process of putting a pre-existing or newly developed software application into a working environment
- Development focuses on the design, coding, and testing of the software itself, whereas implementation involves installation, configuration, and user training for that software within an organization

Project Quality



Project Quality

The Rule: Changing one side affects the others

- **Scope:** What will be delivered
- **Time:** When it will be delivered
- **Cost:** Resources/budget required
- **Quality:** At the center – affected by all three

Project Project Lifecycle: 5 Process Groups

- **Initiation** - Define project value & feasibility
  Deliverable: Project Charter
- **Planning** - Create the detailed roadmap
  Deliverable: Project Management Plan
- **Execution** - Do the work, build the product
  Deliverable: Working product/service
- **Monitoring & Controlling** - Track, review, adjust
  Ongoing through execution
- **Closure** - Formal handover & lessons learned

Lecture 2:
Project Integration Management

Learning Objectives

- Describe an overall framework for project integration management as it relates to the other project management knowledge areas and the project life cycle
- Describe project plan development, including project plan content, using guidelines and templates for developing plans, and performing a stakeholder analysis to help manage relationships
- Explain project plan execution, its relationship to project planning, the factors related to successful results, and tools and techniques to assist in project plan execution
- Understand the integrated change control process, planning for and managing changes on information technology projects, and developing and using a change control system
- Describe how software can assist in project integration management

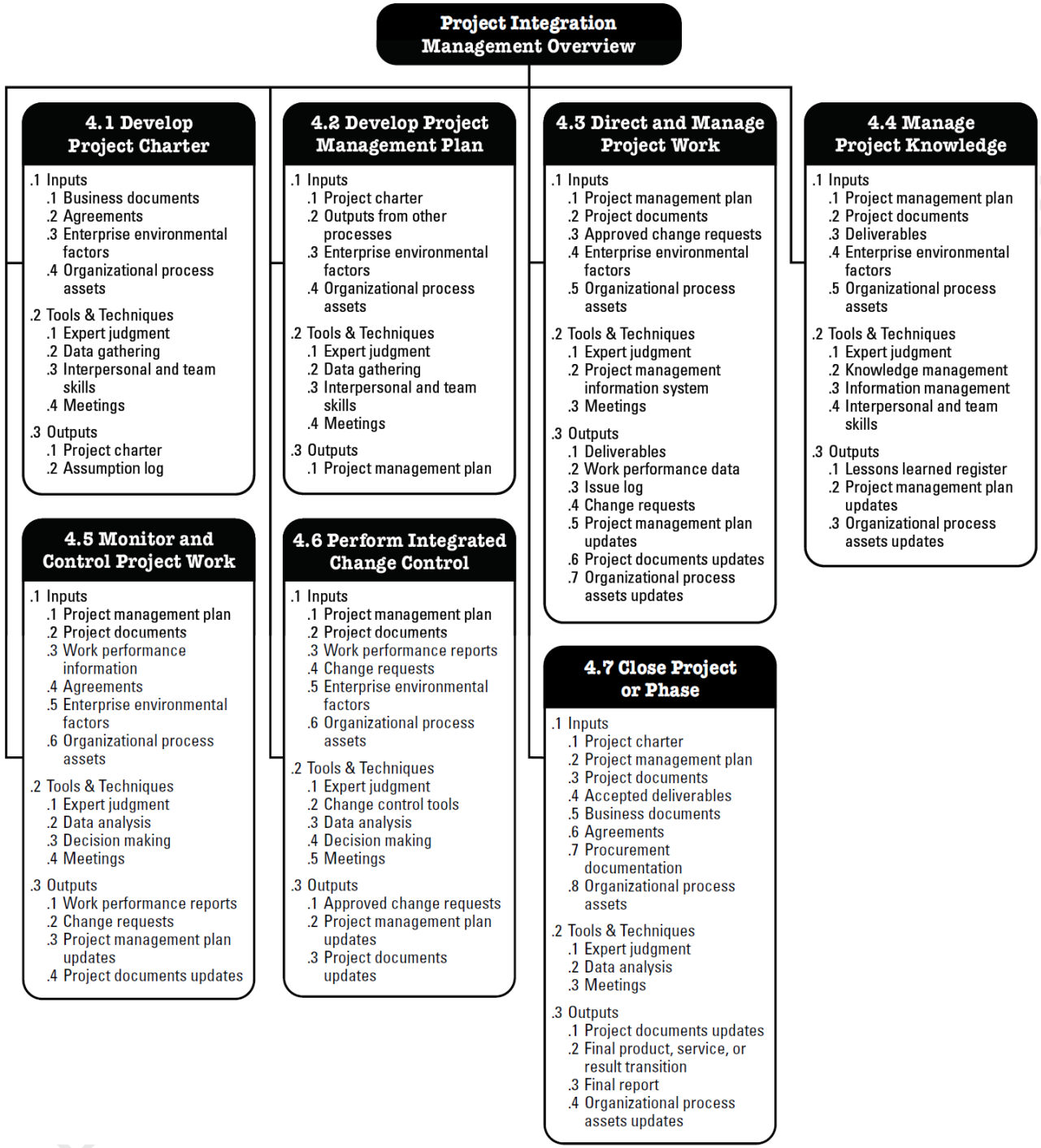
The Key to Overall Project Success: Good Project Integration Management

- Unlike most of the project management knowledge areas, which deal with a specific area of expertise of the project manager,
- the Project Integration Management knowledge area is something of a catch-all, containing many of the everyday actions of a project manager.
- It is the glue that holds the project together.

The Key to Overall Project Success: Good Project Integration Management

The processes within the Project Integration Management knowledge area are:

- Develop Project Charter
- Develop Project Management Plan
- Direct and Manage Project Work
- Manage Project Knowledge
- Monitor and Control Project Work
- Perform Integrated Change Control
- Close Project or Phase


Project Integration Management Overview
4.1 Develop Project Charter
.1 Inputs
.1 Business documents
.2 Agreements
.3 Enterprise environmental factors
.4 Organizational process assets
.2 Tools & Techniques
.1 Expert judgment
.2 Data gathering
.3 Interpersonal and team skills
.4 Meetings
.3 Outputs
.1 Project charter
.2 Assumption log
4.2 Develop Project Management Plan
.1 Inputs
.1 Project charter
.2 Outputs from other processes
.3 Enterprise environmental factors
.4 Organizational process assets
.2 Tools & Techniques
.1 Expert judgment
.2 Data gathering
.3 Interpersonal and team skills
.4 Meetings
.3 Outputs
.1 Project management plan
4.3 Direct and Manage Project Work
.1 Inputs
.1 Project management plan
.2 Project documents
.3 Approved change requests
.4 Enterprise environmental factors
.5 Organizational process assets
.2 Tools & Techniques
.1 Expert judgment
.2 Project management information system
.3 Meetings
.3 Outputs
.1 Deliverables
.2 Work performance data
.3 Issue log
.4 Change requests
.5 Project management plan updates
.6 Project documents updates
.7 Organizational process assets updates
4.4 Manage Project Knowledge
.1 Inputs
.1 Project management plan
.2 Project documents
.3 Deliverables
.4 Enterprise environmental factors
.5 Organizational process assets
.2 Tools & Techniques
.1 Expert judgment
.2 Knowledge management
.3 Information management
.4 Interpersonal and team skills
.3 Outputs
.1 Lessons learned register
.2 Project management plan updates
.3 Organizational process assets updates
4.5 Monitor and Control Project Work
.1 Inputs
.1 Project management plan
.2 Project documents
.3 Work performance information
.4 Agreements
.5 Enterprise environmental factors
.6 Organizational process assets
.2 Tools & Techniques
.1 Expert judgment
.2 Data analysis
.3 Decision making
.4 Meetings
.3 Outputs
.1 Work performance reports
.2 Change requests
.3 Project management plan updates
.4 Project documents updates
4.6 Perform Integrated Change Control
.1 Inputs
.1 Project management plan
.2 Project documents
.3 Work performance reports
.4 Change requests
.5 Enterprise environmental factors
.6 Organizational process assets
.2 Tools & Techniques
.1 Expert judgment
.2 Change control tools
.3 Data analysis
.4 Decision making
.5 Meetings
.3 Outputs
.1 Approved change requests
.2 Project management plan updates
.3 Project documents updates
4.7 Close Project or Phase
.1 Inputs
.1 Project charter
.2 Project management plan
.3 Project documents
.4 Accepted deliverables
.5 Business documents
.6 Agreements
.7 Procurement documentation
.8 Organizational process assets
.2 Tools & Techniques
.1 Expert judgment
.2 Data analysis
.3 Meetings
.3 Outputs
.1 Project documents updates
.2 Final product, service, or result transition
.3 Final report
.4 Organizational process assets updates

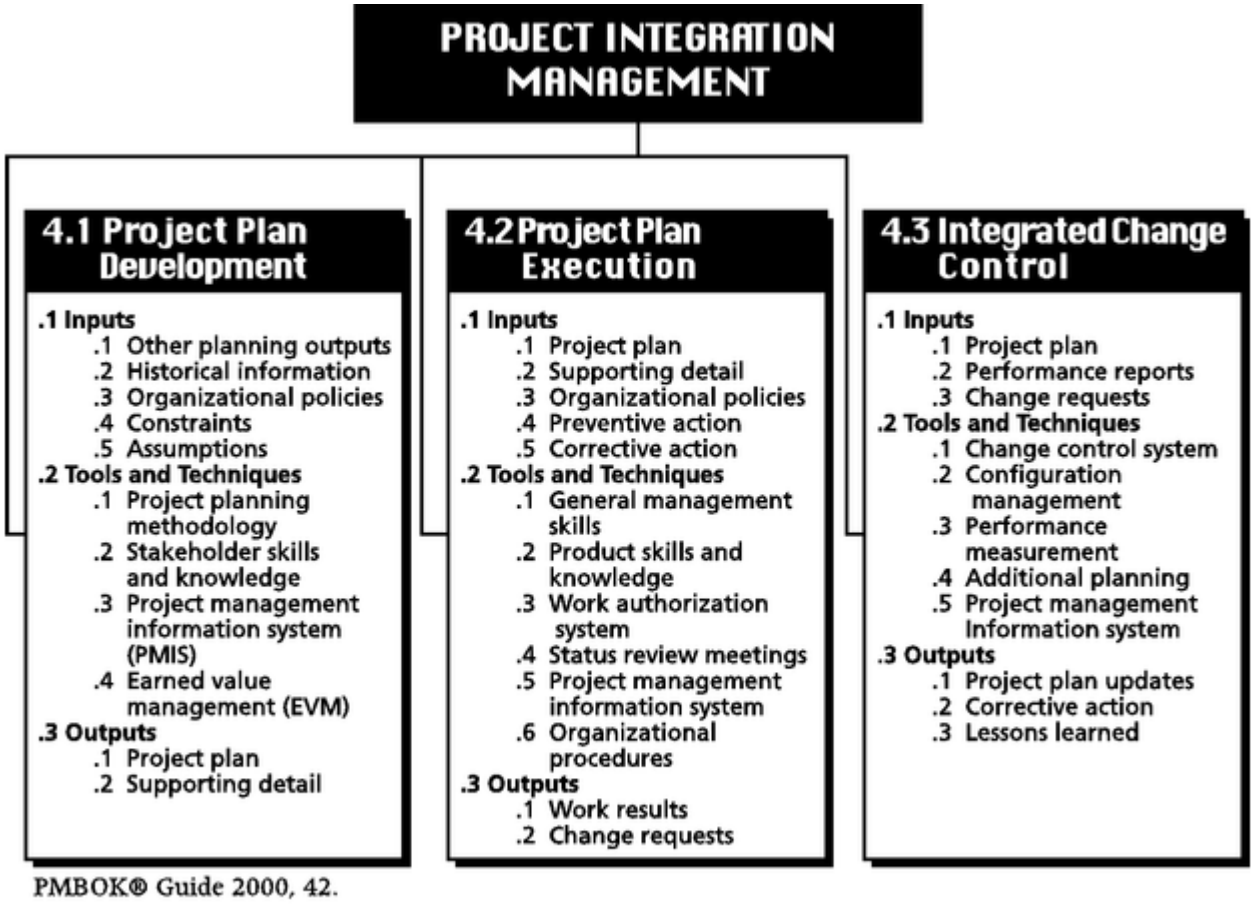
The Key to Overall Project Success: Good Project Integration Management

- Project managers must coordinate all of the other knowledge areas throughout a project's life cycle
- Many new project managers have trouble looking at the "big picture" and want to focus on too many details (See opening case for a real example)
- Project integration management is not the same thing as software integration

Project Integration Management Processes

- Project Plan Development: taking the results of other planning processes and putting them into a consistent, coherent document—the project plan
- Project Plan Execution: carrying out the project plan
- Integrated Change Control: coordinating changes across the entire project

Project Integration Management Overview

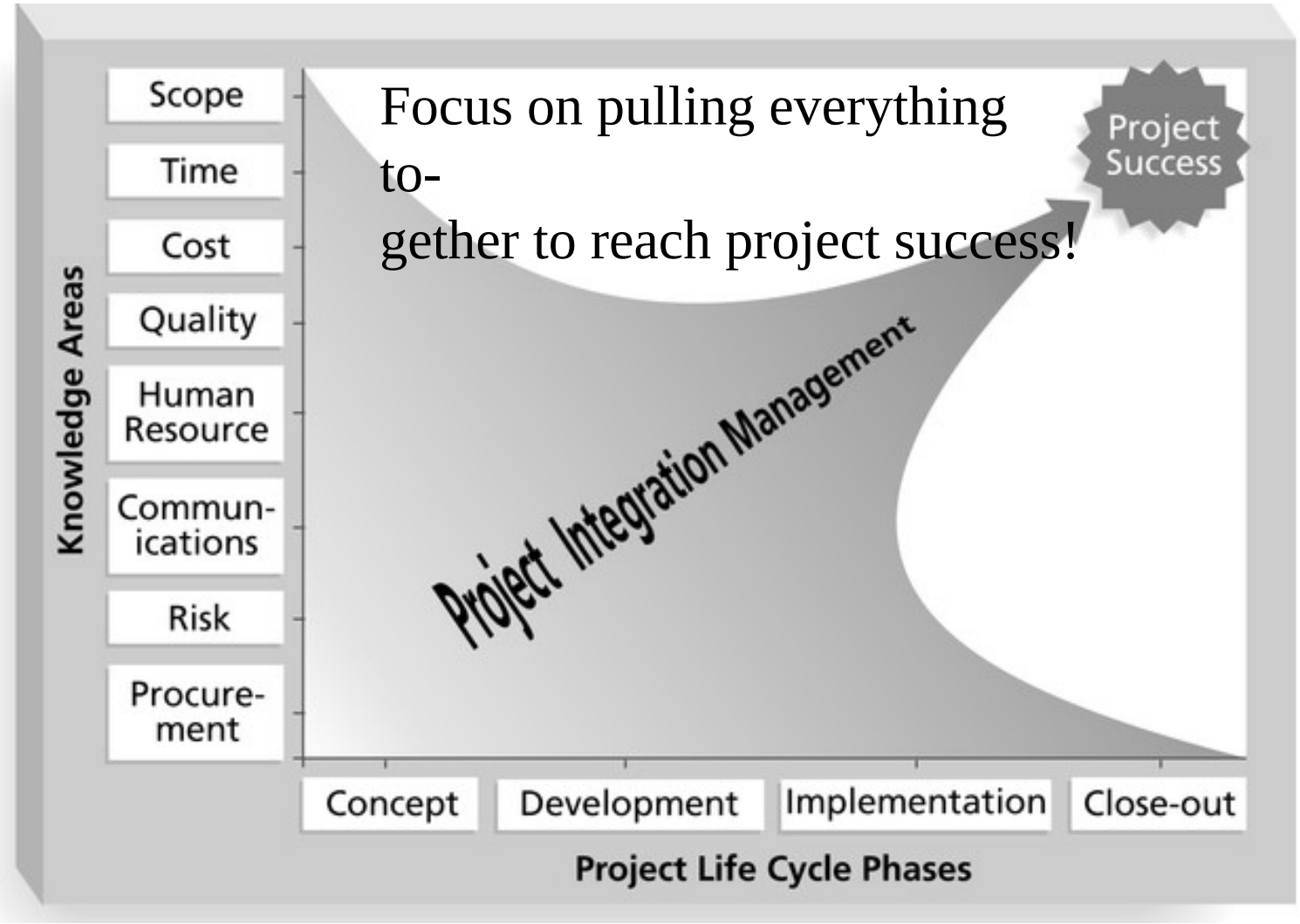


PMBOK® Guide 2000, 42.

Figure 4-1. Project Integration Management Overview

Note: The PMBOK® Guide 2000 includes similar charts for each knowledge area.

Framework for Project Integration Management



Project Plan Development

- A project plan is a document used to coordinate all project planning documents
- Its main purpose is to *guide project execution*
- Project plans assist the project manager in leading the project team and assessing project status
- Project performance should be measured against a baseline plan

Attributes of Project Plans

Just as projects are unique, so are project plans

- Plans should be dynamic
- Plans should be flexible
- Plans should be updated as changes occur
- Plans should first and foremost guide project execution

Common Elements of a Project Plan

- Introduction or overview of the project
- Description of how the project is organized
- Management and technical processes used on the project
- Work to be done, schedule, and budget information

Introduction or overview of the project

- The project name
- A brief description of the project and the need it addresses
- The sponsor’s name
- The names of the project manager and key team members
- Deliverables of the project
- A list of important reference materials
- A list of definitions and acronyms, if appropriate

Description of how the project is organized

- Organizational charts
- Project responsibilities
- Other organizational or process related information

Management and technical processes used on the project

- Management objectives
- Project controls
- Risk management
- Project staffing
- Technical processes

Work to be done

- Major work packages
- Key deliverables
- Other work-related information

Project schedule information

- Summary schedule
- Detailed schedule
- Other schedule-related information

Project budget

- Summary budget
- Detailed budget
- Other budget-related information

Sample Outline for a Software Project Management Plan (SPMP)

Project Management Plan Sections					
	Introduction	Project Organization	Managerial Process	Technical Process	Work Packages, Schedule, and Budget
Section Topics	Project overview; project deliverables; evolution of the SPMP; reference materials; definitions and acronyms	Process model; organizational structure; organizational boundaries and interfaces; project responsibilities	Management objectives and priorities; assumptions, dependencies, and constraints; risk management; monitoring and controlling mechanisms; and staffing plan	Methods, tools, and techniques; software documentation; and project support functions	Work packages; Dependencies; resource requirements; budget and resource allocation; and schedule

IEEE Std 10581-1987

What the Winners Do

"The winners clearly spell out **what** needs to be done in a project, by **whom**, **when**, and **how**. For this they use an integrated toolbox, including PM tools, methods, and techniques…If a scheduling template is developed and used over and over, it becomes a repeatable action that leads to higher productivity and lower uncertainty. Sure, using scheduling templates is neither a breakthrough nor a feat. But laggards exhibited almost no use of the templates. Rather, in constructing schedules their project managers started with a clean sheet, a clear waste of time." *

Milosevic, Dragan and And Ozbay. “Delivering Projects: What the Winners Do.” Proceedings of the Project Management Institute Annual Seminars & Symposium (November 2001)

Stakeholder Analysis

- A stakeholder analysis documents important (often sensitive) information about stakeholders such as
  - stakeholders' names and organizations
  - roles on the project
  - unique facts about stakeholders
  - level of influence and interest in the project
  - suggestions for managing relationships

Sample Stakeholder Analysis

Key Stakeholders					
Organization	Ahmed Internal senior management	Susan Project team	Erik Project team	Mark Hardware vendor	David Project manager for other internal project
Role on project	Sponsor of project and one of the company's founders	DNA sequencing expert	Lead programmer	Supplies some instrument hardware	Competing for company resources
Unique facts	Demanding, likes details, business focus, Stanford MBA	Very smart, Ph.D. in biology, easy to work with, has a toddler	Best programmer I know, weird sense of humor	Start-up company, he knows we can make him rich if this works	Nice guy, one of oldest people at company, has 3 kids in college
Level of interest	Very high	Very high	High	Very high	Low to medium
Level of influence	Very high; can call the shots	Subject matter expert; critical to success	High; hard to replace	Low; other vendors available	Low to medium
Suggestions on managing relationship	Keep informed, let him lead conversations ,do as he says and quickly	Make sure she reviews specifications and leads testing; can do some work from home	Keep him happy so he stays; emphasize stock options; likes Mexican food	Give him enough lead time to deliver hardware	He knows his project takes a back seat to this one, but I can learn from him

Project Plan Execution

- Project plan execution involves managing and performing the work described in the project plan
- The majority of time and money is usually spent on execution
- The application area of the project directly affects project execution because the products of the project are produced during execution

What Went Wrong?

- Many people have a poor view of plans based on past experiences. Senior managers often require a plan, but then no one follows up on whether the plan was followed.
- For example, one project manager said he would meet with each project team leader within two months to review their plans. The project manager created a detailed schedule for these reviews. He cancelled the first meeting due to another business commitment. He rescheduled the next meeting for unexplained personal reasons. Two months later, the project manager had still not met with over half of the project team leaders.
- Why should project members feel obligated to follow their own plans when the project manager obviously did not follow his?

Important Skills for Project Execution

- General management skills like leadership, communication, and political skills
- Product skills and knowledge (see example of “What Went Right?” on p. 120)
- Use of specialized tools and techniques

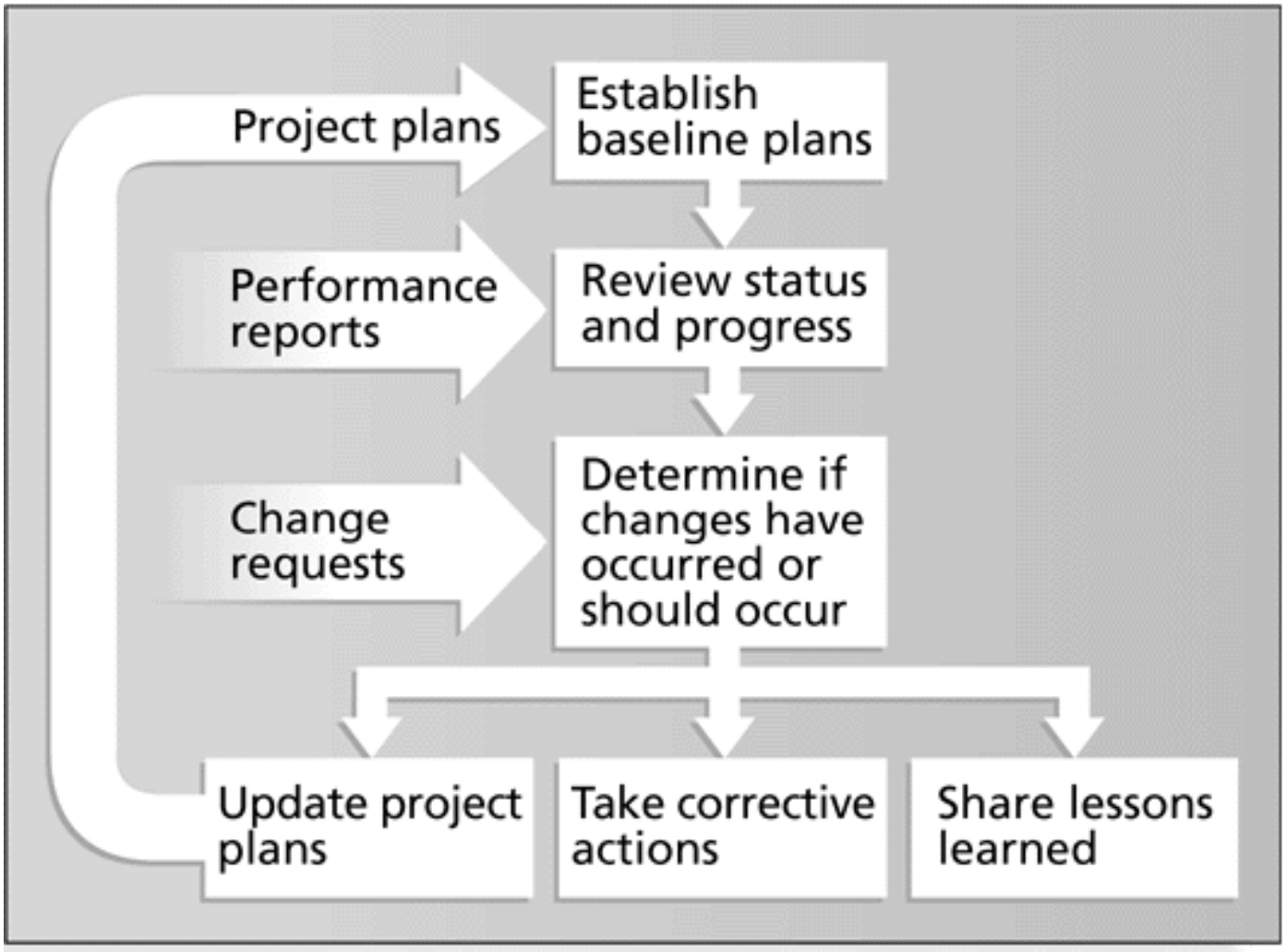
Tools and Techniques for Project Execution

- **Work Authorization System:** a method for ensuring that qualified people do work at the right time and in the proper sequence
- **Status Review Meetings:** regularly scheduled meetings used to exchange project information
- **Project Management Software:** special software to assist in managing projects

Integrated Change Control

- Integrated change control involves identifying, evaluating, and managing changes throughout the project life cycle (Note: 1996 PMBOK called this process “overall change control”)
- Three main objectives of change control:
  - Influence the factors that create changes to ensure they are beneficial
  - Determine that a change has occurred
  - Manage actual changes when and as they occur

Integrated Change Control Process



Change Control on Information Technology Projects

- Former view: The project team should strive to do exactly what was planned on time and within budget
- Problem: Stakeholders rarely agreed up-front on the project scope, and time and cost estimates were inaccurate
- Modern view: Project management is a process of constant communication and negotiation
- Solution: Changes are often beneficial, and the project team should plan for them

Change Control System

- A formal, documented process that describes when and how official project documents and work may be changed
- Describes who is authorized to make changes and how to make them
- Often includes a change control board (CCB), configuration management, and a process for communicating changes

Change Control Boards (CCBs)

- A formal group of people responsible for approving or rejecting changes on a project
- CCBs provide guidelines for preparing change requests, evaluate change requests, and manage the implementation of approved changes
- Includes stakeholders from the entire organization

Making Timely Changes

- Some CCBs only meet occasionally, so it may take too long for changes to occur
- Some organizations have policies in place for time-sensitive changes
  - "48-hour policy" allows project team members to make decisions, then they have 48 hours to reverse the decision pending senior management approval
  - Delegate changes to the lowest level possible, but keep everyone informed of changes

Configuration Management

- Ensures that the products and their descriptions are correct and complete
- Concentrates on the management of technology by identifying and controlling the functional and physical design characteristics of products
- Configuration management specialists identify and document configuration requirements, control changes, record and report changes, and audit the products to verify conformance to requirements

Suggestions for Managing Integrated Change Control

- View project management as a process of constant communications and negotiations
- Plan for change
- Establish a formal change control system, including a Change Control Board (CCB)
- Use good configuration management
- Define procedures for making timely decisions on smaller changes
- Use written and oral performance reports to help identify and manage change
- Use project management and other software to help manage and communicate changes

Using Software to Assist in Project Integration Management

- Several types of software can be used to assist in project integration management
  - Documents can be created with word processing software
  - Presentations are created with presentation software
  - Tracking can be done with spreadsheets or databases
  - Communication software like e-mail and Web authoring tools facilitate communications
  - Project management software can pull everything together and show detailed and summarized information (see Appendix A for details)

ResNet Summary Gantt Chart

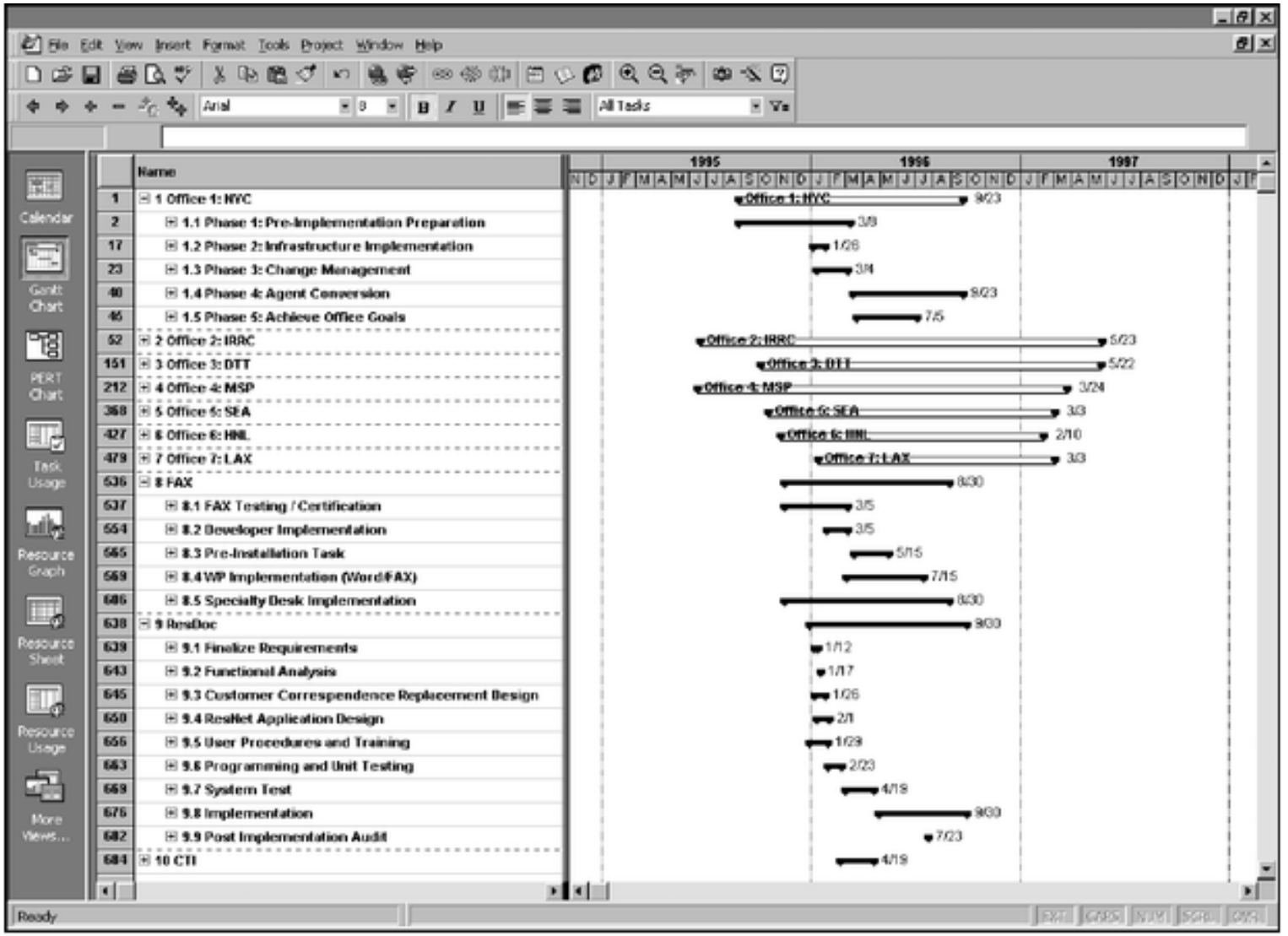


Figure 4-4. Summary Gantt Chart for ResNet 1996 Office Implementation

Questions for discussion

- Review and discuss the opening case. Answer the following questions:
  - What was the real problem in this case?
  - Was Nick a good project manager?
  - Why or why not?
  - What could he have done to improve the situation, and what should senior management have done?

Project Scope Management

By

Casmir, R. (Ph.D)

What is a Project Scope?

- The project scope is a document that defines the following parameters:

  —factors that define a system and determine behaviour of the project,

  —what work is done within the boundaries of the project, and

  — the work that is outside the project boundaries

- The scope of work (SOW) is typically a written document that defines

  —what work will be accomplished by the end of the project

  —the deliverables of the project.

- NOTE: No template works for all projects. Some projects have a very detailed scope of work, and some have a short summary document.

What is the Quality of a Project Scope?

- The quality of the scope is measured by — the ability of the project manager and project stakeholders to develop and maintain a common understanding of what products or services the project will deliver.
- The size and details of the project scope is related to the complexity profile of the project.

Overview of a Project Scope

According to the PMI, the scope statement should include the following:

- Description of the scope
- Product acceptance criteria
- Project deliverables
- Project exclusions
- Project constraints
- Project assumptions

Overview of a Project Scope

- The scope document is the basis for agreement by all parties.
- A clear project scope document is also critical to managing change on a project.
- Since the project scope reflects what work will be accomplished on the project, any change in expectations that is not captured and documented creates the opportunity for confusion.

Overview of a Project Scope

- One of the most common trends on projects is the incremental expansion in the project scope.
- This trend is labeled “scope creep.”
- Scope creep threatens the success of a project because the small increases in scope require additional resources that were not in the plan.
- Increasing the scope of the project is a common occurrence, and
- adjustments are made to the project budget and schedule to account for these changes
- Scope *creep* occurs when these changes are not recognized or not managed.
- The ability of a project manager to identify potential changes is often related to the quality of the scope documents.

Overview of a Project Scope

- Events do occur that require the scope of the project to change.
- Changes in the marketplace may require change in a product design or the timing of the product delivery.
- Changes in the client's management team or the financial health of the client may also result in changes in the project scope.
- Changes in the project schedule, budget, or product quality will have an effect on the project plan.
- Generally, the later in the project the change occurs, the greater the increase to the project costs.

Overview of a Project Scope

- Establishing a change management system for the project that captures changes to the project scope and assures that these changes are authorized by the appropriate level of management in the client's organization. This is a responsibility of the project manager.
- The project manager also analyzes the cost and schedule impact of these changes and adjusts the project plan to reflect the changes authorized by the client.
- Changes to the scope can cause costs to increase or decrease.

Overview of a Project Charter

- A project charter should only include three major elements: that is

1) project objectives,
2) Project scope, and
3) Project Responsibilities.

- Once your charter has been approved, you should then create a project plan.
- Your project plan builds on your project charter to provide a more in-depth blueprint of the key elements of your project.

Overview of a Project Constraints

The PMBOK Guide recognizes six project constraints:

- **scope,**
- **quality,**
- **schedule,**
- **budget,**
- **resources, and**
- **risk**. Out of these six, scope, schedule, and budget are known as the triple constraints and constitutes to the project quality.

Overview of a Project Assumptions

- An assumption is what you believe to be true.
- These are anticipated events or circumstances that are expected during your project’s life cycle.
- You make assumptions based on your experience or the information available on hand.
- Assumptions may not end up being true.
- In fact, they can sometimes be false and may negatively affect your project, adding risk to the project.

Overview of a Project Deliverables

- A deliverable is a tangible or intangible good or service produced as a result of a project that is intended to be delivered to a customer or client.
- A deliverable could be a report, a document, a software product, a server upgrade or any other building block of an overall project

Project Time Management

By

Casmir, R. (Ph.D)

Project Time Management

Time management is crucial in project management because it ensures projects are completed on time, within budget, and to the satisfaction of stakeholders

Project Time Management includes the processes required to manage timely completion of the project.

- **Plan schedule** management – The process of establishing the policies, procedures and documentation for planning, developing, managing, executing and controlling the project schedule
- **Define Activities**—The process of identifying the specific actions to be performed to produce the project deliverables.
- **Sequence Activities** —The process of identifying and documenting relationships among the project activities.
- **Estimate Activity** Resources—The process of estimating the type and quantities of material, people, equipment, or supplies required to perform each activity.
- **Estimate Activity Durations**—The process of approximating the number of work periods needed to complete individual activities with estimated resources.
- **Develop Schedule—The** process of analyzing activity sequences, durations, resource requirements, and schedule constraints to create the project schedule.
- **Control Schedule** —The process of monitoring the status of the project to update project progress and managing changes to the schedule baseline.

Plan Schedule Management

Plan schedule Management is the process of establishing the policies, Procedures, and documentation for planning, developing, managing and controlling the project schedule

The key benefit of this process is that it provides direction on how the project schedule will be managed throughout the project

Plan Schedule Management

inputs	Tools and techniques	outputs
❑ Project Management Plan ❑ Project Charter ❑ Enterprise environmental factors ❑ Organization process	❑ Expert Judgment ❑ Analytical techniques ❑ Meetings	❑ Schedule management plan

Plan Schedule Management

The schedule management plan can establish the following:

- Project schedule model development. The scheduling methodology and the scheduling tool to be used
- Level of accuracy. The acceptable range used in determining realistic activity duration estimates is
- Units of measure. Each unit used in measurements (such as staff hours, staff days, or weeks for tim
- Organizational procedures links. The WBS provides the framework for the schedule
- Project schedule model maintenance. The process used to update the status and record progress of

Plan Schedule Management

- Control thresholds. Variance thresholds for monitoring schedule performance may be specified to indicate
- An agreed-upon amount of variation to be allowed before some action needs to be taken. Thresholds are typically expressed as percentage deviations from the parameters established in the baseline plan.

Define Activities

Define Activities is the process of identifying and documenting the specific actions to be performed to produce the project deliverables.

The key benefit of this process is to break down work packages into Activities that provide a basis for estimating, scheduling, executing, Monitoring , and controlling the project work.

Define Activities

Inputs	Tools and Techniques	Outputs
❑ Schedule management plan ❑ Scope Baseline ❑ Enterprise Environment Factors ❑ Organization Process Assets	❑ Decomposition ❑ Rolling Wave Planning ❑ Expert Judgment	❑ Activity List ❑ Activity Attributes ❑ Milestone List

Define Activities : Inputs

Schedule Management Plan - provides direction on how the project schedule will be managed throughout the project.

Scope Baseline provides the project deliverables, assumptions and constraints

Enterprise Environmental Factors that can influence the Define Activities process include, but are not limited to, the project management information system (PMIS).

Organizational Process Assets that can influence the Define Activities process include, but are not limited to:

- Existing formal and informal activity planning-related policies, procedures, and guidelines, such as the scheduling methodology, that are considered in developing the activity definitions, and
- Lessons-learned knowledge base containing historical information regarding activities lists used by previous similar projects.

Define Activities : Tools and Techniques

Decomposition involves subdividing the project work packages into smaller, more manageable components called activities.

Rolling Wave Planning is a form of progressive elaboration planning where the work to be accomplished in the near term is planned in detail and future work is planned at a higher level of the WBS.

Templates is a standard activity list or a portion of an activity list from a previous project is often usable as a template for a new project. The related activity attributes information in the templates can also contain other descriptive information useful in defining activities.

Define Activities : Outputs

Activity List is a comprehensive list including all schedule activities required on the project.

Activity attributes depicts the multiple components associated with each activity

- Activity ID, WBS ID, Activity Name, activity description, predecessor activities, successor activities, logical relationships, leads and lags, resource requirements, imposed dates, constraints, and assumptions.
- Activity attributes can be used to identify the person responsible for executing the work, geographic area, or place where the work has to be performed, and activity type such as level of effort (LOE), discrete effort, and apportioned effort (AE).

Milestone List identifies all milestones and indicates whether the milestone is mandatory, such as those required by contract, or optional, such as those based upon historical information.

Sequence Activities

Sequence Activities is the process of Identifying and documenting relationships among the project activities.

The key benefit of this Process is that it defines the logical sequence of Work to obtain the greatest efficiency given all project constraint.

Sequence Activities

❑ Schedule management plan ❑ Activity List ❑ Activity Attributes ❑ Milestone List ❑ Project Scope Statement ❑ Enterprise environmental Factors ❑ Organization Process Assets	❑ Precedence Diagramming Method (PDM) ❑ Dependency determination ❑ Leads and Lags	❑ Project Schedule Network Diagrams ❑ Project Document Updates

Sequence Activities - Inputs

Schedule Management Plan - provides direction on how the project schedule will be managed throughout the project.

Activity List – Output of Define Activities process

Activity Attributes – Output of Define Activities process

Milestone List - Output of Define Activities process

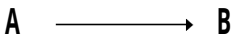
Project Scope Statement – Output of Define Scope Process

Organizational Process Assets - The organizational process assets that can influence the Sequence Activities process include, but are not limited to, project files from the corporate knowledge base used for scheduling methodology.

Sequence Activities – Tools and Techniques

Precedence Diagramming Method (PDM)- Also known as activity on node (AON) as the activities are represented on nodes and the logical relationship between activities is represented through arrows

The relationship in the figure below represents that the Activity B is dependent on Activity A. i.e. Activity B can only start after the completion of activity A



Sequence Activities – Tools and Techniques

Types of dependencies in PDM

Finish – to – Start (FS) – This relationship means Task A must complete before Task B can begin. This is the most common relationship.

Finish – to – Finish (FF) - Finish of successor activity depends on finish of the predecessor

Start-to-Start (SS) – Start of a successor activity depends on start of a predecessor activity.

Start- to – Finish (SF) – Finish of Successor depends on the start of preceding activity.

Sequence Activities – Tools and Techniques

Dependency Determination

There are three kinds of dependencies, Mandatory dependency, Discretionary dependency and external dependency

Mandatory dependencies referred as hard logic are natural order of activities or which are contractually required.

Discretionary dependencies referred as preferred logic, soft logic or preferential logic are defined as best practices and are considered the preferred choice among the other alternatives.

External dependencies are external to the project and are not in project's control. E.g. Deliverable from other project or expected new law etc

Estimate Activity Resources

Estimate Activity Resources is the process of estimating the type and quantities of material, human resources, Equipment, or supplies required to perform each activity.

The key benefit of this process is that it identifies the type, Quantity, and characteristics of resources required to Complete the activity which allows more accurate cost and duration estimate

Estimate Activity Resources

Inputs	Tools and techniques	Output
❑ Schedule Management Plan ❑ Activity List ❑ Activity Attributes ❑ Resource Calendars ❑ Risk Register ❑ Enterprise Environmental Factors ❑ Organization Process Assets	❑ Expert judgment ❑ Alternatives analysis ❑ Published estimating data ❑ Bottom-up estimating ❑ Project management software	❑ Activity resource requirements ❑ Resource breakdown structure ❑ Project management updates

Estimate Activity Resources - Inputs

Schedule Management Plan – Identifies the level of accuracy and the units of measure for the resource to be estimated

Activity List – Identifies the activities which will need resources

Activity Attributes – Provide the primary data input for use in estimating those resources required for each activity in the activity list.

Resource Calendar – Resource calendars specify when and how long identified project resources will be available during the project

Enterprise Environmental Factors- The enterprise environmental factors that can influence the Estimate Activity Resources process include but are not limited to resource availability and skills

Organizational Process Assets – that influence this process include but are not limited to; Policies and Procedures related to staffing, policies and procedures related to rental purchase of supplies and equipments, Historical information regarding types of resources used for similar work on previous projects

Estimate Activity Resources – Tools and Techniques

Expert Judgment is often required to assess the resource-related inputs to this process. Any group or person with specialized knowledge in resource planning and estimating can provide such expertise.

Alternatives Analysis

Many schedule activities have alternative methods of accomplishment. They include using various levels of resource capability or skills, different size or type of machines, different tools (hand versus automated), and make-or-buy decisions regarding the resource.

Published Estimating Data

Several companies routinely publish updated production rates and unit costs of resources for an extensive array of labor trades, material, and equipment for different countries and geographical locations within countries.

Estimate Activity Resources – Tools and Techniques

Bottom-Up Estimating When an activity cannot be estimated with a reasonable degree of confidence, the work within the activity is decomposed into more detail. The resource needs are estimated. These estimates are then aggregated into a total quantity for each of the activity's resources. Activities may or may not have dependencies between them that can affect the application and use of resources. If there are dependencies, this pattern of resource usage is reflected and documented in the estimated requirements of the activity.

Project Management Software Project management software has the capability to help plan, organize, and manage resource pools and develop resource estimates. Depending on the sophistication of the software, resource breakdown structures, resource availability, resource rates and various resource calendars can be defined to assist in optimizing resource utilization.

Estimate Activity Resources – Outputs

Activity Resource Requirements identifies the types and quantities of resources required for each activity in a work package. These requirements can then be aggregated to determine the estimated resources for each work package. The amount of detail and the level of specificity of the resource requirement descriptions can vary by application area. The resource requirements documentation for each activity can include the basis of estimate for each resource, as well as the assumptions that were made in determining which types of resources are applied, their availability, and what quantities are used.

Resource Breakdown Structure is a hierarchical structure of the identified resources by resource category and resource type. Examples of resource categories include labor, material, equipment, and supplies. Resource types can include the skill level, grade level or other information as appropriate to the project. The resource breakdown structure is useful for organizing and reporting project schedule data with resource utilization information.

Project Documents may be updated include, but are not limited to:

- Activity list,
- Activity attributes, and
- Resource calendars.

Estimate Activity Duration

Estimate Activate Durations is the process of estimating the Number of work periods needed to complete individual Activities with estimated resources.

The key benefit of this process is that it provides the amount of time each activity will take to complete, which is a major input to Develop schedule process.

Inputs	Tools and Techniques	Outputs
❑ Schedule management plan ❑ Activity List ❑ Activity Attributes ❑ Activity Resource Requirements ❑ Resource Calendars ❑ Project Scope Statement ❑ Risk Register ❑ Resource Breakdown Structure ❑ Enterprise Environmental Factors ❑ Organization Process Assets	❑ Expert judgment ❑ Analogous Estimating ❑ Parametric Estimating ❑ Three point estimates ❑ Group decision making techniques ❑ Reserve Analysis	❑ Activity duration updates ❑ Project document updates

Estimate Activity Duration - Inputs

Schedule Management Plan – defines the method used and the level of accuracy along with other criteria required to estimate activity durations including the project update cycle**.**

Activity List – Defines activity list that will need duration estimates.

Activity Attributes- Output of Define Activities process

Activity Resource Requirements –The number of resources allocated will impact the duration of most of the activities.

Resource Calendars - can include the type, availability, and capabilities of human resources. The type, quantity, availability, and capability, when applicable, of both equipment and material resources, which could significantly influence the duration of schedule activities, are also considered. Examples such as Existing conditions, Availability information and length of the reporting periods. Examples of constraints include , Available skilled resources and contract terms and requirement.

Estimate Activity Duration - Inputs

Project Scope Statement- The assumptions and constraints from the project scope statement are considered when estimating the activity durations. This includes

- Existing conditions
- Availability of information
- Length of the reporting periods

Examples of constraints include, but are not limited to

- Available skilled resources
- Contract terms and requirements

Enterprise Environmental Factors that can influence the Estimate Activity Durations process include, but are not limited to: Duration estimating databases and other reference data, Productivity metrics, and Published commercial information.

Organization Process Assets that can influence the Estimate Activity Durations process include but are not limited to: Historical duration information, Project calendars, Scheduling methodology and Lessons learned.

Estimate Activity Duration – Tools and Techniques

- **Expert Judgment** guided by historical information, can provide duration estimate information or recommended maximum activity durations from prior similar projects. Expert judgment can also be used to determine whether to combine methods of estimating and how to reconcile differences between them.

- **Analogous estimating** uses parameters such as duration, budget, size, weight, and complexity, from a previous, similar project, as the basis for estimating the same parameter or measure for a future project.

- **Parametric Estimating** Parametric estimating uses a statistical relationship between historical data and other variables (e.g., square footage in construction) to calculate an estimate for activity parameters, such as cost, budget, and duration.

- **Reserve Analysis** include contingency reserves, (sometimes referred to as time reserves or buffers) into the overall project schedule to account for schedule uncertainty. The contingency reserve may be a percentage of the estimated activity duration, a fixed number of work periods, or may be developed by using quantitative analysis methods.

Estimate Activity Duration – Tools and Techniques

Three-Point Estimates (PERT) is based on statistical methods in particular normal distribution. The estimation of activity duration is done based on the following three points

Most likely (M). The duration of the activity, given the resources likely to be assigned, their productivity, realistic expectations of availability for the activity, dependencies on other participants, and interruptions

Optimistic (O) . The activity duration is based on analysis of the best-case scenario for the activity.

Pessimistic (P). The activity duration is based on analysis of the worst-case scenario for the activity.

PERT analysis calculates an **Expected** (E) activity duration using a weighted average of these three estimates:

Double Triangular or Beta Distribution - E = O + 4M + P/6 , Standard Deviation - SD= P-0/6
Triangular Distribution – E= 0+M+P/3 , Standard Deviation =√ 02+M2+P2-OM-MP-PO/18

Develop Schedule

Develop schedule is the process of analyzing activity sequences, durations, resource requirements, and schedule constraints to create the project schedule model.

The key benefits of this process is that by entering schedule activities , durations, resources, resource availabilities and Logical relationships into the scheduling tool , it generates a schedule model with planned dates for completing project activities

Develop Schedule

Inputs	Tools and Techniques	Outputs
❑ Schedule management ❑ Activity List ❑ Activity attributes ❑ Project schedule network diagrams ❑ Activity resource requirements ❑ Resource calendars ❑ Activity duration estimates ❑ Project scope statement ❑ Project staff assignments ❑ Resource Breakdown Structure ❑ Risk Register ❑ Enterprise Environmental Factors ❑ Organizational Process Assets	❑ Schedule network analysis ❑ Critical path method ❑ Critical chain method ❑ Resource optimization techniques ❑ Modeling techniques ❑ Leads and lags ❑ Schedule compression ❑ Scheduling tool	❑ Project schedule ❑ Schedule baseline ❑ Schedule data ❑ Project Calendars ❑ Project Management Plan updates ❑ Project document updates

Develop Schedule - Inputs

Schedule Management Plan – Identifies the scheduling method and tool used to create the schedule and how schedule is to be calculated.

Activity List – Identifies the activities that will be included in the schedule model **Activity attributes** – Provide the details used to build the schedule model.

Project schedule network diagram – contains logical relationship between predecessor and successor that will be used to build the schedule.

Activity resource requirements – Identify the types and quantities of resources required for each activity used to create the schedule model.

Resource calendars- specify when and how long identified project resources will be available during the project

Activity duration estimates contain the quantitative assessment of the likely number of work periods that will be required to complete an activity that will be used to calculate the schedule.

Develop Schedule - Inputs

Project scope statement- contains assumptions and the constraints that can impact the development of the schedule.

Enterprise Environmental Factors that can influence the Develop Schedule process include, but are not limited to, a scheduling tool that can be used in developing the schedule.

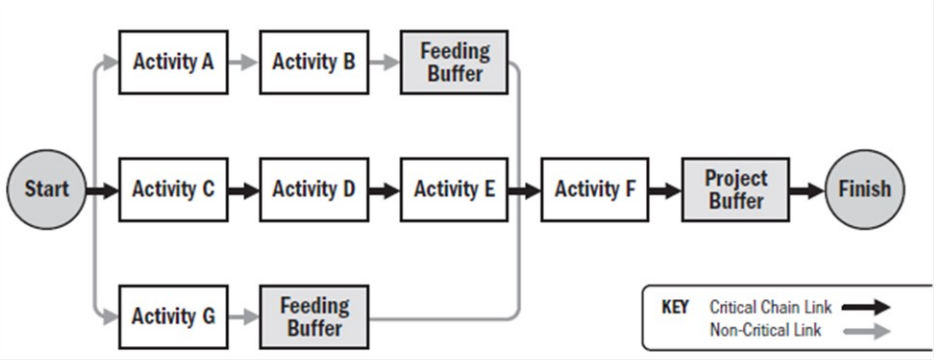
Organizational Process Assets The organizational process assets that can influence the Develop Schedule process include, but are not limited to:

Scheduling methodology and Project calendar.

Develop Schedule Tools and Techniques

The critical path method calculates the theoretical early start and finish dates, and late start and finish dates, for all activities without regard for any resource limitations, by performing a forward and backward pass analysis through the schedule network.

Critical Chain Method is a schedule network analysis technique that modifies the project schedule to account for limited resources. The resource-constrained critical path is known as the critical chain. The critical chain method adds duration buffers that are non-work schedule activities to manage uncertainty.



Develop Schedule Tools and Techniques

Resource Optimization Techniques – that can be used to Adjust the schedule model due to demand and supply of resources include.

Resource Levelling - Is necessary when resources have been over-allocated, such as when a resource has been assigned to two or more activities during the same time period, when shared or critical required resources are only available at certain times or are only available in limited quantities. Resource leveling can often cause the original critical path to change.

Resource Smoothing – A technique that adjusts the activities of a schedule model such that the requirements for resources on the project do not exceed certain predefined resource limits. There is no change in the project completion date the activities may only be delayed within their free and total float.

Develop Schedule Tools and Techniques

Modeling Techniques include

What -If Scenario Analysis- A schedule network analysis is performed using the schedule to compute the different scenarios, such as delaying a major component delivery, or introducing external factors, such as a strike The outcome of the what if scenario analysis can be used to assess the Feasibility of the project schedule under adverse conditions

Simulation involves calculating multiple project durations with different sets of activity assumptions. The most common technique is Monte Carlo Analysis in which a distribution of possible activity durations is defined for each activity and used to calculate a distribution of possible outcomes for the total project.

Develop Schedule Tools and Techniques

Applying Leads and Lags - Leads and lags are refinements applied during network analysis to develop a viable schedule.

Schedule Compression

Crashing. A schedule compression technique in which cost and schedule tradeoffs are analyzed to determine how to obtain the greatest amount of compression for the least incremental cost. Examples of crashing could include approving overtime, bringing in additional resources, or paying to expedite delivery to activities on the critical path. Crashing only works for activities where additional resources will shorten the duration. Crashing does not always produce a viable alternative and may result in increased risk and/or cost.

Fast tracking. A schedule compression technique in which phases or activities normally performed in sequence are performed in parallel. An example is constructing the foundation for a building before completing all of the architectural drawings. Fast tracking may result in rework and increased risk. Fast tracking only works if activities can be overlapped to shorten the duration

Develop Schedule Tools and Techniques

Scheduling Tool - Automated scheduling tools expedite the scheduling process by generating start and finish dates based on the inputs of activities, network diagrams, resources and activity durations. A scheduling tool can be used in conjunction with other project management software applications as well as manual methods.

Develop Schedule Outputs

- Schedule Baseline - is a real or theoretical construct that captures the approved schedule. It is used to provide a comparison or contrast with the actual progress of work against the schedule and to determine if performance to date is within acceptable parameters.

- Project schedule displays the start and end dates of all the activities in the project and the same for overall project. It can be represented in the form of Gantt Chart, Milestone Chart or a project network diagram.

- Schedule data is the data on the basis of which schedule was developed. It includes resource requirements by time period, alternative schedules such as best-case or worst case, not resource leveled or resource leveled with, or without imposed dates and scheduling of contingency reserves

Develop Schedule Outputs

Project Calendars – A project calendar identifies working days and shifts that are available for scheduled activities. It distinguishes time period in days or part of days that are available to complete scheduled activities from time period that are not available.

Project Management Plan Updates – Elements of the project management plan that may be updated include

- Schedule baselines
- Schedule management plan

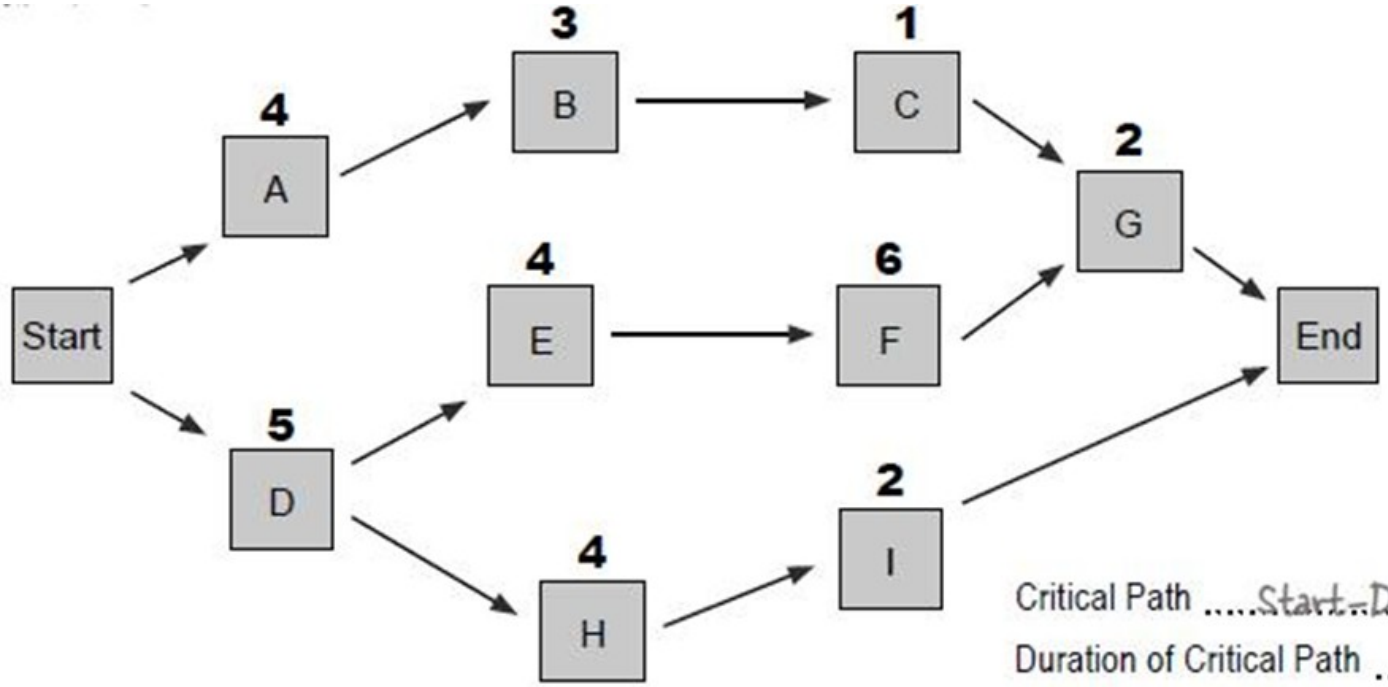
Project document updates includes updates to resource requirement, activity attributes, risk registers and calendars

The **critical path** method is a technique that allows you to identify tasks that are necessary for project completion. The critical path in project management is the longest sequence of activities that must be finished on time to complete the entire project. Any delays in critical tasks will delay the rest of the project.

Float, or slack, refers to the amount of flexibility of a given task. It indicates how much the task can be delayed without impacting subsequent tasks or the project end date.

Exercise : Critical Path

For the network diagram below find Critical Path, float for activity A and E

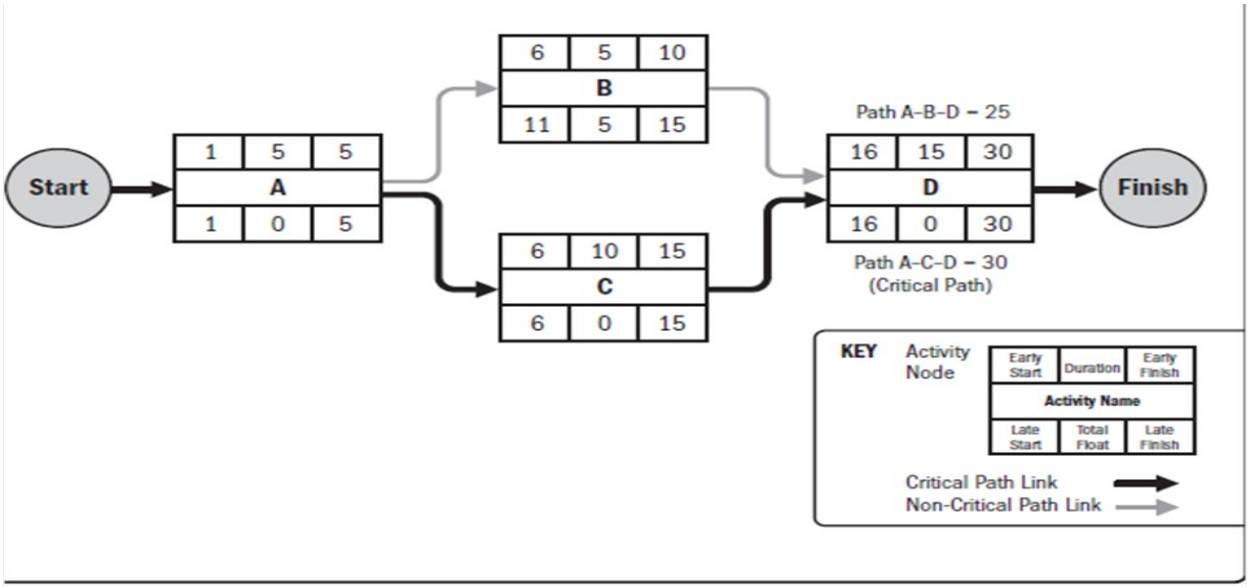


Critical Path ....Start-D-E-F-G-End..

Duration of Critical Path .....17.....

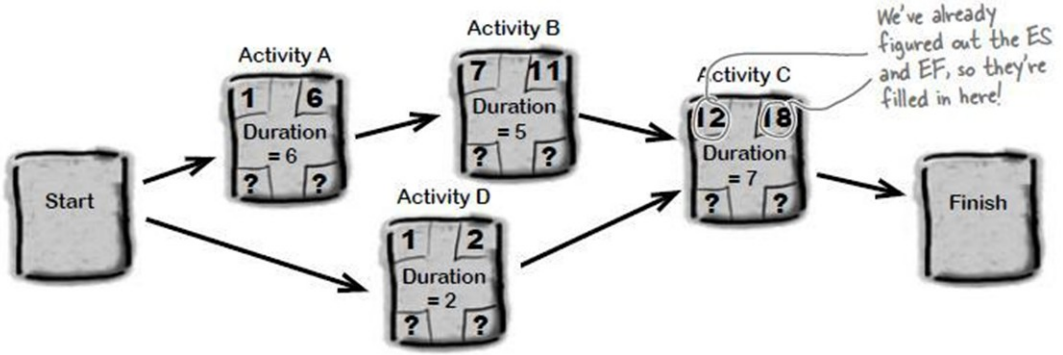
Total Number of Paths .....3.....

Exercise : Forward and Backward Pass



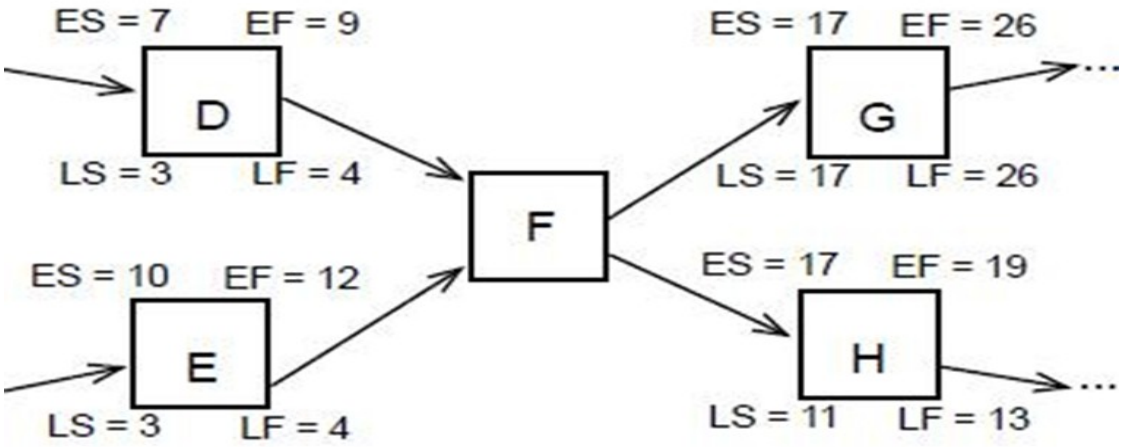
Exercise : Critical Path

Complete the following Project Network Diagram



Exercise : Critical Path

Based on the portion of network diagram shown below find ES and LF for activity F



Control Schedule (Monitoring and Controlling Process Group)

Control Schedule is the process of monitoring the status of the project to update project progress and manage changes to the schedule baseline.

Control Schedule

Control Schedule is the process of monitoring the status of project activities to update project progress and manage changes to the schedule baseline to achieve the plan.

The key benefit of this process is that it provides the means to recognize deviation from the plan and take corrective action and preventive actions and thus minimizes risk.

Control Schedule

Inputs	Tools and Techniques	Outputs
❑ Project management plan ❑ Project schedule ❑ Work performance data ❑ Organizational process assets	❑ Performance reviews ❑ Project management Software ❑ Resource optimization techniques ❑ Modeling techniques ❑ Leads and lags ❑ Schedule compression ❑ Scheduling tool	❑ Work performance information ❑ Schedule forecasts ❑ Change Requests ❑ Project Management Plan updates ❑ Project document updates ❑ Organization Process asset updates

Control Schedule- Inputs

Project Management Plan is a formal, approved document that defines how the project is executed, monitored and controlled.

Project Schedule is the primary input to this process, since it is the source against which the results are controlled.

Work Performance Data details the work that is being executed, recently completed, and next steps. This information is gathered from the Direct and Manage Project execution process.

Project Calendars - A schedule model may require more than one project calendar to allow for different work periods for some activities to calculate the schedule forecasts.

Control Schedule- Inputs

Schedule Data will be reviewed and updated in the Control Schedule process.

Organizational Process Assets can be a system for formally evaluating schedule change, or a software system to help detect changes or measuring variance

Control Schedule- Tools and Techniques

Performance Reviews keeps project manager updated on how project is progressing against the base lined schedule. Some of the techniques that can be used for performance reviews are :

- Trend analysis – examines project performance over time to determine whether performance is improving or deteriorating

- Critical path method – Comparing the progress along the critical path can help determine schedule status. The variance on the critical path will have a direct impact on the project end data. Evaluating the progress of activities on near critical paths can identify schedule risk

- Critical Chain method- Comparing the amount of buffer remaining to the amount of buffer needed to protect the delivery date can help determine schedule status. The difference between the buffer needed and the buffer remaining can determine whether corrective action is appropriate.

- Earned Value Management - Measurements such as SV and SPI helps you to know whether you are ahead or behind in terms of the schedule.

Control Schedule- Tools and Techniques

- **Project Management Software** automates the other tools and techniques in control schedule

- **Resource optimization techniques** involves the scheduling of activities and resources required by those activities while taking into consideration both the resource availability and the project time.

- **Modeling techniques** are used to review various scenarios guided by risk monitoring to bring the schedule model into alignment with the project management plan and approved baseline.

Control Schedule- Tools and Techniques

Leads and lags have impact on the schedule and as the project progresses lead and lags need to be adjusted to reflect reality so as to provide the most realistic view of the schedule possible

Schedule compression includes shortening the duration of the plan through Crashing and Fast tracking

Scheduling tool provides various features (Views, reports etc) that helps in managing the schedule

Control Schedule- Outputs

Work Performance Information– The updated schedule performance index (SPI)

Schedule variance (SV) and other earned value measurements relevant to the schedule need to be calculated and communicated out.

Change Requests – Any change in the schedule due to change in scope or due to any other valid reason requires Change Request.

Project Management Plan Updates - As the schedule changes, the project management plan is updated.

Project Document Updates – This includes any project document (Other than project management plan) that has changed as a result of Control Schedule. For example Scheduled data such as revised planned start and finish dates, Project schedule and the Risk register

Organization Process Updates – Any times a lesson is learned in Control Schedule, it becomes a reusable asset for the performing organization.

Exercise 3: Time Management

Complete the following

1.Adding more resources to a project so you can get it done faster is called_____ .

2._____ is the process where you put the activities in order.

Answers to Exercise 3: Time Management

1. Adding more resources to a project so you can get it done faster is called Crashing .

2. Sequence Activities is the process where you put the activities in order.

Project Time Management

By

Casmir, R. (Ph.D)

Critical Path Method (CPM)

- ***Critical Path Method (CPM)*** is a project schedule modeling technique.
- ***Schedule modeling*** allows one to see the big picture and focus on the primary task to be completed
- A Project ***network diagram*** has many paths originating from one point and ending at another point.
- ***Every path has a duration***, and the one with the longest duration is the critical path.

Critical Path Method (CPM)

- In the *critical path network method*, you focus on managing the critical path.

Critical path may be defined as:

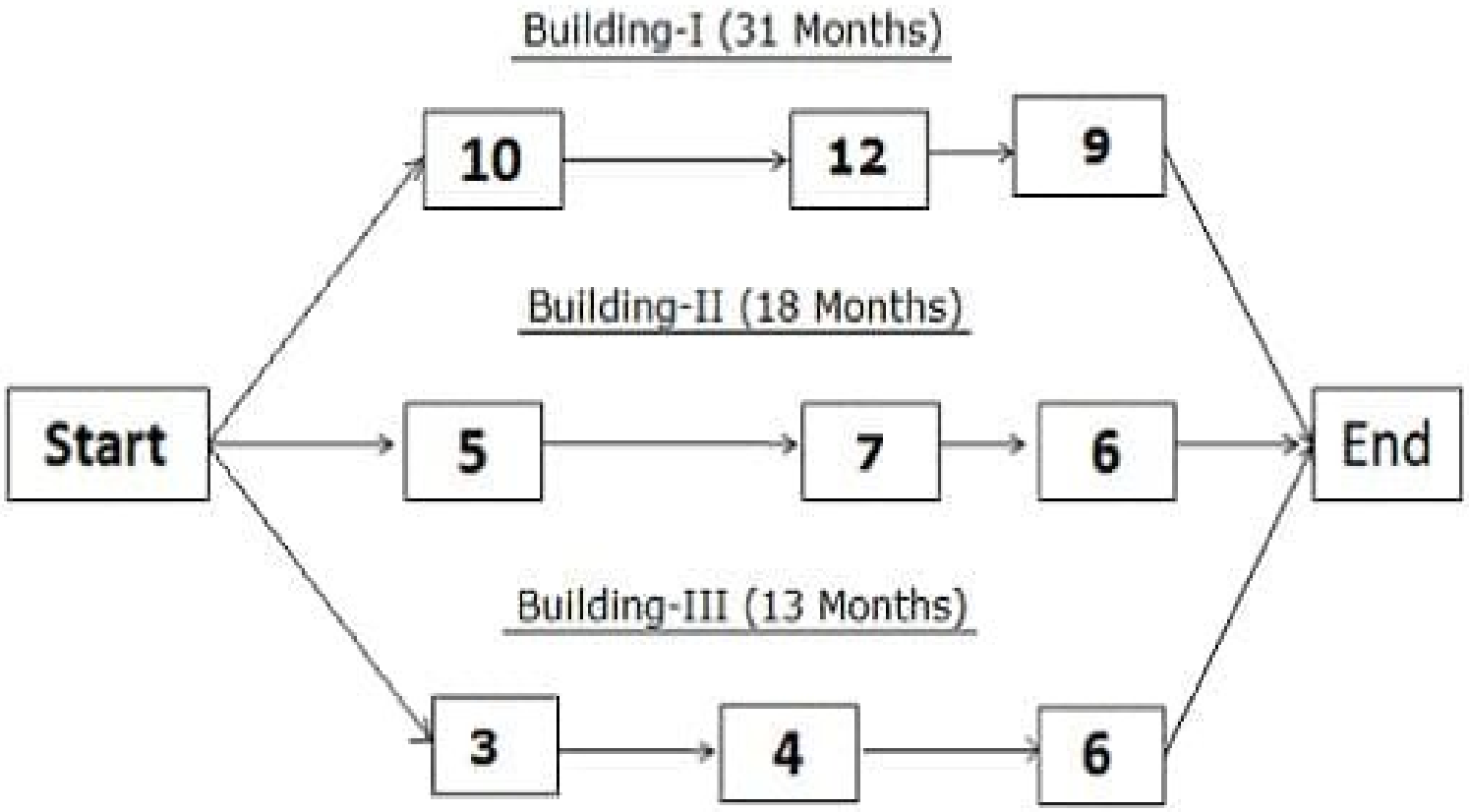
- *The longest path in the network diagram*, or
- *The shortest duration to complete the project*.

- They may appear to be opposites, but they convey the same message.

Critical Path Method (CPM)

- For example, let’s say you have a project to construct three buildings.
- The first is the largest, the second is medium-sized, and the third is the smallest.
- You develop the network diagram, which comprises three paths, each representing a building.

Critical Path Method (CPM)



Critical Path Method (CPM)

Now, let us review the diagram for the critical path analysis.

- Did you notice that the first path is the longest?
- It is 13 months longer than the second and 18 months longer than the third.
- This means that you can wait 13 months and then start working on the second building because you can complete the second building in 18 months.

Critical Path Method (CPM)

- Likewise, you could wait 18 months and then start working on the third building because it will take 13 months to complete.
- This means that, even if you start work on the third building after 18 months from the project start date, you can finish it on time.
- This waiting period is known as float or slack.
- So, which is the critical path in this network diagram?

Critical Path Method (CPM)

- It is the longest path on the network diagram because you cannot complete your project before finishing the first building.
- Although you can complete the other two buildings quickly, your project is not considered complete until the first building is completed.
- This confirms the first statement that says, “the critical path is the longest path on the network diagram.”

Critical Path Method (CPM)

- Now, what is the shortest duration to complete the project?
- It is 31 months because you cannot complete your project in less time, and this is the duration of the critical path.
- This bears out the second statement that says, “The critical path is the shortest duration in which you can complete the project.”
- So, both definitions are the same.

Critical Path Method (CPM)

- You can define the critical path as the sequence of activities from start to end, and it has the longest duration of all paths in a network diagram.
- In ideal conditions, a network diagram should have one critical path.
- Multiple critical paths will put you in a difficult situation.

Critical Path Method (CPM)

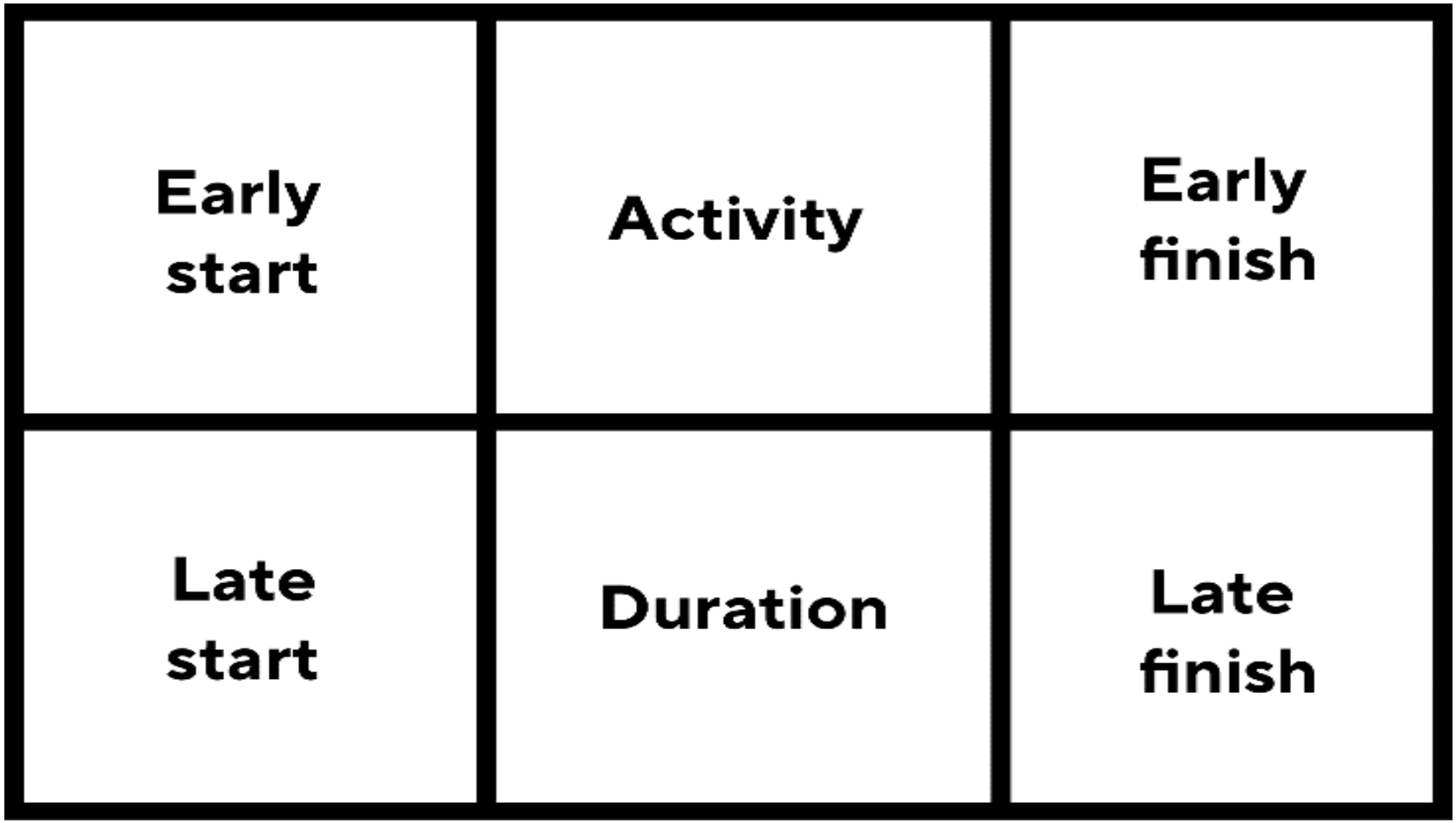
- **Earliest Start Time or Early Start (ES):** This is the earliest time an activity can be started in your project.

- **Latest Start Time or Late Start (LS):** This is the latest time that an activity can be started on your project. If you start the activity beyond this time, it will affect your critical path.

- **Earliest Finish Time or Early Finish (EF):** This is the earliest time an activity is completed in your project.

- **Latest Finish Time of Late Finish (LF):** This is the latest time you can complete the activity on your project. If your activity crosses this time, your project will be delayed**.**

Critical Path Method (CPM)



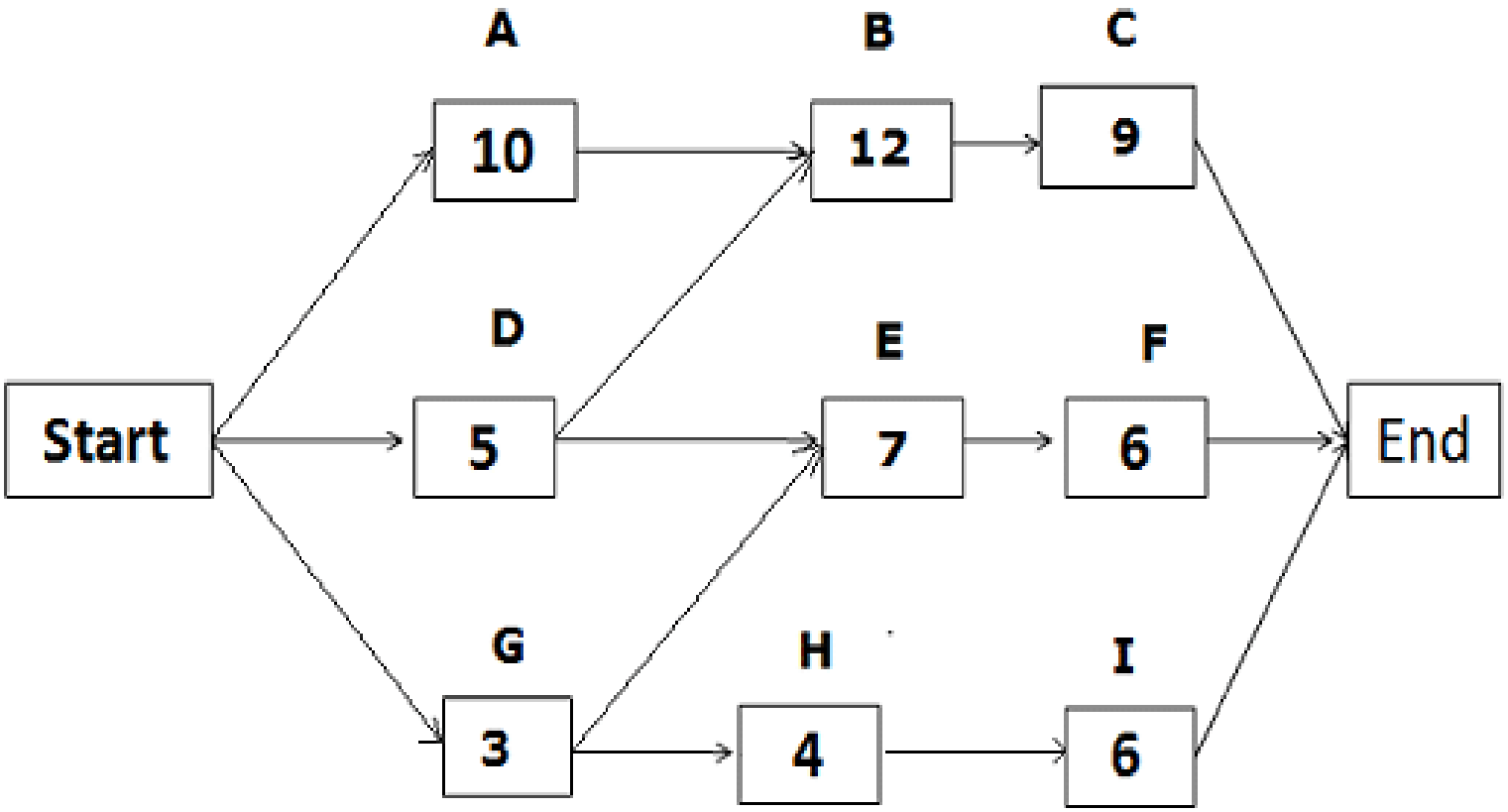
How to Find the Critical Path in a Network Diagram

- The first step is to collect all project activities.
- You can use the work breakdown structure to find tasks for your project.
- After listing project activities, find the task dependencies among the activities.
- Find which activity will follow what activity and task relationship, such as finish to start, finish to finish, start to start, or start to finish.
- Now, draw the network diagram and assign durations to each activity.
- Identify all paths in the network diagram.
- Find the duration of each path.
- The path with the largest duration is the critical path.

How to Find the Critical Path in a Network Diagram

- Use forward pass to calculate the earliest start or early start and earliest finish or early finish of each activity.
- Use backward pass to calculate the latest start or late start and the latest finish or late finish for each activity.
- Now you can calculate the float for all activities.
- **Note** that on a critical path, the early start and late start times are equal, and the early finish and late finish times are also equal for all activities. Also, all activities on the critical path have zero float.
- **Float** is the amount of time an activity can be delayed without affecting the **project completion date**.

Critical Path Method Example



How to Find the Critical Path in a Network Diagram- Example

- Based on the network diagram above, identify the total paths, critical path, and float for each path.
- The above network diagram has five paths. The paths and their durations are as follows:

1. Start -> A -> B -> C-> End {duration: 31 days.}
2. Start ->D -> E ->F -> End {duration: 18 days.}
3. Start -> D -> B -> C -> End {duration: 26 days.}
4. Start -> G ->H ->I -> End {duration: 13 days.}
5. Start -> G -> E ->F -> End {duration: 16 days.}

- Since the duration of the first path is the longest, it is the critical path. The float on the critical path is zero.

How to Find the Critical Path in a Network Diagram- Example

- The float for the second path “Start ->D -> E ->F -> End” = duration of the critical path – duration of the path “Start ->D -> E ->F -> End”
- = 31 – 18 = 13
- Hence, the float for the second path is 13 days.
- Using the same process, we can also calculate the float for other paths.
- Float for the third path = 31 – 26 = 5 days.
- Float for the fourth path = 31 – 13 = 18 days.
- Float for the fifth path = 31 – 16 = 15 days.

Calculating Early Start (ES) and Early Finish (EF)

- To calculate the Early Start and Early Finish dates, we use the forward pass; we will start from the beginning and proceed to the end.

- The Early Start (ES) for the first activity on any path will be 1 because you cannot start the activity before the first day of your project.

- The starting point for any activity is the endpoint of the predecessor activity on the same path (plus one).

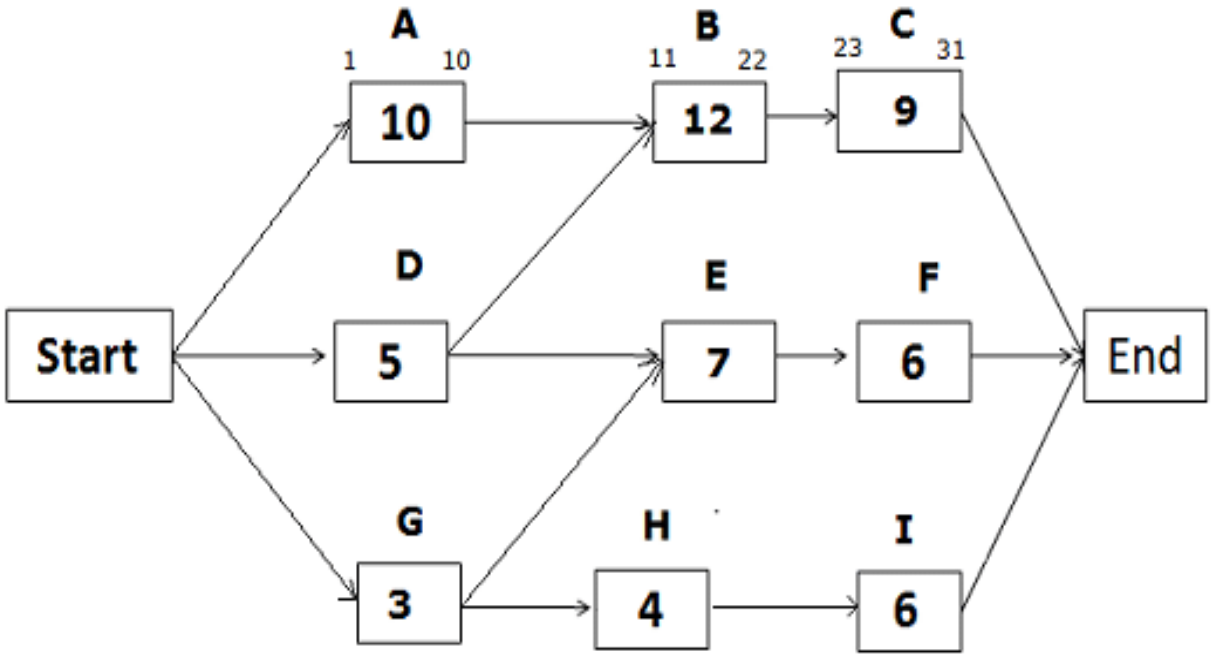
Calculating Early Start (ES) and Early Finish (EF)

- The formula used for calculating Early Start and Early Finish dates:
- Early Start of the activity = Early Finish of predecessor activity + 1
- Early Finish of the activity = Activity duration + Early Start of activity – 1

- *Early Start and Early Finish Dates for the path Start -> A -> B -> C -> End*

Calculating Early Start (ES) and Early Finish (EF)

- The formula used for calculating Early Start and Early Finish dates:
- Early Start of the activity = Early Finish of predecessor activitv + 1

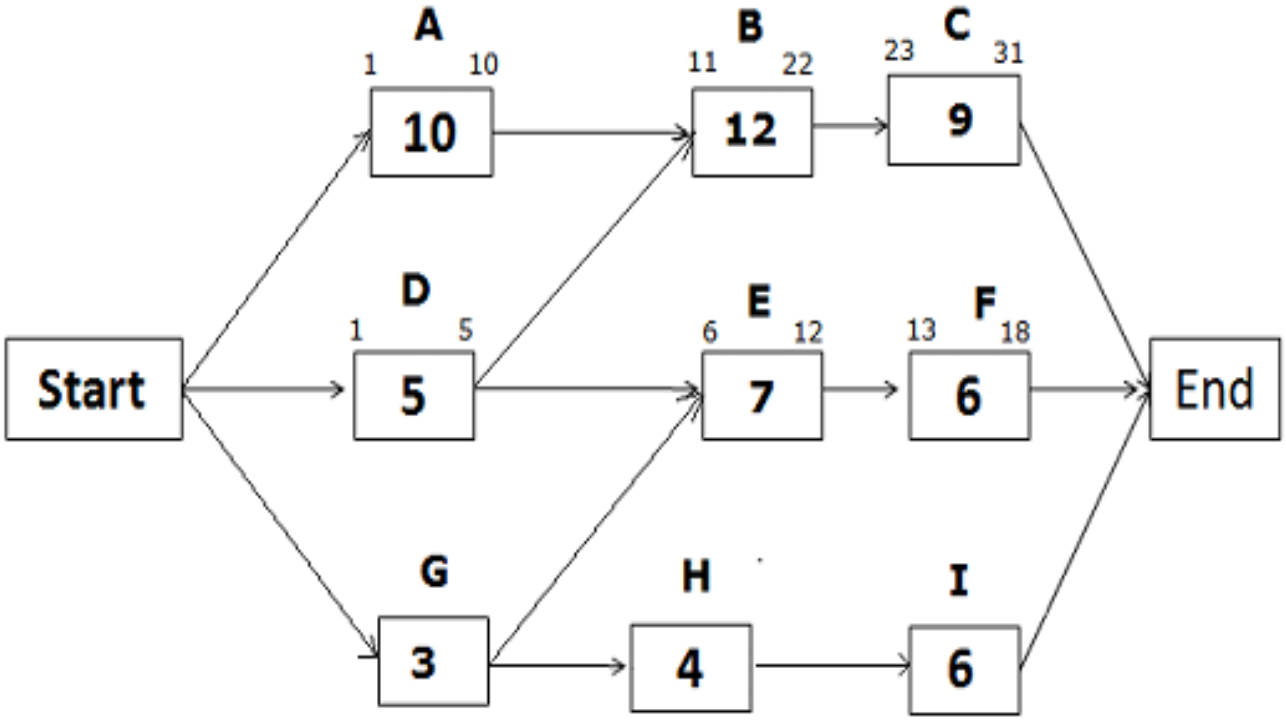


Calculating Early Start (ES) and Early Finish (EF)

- Early Start of activity A = 1 (Since this is the first activity of the path)
- Early Finish of activity A = ES of activity A + activity duration – 1
- = 1 + 10 – 1 = 10
- Early Start of activity B = EF of predecessor activity + 1
- = 10 +1 = 11
- Early Finish of activity B = ES of activity B + activity duration – 1
- = 11 + 12 – 1 = 22
- Early Start of activity C = EF of predecessor activity + 1
- = 22 +1 = 23
- Early Finish of activity C = ES of activity C + activity duration – 1
- = 23 + 9 – 1 = 31

Calculating Early Start (ES) and Early Finish (EF)

- Early Start and Early Finish Dates for the path Start -> D -> E -> F -> End



Calculating Early Start (ES) and Early Finish (EF)

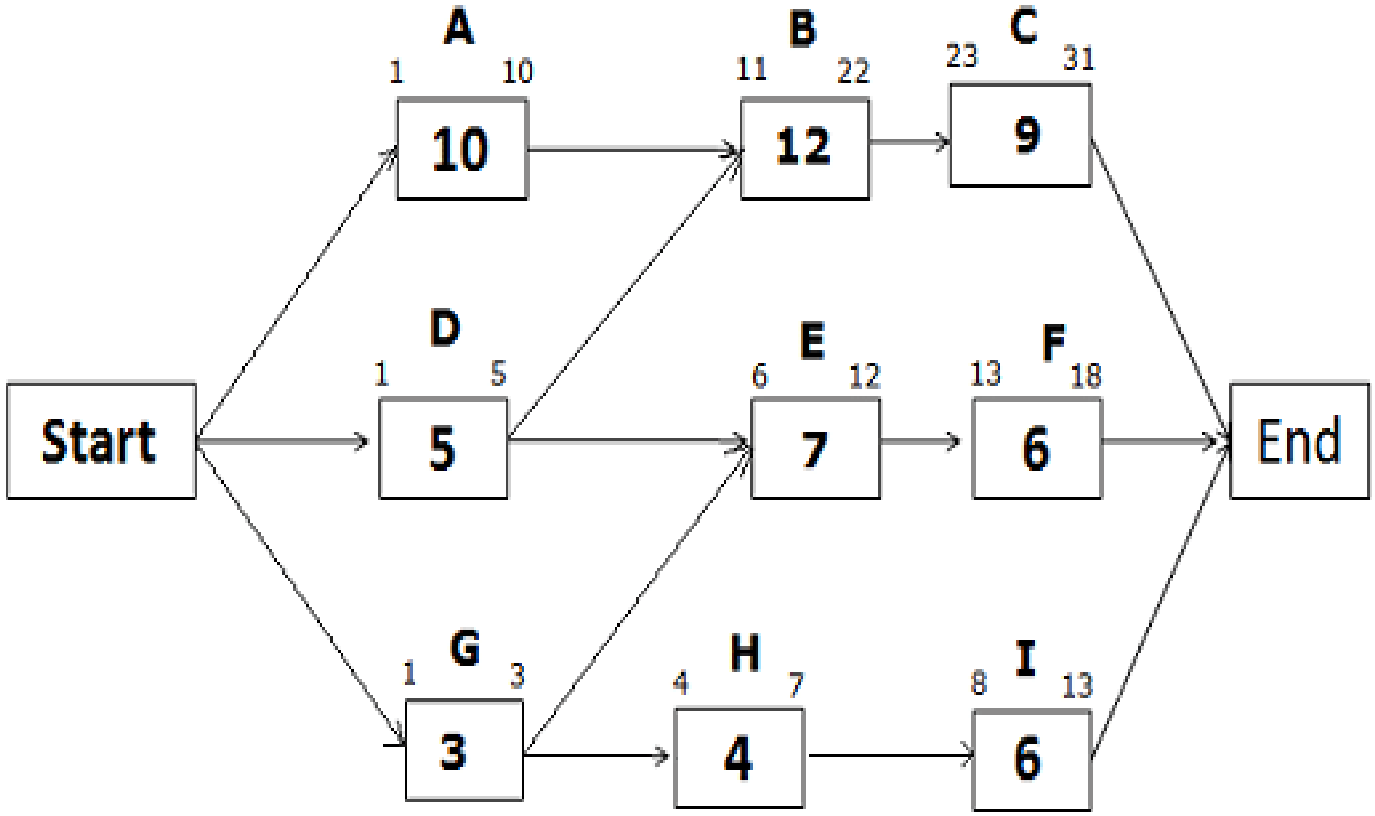
- Early Start of activity D = 1 (Since this is the first activity of the path)
- Early Finish of activity D = 1 + 5 – 1 = 5
- Early Start of activity E = EF of predecessor activity + 1
- Since activity E has two predecessor activities, which one will you select?
- The answer is the activity with the greater Early Finish date. The Early Finish of activity D is 5, and the Early Finish of activity G is 3 (we will calculate it later).

Calculating Early Start (ES) and Early Finish (EF)

- Therefore, we will select the Early Finish of activity D to find the Early Start of activity E.
- Early Start of activity E = EF of predecessor activity + 1
- = 5 + 1 = 6
- Early Finish of activity E = 6 + 7 – 1 = 12
- Early Start of activity F = 12 + 1 = 13
- Early Finish of activity F = 13 + 6 -1 = 18

Calculating Early Start (ES) and Early Finish (EF)

- Early Start and Early Finish Dates for the path Start -> G -> H -> I -> End



Calculating Early Start (ES) and Early Finish (EF)

- Early Start of activity G = 1 (Since this is the first activity of the path)
- Early Finish of activity G = 1 + 3 – 1 = 3
- Early Start of activity H = 3 + 1 = 4
- Early Finish of activity H = 4 + 4 – 1 = 7
- Early Start of activity I = 7 +1 = 8
- Early Finish of activity I = 8 + 6 – 1 = 13

The Key Concepts

Calculating Early Start (ES) and Early Finish (EF)

The Concepts:

- The formula used for calculating Early Start and Early Finish dates:
- **Early Start of the activity** = Early Finish of predecessor activity + 1
- **Early Finish of the activity** = Activity duration + Early Start of activity – 1

Calculating Late Start (LS) and Late Finish (LF)

The Concepts:

- The **Late Finish date of the last activity on all paths will be the same** because no activities can continue once the project is completed.
- The formula used for Late Start and Late Finish dates:
- **Late Start of Activity** = Late Finish of activity – activity duration + 1
- **Late Finish of Activity** = Late Start of successor activity – 1
- To calculate the Late Start and Late Finish, we use the backward pass; i.e., we will start from the last activity and move back towards the first activity.

Calculating Late Start (LS) and Late Finish (LF)

- We have calculated the Early Start and Early Finish dates of all activities. Now it is time to calculate the Late Start and Late Finish dates.
- The Late Finish date of the last activity on all paths will be the same because no activities can continue once the project is completed.
- The formula used for Late Start and Late Finish dates:
- Late Start of Activity = Late Finish of activity – activity duration + 1
- Late Finish of Activity = Late Start of successor activity – 1
- To calculate the Late Start and Late Finish, we use the backward pass; i.e., we will start from the last activity and move back towards the first activity.

Calculate the Free Float

- Total Float represents the amount of time a task can be delayed without affecting the completion date of the project.

while

- Free Float represents the amount of time a task can be delayed without affecting the start date of its subsequent tasks.

Calculate the Free Float

- Formulas for calculating Total Float and Free Float are as follows:
- **Total Float** = LS – ES (it is also calculated by LF – EF)Free Float = Lowest ES of successors – EF
- **Free Float** = Lowest ES of successors – EF

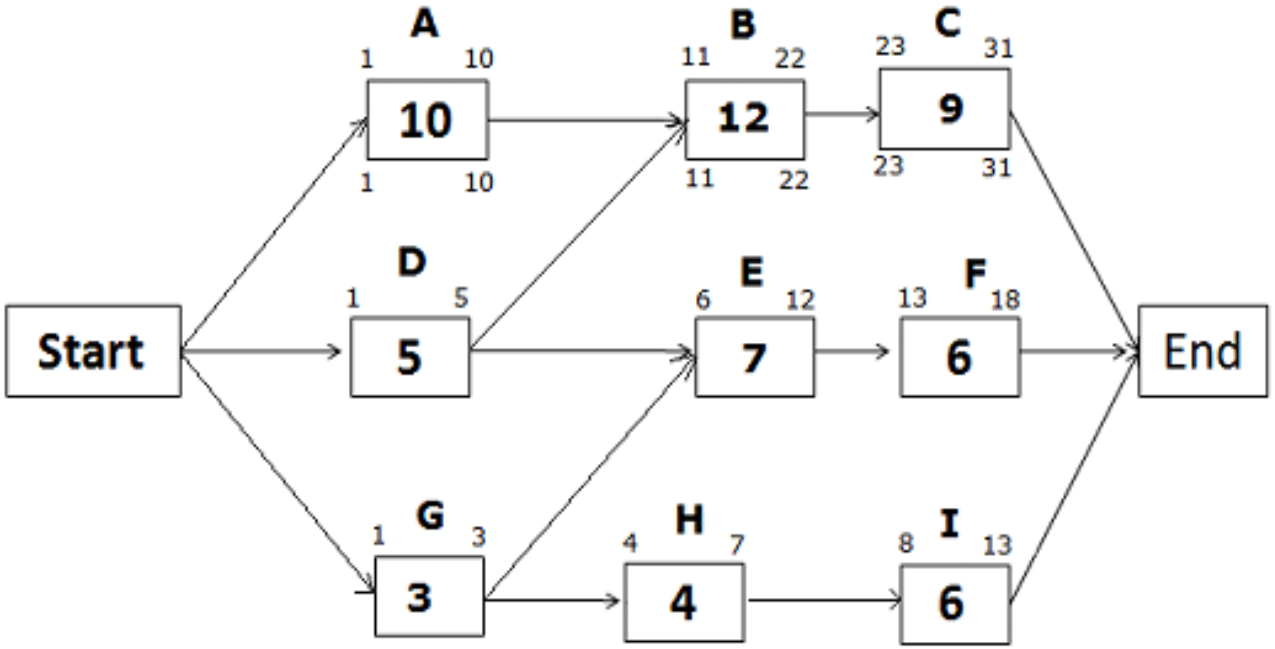
Or

- Free Float = ES of Next Activity – EF of Current Activity

End of The Key Concepts

Calculating Late Start (LS) and Late Finish (LF)

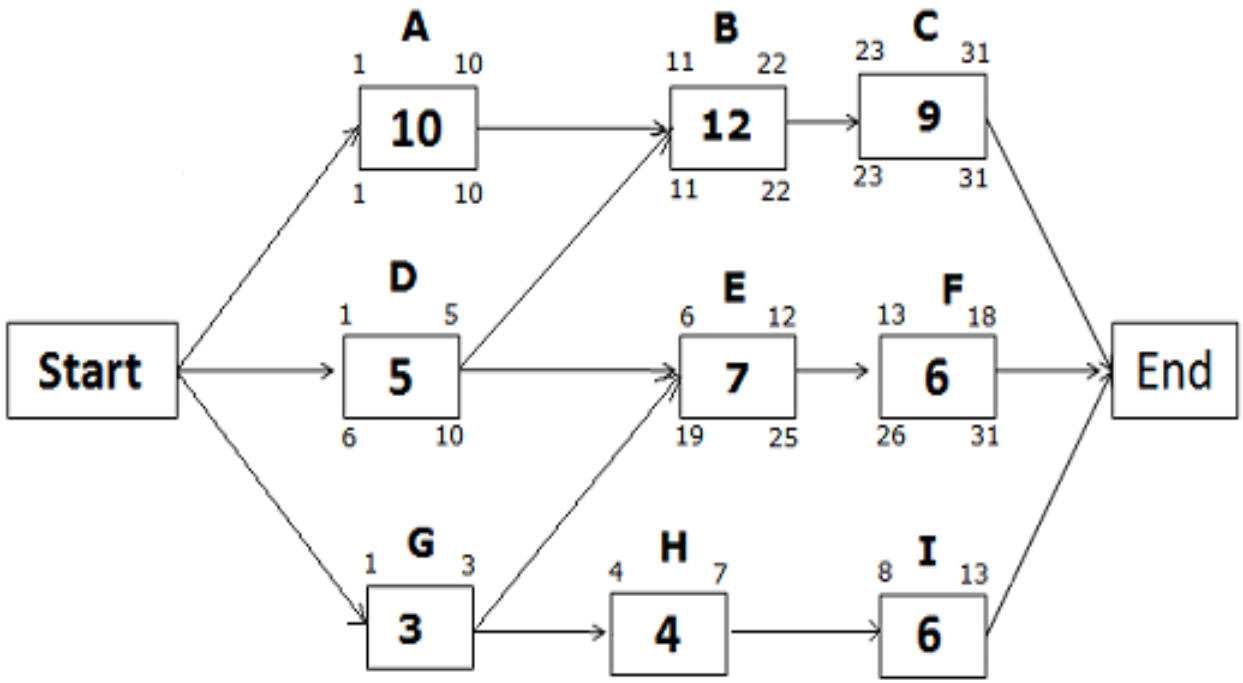
- Late Start and Late Finish Dates for the path Start -> A -> B -> C -> End



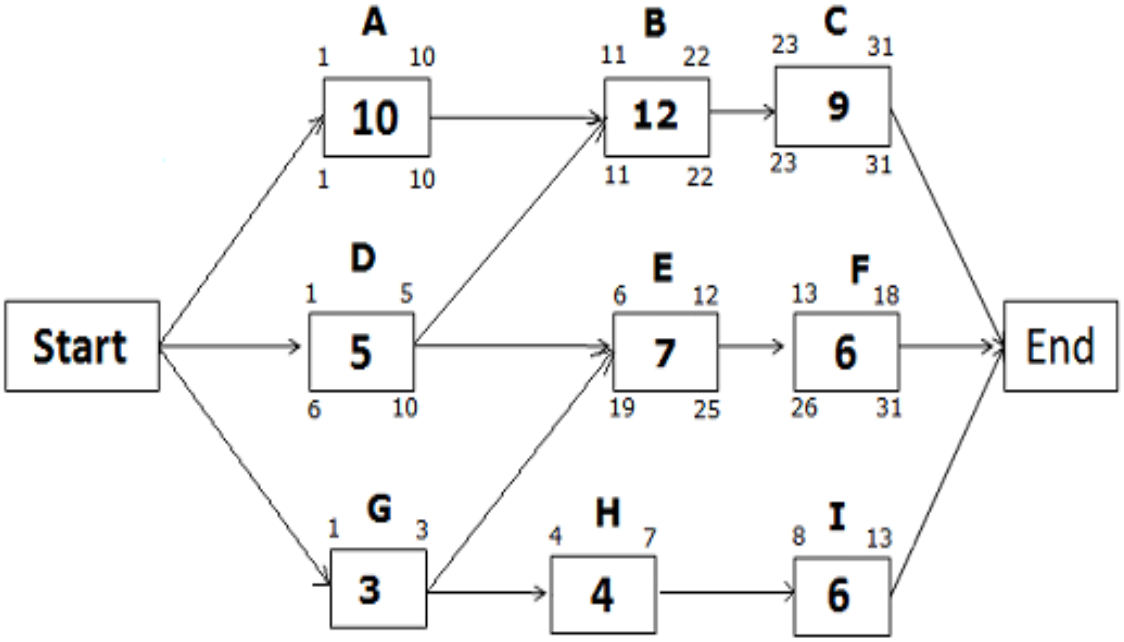
- On a critical path, the Late Start, and Late Finish dates will be the same as the Early Start and Early Finish dates

Calculating Late Start (LS) and Late Finish (LF)

- Late Start and Late Finish Dates for the path Start -> D -> E -> F -> End



Calculating Late Start (LS) and Late Finish (LF)



Late Finish of activity F = 31 (because you cannot allow any activity to pass the project completion date)

Calculating Late Start (LS) and Late Finish (LF)

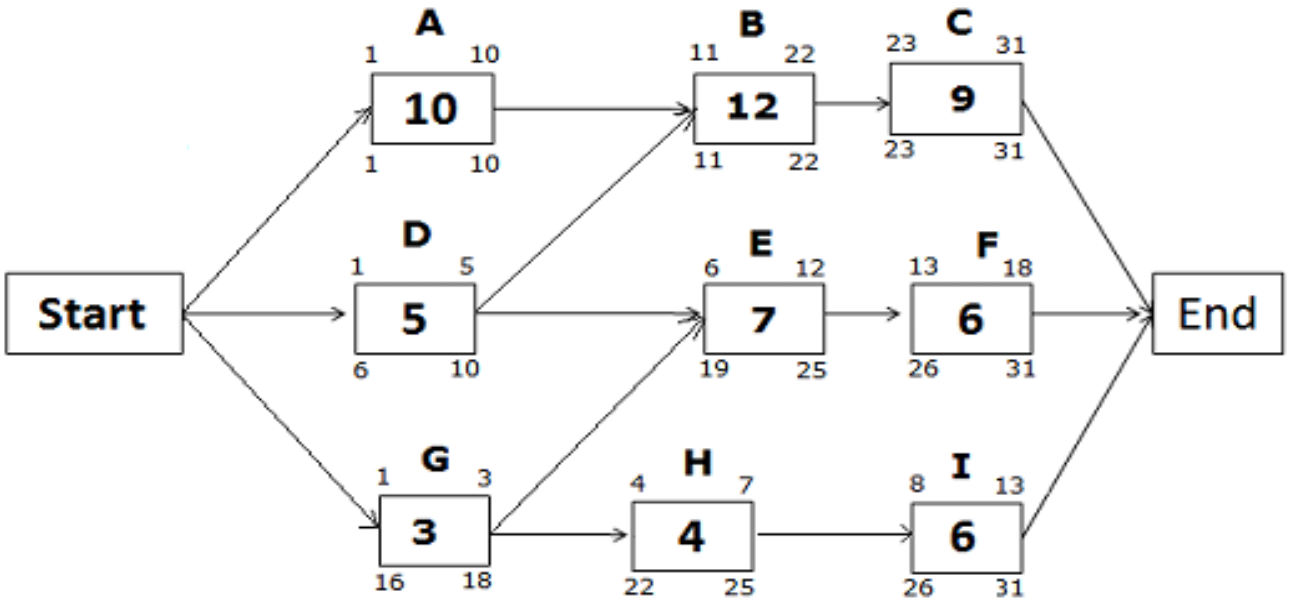
- Late Start of activity F = LF of activity F – activity duration + 1
- = 31 – 6 +1 = 26
- Late Finish of Activity E = LS of successor activity – 1
- = LS of Activity F – 1
- = 26 – 1 = 25
- Late Start of Activity E = LF of activity E – activity duration + 1
- = 25 – 7 + 1 = 19
- Late Finish of activity D = LS of successor activity – 1

Calculating Late Start (LS) and Late Finish (LF)

- If you look at the network diagram, you will notice that activity D has two successor activities, B and E. So, which activity would you select?
- You will select the activity with the earlier (least) Late Start date. Here, the Late Start of activity B is 11, and the Late Start of activity E is 19.
- Therefore, you will select activity B, which has the earlier Late Start date.
- Hence,
- Late Finish of activity D = LS of activity B – 1
- = 11 – 1 = 10
- Late Start of Activity D = LF of activity D – activity duration + 1
- = 10 – 5 + 1 = 6

Calculating Late Start (LS) and Late Finish (LF)

- Late Start and Late Finish Dates for the path Start -> G -> H -> I -> End



- Late Finish of activity I = 31 (because you cannot allow any activity to pass the project completion date)

Calculating Late Start (LS) and Late Finish (LF)

- Late Start of activity I = 31 – 6 + 1 = 26
- Late Finish of activity H = 26 – 1 = 25
- Late Start of activity H = 25 – 4 + 1 = 22
- Late Finish of Activity G = 19 – 1= 18 (we will choose the late start of activity E, not activity H, because the Late Start of activity E is earlier than the Late Start of activity H).
- Late Start of activity G = 18 – 3 + 1
- = 16

Total Float Vs Free Float

- Total float is also known as “float.”
- Total float is how long an activity can be delayed without putting off the project completion date.
- On a critical path, the total float is zero. Total float is often known as the slack.
- You can calculate it by subtracting the Early Start date of the activity from its Late Start Date.
- Total Float = Late Start date – Early Start date

Total Float Vs Free Float

Or

- You can get it by subtracting the activity's Early Finish date from its Late Finish date.
- Total Float = Late Finish date – Early Finish date

Total Float Vs Free Float

Or

- You can get it by subtracting the activity's Early Finish date from its Late Finish date.
- Total Float = Late Finish date – Early Finish date

Calculate the Free Float

- Free float is how long an activity can be delayed without delaying the Early Start of its successor.
- You can calculate the free float by subtracting the Early Finish Date of the activity from the Early Start Date of the next activity.

Calculate the Free Float

- Formulas for calculating Total Float and Free Float are as follows:
- **Total Float** = LS – ES (it is also calculated by LF – EF)Free Float = Lowest ES of successors – EF
- **Free Float** = Lowest ES of successors – EF

Project Cost Management

By

Casmir, R. (Ph.D)

Project Cost Management

- Project Cost Management includes the processes involved in planning, estimating, budgeting, financing, funding, managing, and controlling costs so that the project can be completed within the approved budget
- Funding refers to the source of resources, while financing describes the act/process of acquiring or furnishing those resources.

Project Cost Management

Project Cost Management involve the following processes:

- **Plan cost Management—**The process that establishes the policies, procedures, and documentation for planning, managing, expending, and controlling project costs.
- **Estimate costs—**The process of developing an approximation of the monetary resources needed to complete project activities.
- **Determine Budget—**The process of aggregating the estimated costs of individual activities or work packages to establish an authorized cost baseline.
- **Control costs—**The process of monitoring the status of the project to update the project costs and managing changes to the cost baseline.

Project Cost Estimate

Three-Point Estimating

The accuracy of single-point activity cost estimates may be improved by considering estimation uncertainty and risk and using three estimates to define an approximate range for an activity's cost:

- **Most likely (cM)**. The cost of the activity, based on realistic effort assessment for the required work and any predicted expenses.
- **Optimistic (cO)**. The activity cost based on analysis of the best-case scenario for the activity.
- **Pessimistic (cP)**. The activity cost based on analysis of the worst-case scenario for the activity.

Project Cost Estimate

- Depending on the assumed distribution of values within the range of the three estimates the expected cost, cE, can be calculated using a formula. Two commonly used formulas are triangular and beta distributions. The formulas are:
- **triangular distribution.** cE = (cO + cM + cP) / 3
- **Beta distribution** (from a traditional Program Evaluation and Review Technique (PERT) analysis). cE = (cO + 4cM + cP) / 6
- Cost estimates based on three points with an assumed distribution provide an expected cost and clarify the range of uncertainty around the expected cost.

Project Cost Estimate

- Triangular distribution treats all three estimates equally

while

- The beta distribution (also known as PERT) gives the most likely estimate a greater weight

Cost Management Processes

Process	Process Group
Plan Cost Management	Planning
Estimate Costs	Planning
Determine Budget	Planning
Control Costs	Monitoring and Controllir

Estimate Costs

- Estimate Costs is the process of developing an approximation of the monetary resources needed to complete project activities.
- The key benefit of this process is that it determines the amount of cost required to complete project work. The table below shows five different types of estimates that can be used based on the need.

Estimate Type	Range
Order of Magnitude Estimates	-50% to +50%
Conceptual Estimate	-30% to +50%
Preliminary Estimate	-20% to +30%
Definitive Estimate	-15% to +20%
Control Estimate	-10% to +15%

Types of Cost

- **Direct Cost-** Any cost that is identified specifically with a particular final cost objective.
- **Indirect Cost**- Any cost not directly identified with a single, final cost objective (more than one project)
- **Fixed Cost -** Periodic charge that does not vary with business volume
- **Variable Cost -** Cost that fluctuates based on the business volume
- **Opportunity Cost**: When there is a decision to be made between two opportunities, the opportunity cost is the value / money forgone for the not doing the project.

Types of Cost

- **Lifecycle Costing**: Takes into account cost of development of a project and also the cost of supporting during the lifecycle.

- **Sunk Cost** - Cost already incurred which cannot be recovered regardless of future events

Plan Cost Management

The process that establishes the policies, procedures , and documentation for planning , managing , expending , and controlling project costs.

Plan Cost Management

Inputs	Tools and Techniques	Outputs
❑ Project Management Plan ❑ Project Charter ❑ Enterprise Environmental Factors ❑ Organizational Process Assets	❑ Expert Judgment ❑ Analytical Techniques ❑ Meetings	❑ Cost Management Plan

Plan Cost Management- Inputs

Project Management Plan component that is important for this process are primarily the scope and cost baselines.

Project Charter provides the summary budget from which the detailed project costs are developed . It also defines the project approval requirements that will influence the management of the project costs.

Enterprise Environmental Factors include factors such as Organization structure/culture, Market conditions, currency exchange rates (for project sourced from more than one country) that influence the project cost.

Organizational Process Assets include financial control procedures , historical information . Existing formal and informal cost estimating and budgeting related policies, procedures and guidelines.

Plan Cost Management- Tools and Techniques

Expert Judgment – Guided by historical information, provides valuable insight about the environment and information from prior similar projects. Judgment based upon expertise in an application area, knowledge area , discipline, industry can be useful in this process.

Analytical Techniques – Some of the techniques payback period, return on investment, internal rate of return, discounted cash flow and net present value can be useful in this process

Meetings - Project teams may hold meetings planning meetings to develop the cost management plan. These meetings may include, project sponsor, project manager, selected team members , selected stakeholders, anyone with responsibility for project costs.

Plan Cost Management- Outputs

- **Cost Management Plan –** is a component of the project management plan and describes how the project costs will be planned, structured , and controlled. The cost management plan can establish the following.
- **Units of measure -** Each unit of measure for each resource.
- **Level of precision** – The degree to which activity cost estimates will be rounded up or down.
- **Level of accuracy** - The acceptable range (e.g. +_ 100%) used in determining realistic activity cost estimates is specified , may include an amount for contingencies.
- **Control threshold** – Variance threshold for monitoring cost performance may be specified to indicate an agreed – upon amount of variation to be allowed before some action need to be taken.

Plan Cost Management-Outputs

- The cost management plan also contains.
- **Rules of performance measurement –** Earned value management (EVM) rules of performance measurement are set. For example
  - Define the points in the WBS at which measurement of control accounts will be performed
  - Establish the earned value measurement techniques (e.g. weighted milestones, fixed formula, percent complete, etc) to be used and
  - Specify tracking methodologies and the earned value management computation equations for calculating projected estimate at completion (EAC) forecasts to provide a validity check on the bottom-up EAC

Estimate Cost

Estimate cost is the process of developing an approximation of the monetary resources needed to complete project activities. The key benefit of this process is that it determines the amount of cost required to complete project work.

Estimate Costs

Inputs	Tools and Techniques	Outputs
❑ Cost Management Plan ❑ Human Resource Management Plan ❑ Scope baseline ❑ Project Schedule ❑ Risk Register ❑ Enterprise Environmental Factors ❑ Organizational Process Assets	❑ Expert Judgment ❑ Analogous Estimating ❑ Parametric estimating ❑ Bottom-up estimating ❑ Three-point estimates ❑ Reserve Analysis ❑ Cost of Quality ❑ Project management estimating software ❑ Vendor bid analysis ❑ Group decision – making techniques	❑ Activity Cost estimates ❑ Basis of estimates ❑ Project document updates

Estimate Costs - Inputs

Cost management plan – The cost management plan defines how project costs will be managed and controlled . It includes the method used and the level of accuracy required to estimate activity cost.

Human Resource Plan provides detail on Project staffing attributes, personnel rates, and related rewards/recognition, these are necessary components for developing the project cost estimates.

Scope baseline include inputs from scope statement, Work Breakdown Structure and WBS dictionary (i.e. *A WBS dictionary is a document that provides detailed information about each element in the WBS, including work packages and control accounts*).

Project Schedule provides detail on the type and quantity of resources and the amount of time which those resources are applied to complete the work of the project, all these factors directly impact the cost

Risk Register provides input related to costs associated with risk response

Estimate Costs - Inputs

Enterprise Environmental Factors include availability of resources, published commercial information

Organizational Process Assets – include cost estimating policies, historical information, cost estimating templates and lesson learned

Estimate Costs -Tools and Techniques

Expert Judgment guided by historical information, provides valuable insight about the environment and information from prior similar projects. Expert judgment can also be used to determine whether to combine methods of estimating and how to reconcile differences between them.

Analogous Estimating relies on historical information (Previous projects executed by the organization) to predict the cost of current project. This is a top-down approach is good for fast estimates to get a general idea of what the project may cost. The accuracy depends upon how similar the two projects actually are.

Parametric estimating uses a mathematical model based on known parameters to predict the cost of a Project. A parameter can be cost per unit, cost per cubic yard etc. Two Parametric estimating techniques are Regression Analysis and Leaning curve.

Estimate Costs -Tools and Techniques

Bottom -up estimating produces a separate estimate for each scheduled activity. The detailed cost is then summarized or “rolled up” to higher levels for subsequent reporting and tracking purposes. The cost and accuracy of bottom-up cost estimating is typically influenced by the size and complexity of the individual activity or work package.

Three-point estimates concept originated with the program evaluation and review technique (PERT). PERT uses three estimates to define an approximate range for an activity’s cost:

Triangular Distribution

Estimated Cost = Optimistic + Most Likely + (Pessimistic)/ 3

Beta Distribution

ESTIMATED COST = (Optimistic + 4*(Most Likely) + Pessimistic) / 6

Estimate Costs -Tools and Techniques

Reserve Analysis adds reserve amount (Contingency) to the cost. This is done to take care of uncertainties in the project.

Cost of Quality includes assumptions related to Cost of Quality

Project management software is helpful in performing calculations, Organizing and storing large amount of information needed for preparing cost estimates and for reporting purpose

Vendor bid analysis may include analysis of what the project should cost, based on the responsive bids from qualified vendors.

Group Decision Making Techniques Team – based approaches , such as brainstorming, the Delphi or nominal group techniques, are used for engaging team members to improve estimate accuracy and commitment to the emerging estimates.

Estimate Costs - Outputs

Activity Cost estimates details how much it would cost to complete each schedule activity on the project

Basis of estimates include supporting documentation of the cost estimates and it may include documentation of the basis of estimates , documentation of all assumptions made , documentation of any known constraint , indication of the range of possible estimates and indication of the confidence level of the final estimates

Project document updates include project risk register and cost management plan

Exercise : Estimate Cost

Revise Tools to Estimate Cost

1.You figure out a best case scenario and , a likely scenario and a worst case scenario. Then you use a formula to come up with an expected cost for the project. The tool used by you is _____

2.John creates a document with all the historical information from similar project he had done in past and also takes input from a contractor who provides his input. The tool used by John is_____

Determine Budget

Determine Budget is the process of aggregating the estimated costs of individual activities or work packages to establish an authorized cost baseline. This baseline includes all authorized budgets, but excludes management reserves.

Determine Budget

Inputs	Tools and Techniques	Outputs
❑ Cost management plan ❑ Activity cost estimates ❑ Basis of estimate ❑ Scope baseline ❑ Project Schedule ❑ Resource Calendars ❑ Risk register ❑ Agreements ❑ Organizational Process Assets	❑ Cost Aggregation ❑ Reserve Analysis ❑ Expert Judgment ❑ Historical relationship ❑ Funding limit reconciliation	❑ Cost baseline ❑ Project funding requirements ❑ Project document updates

Determine Budget - Inputs

Cost Management Plan - describes how the project costs will be managed and controlled

Scope baseline refers to collection of approved project scope statement and its associated WBS and WBS dictionary.

Activity cost estimates of a work package is a summation of the activity cost estimates of activities under the work package

Basis of estimate includes supporting detail for cost estimates contained in the basis for estimates should specify any basic assumptions

Project Schedule helps in mapping the cost back to time for planning purpose

Determine Budget - Inputs

Resource Calendars along with Project schedule will help plan for when the costs will be incurred

Agreements Applicable agreement information and costs relating to products, services or results that have been or will be purchased are included when determining the budget.

Organizational Process Assets include cost budgeting policies procedures , guidelines and reporting method

Determine Budget – Tools and Techniques

Cost Aggregation involves aggregating the costs of individual activities to work packages that are further aggregated to higher levels of WBS and finally into the project cost

Reserve Analysis include contingency reserve and management reserves for the project. Contingency reserves are allowances for response plan strategized for the identified risks. These are also known as “known unknowns” . Management reserves are allowance for unplanned changes to thee project scope and cost. The are also known as “unknown unknowns”. Reserves are not the part of the project cost baseline.

Expert Judgment can be taken from experienced consultant in the project application area. This can be from an internal organizational resource or external sources

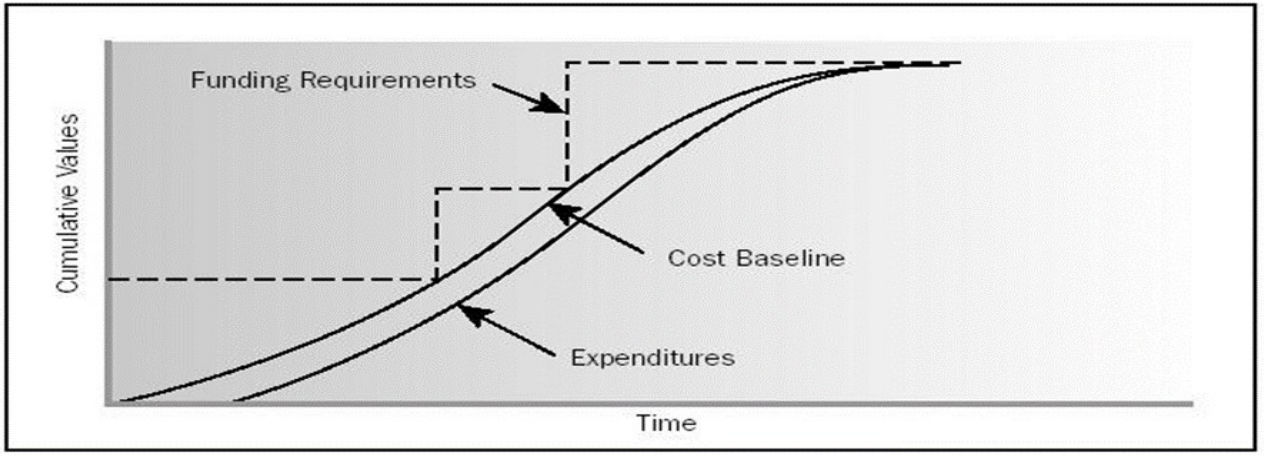
Determine Budget – Tools and Techniques

Historical relationship result in parametric estimates or analogous estimates that involve parameter to develop mathematical models to predict total project costs.

Funding limit reconciliation refers to reconciliation of planned spending with the funding limit

Determine Budget – Outputs

Cost performance baseline is an authorized time-phased budget at completion (BAC) used to measure, monitor, and control overall cost performance on the project. It is developed as a summation of the approved budgets by time period and is typically displayed in the form of an S -curve, as is illustrated in Figure below (Reference PMBOK 4th edition). In the earned value management technique the cost performance baseline is referred to as the performance measurement baseline (PMB).



Determine Budget – Outputs

Project funding requirements are derived from cost baselines. Total funding requirement is sum of cost baseline and management reserve.

Project document updates include updates to risk register, cost estimates and project schedule

Project Budget Components

Planned value. Planned value (PV) is the authorized budget assigned to scheduled work. It is the authorized budget planned for the work to be accomplished for an activity or work breakdown structure component, not including management reserve. This budget is allocated by phase over the life of the project, but at a given moment, planned value defines the physical work that should have been accomplished. The total of the PV is sometimes referred to as the performance measurement baseline (PMB). The total planned value for the project is also known as budget at completion (BAC).

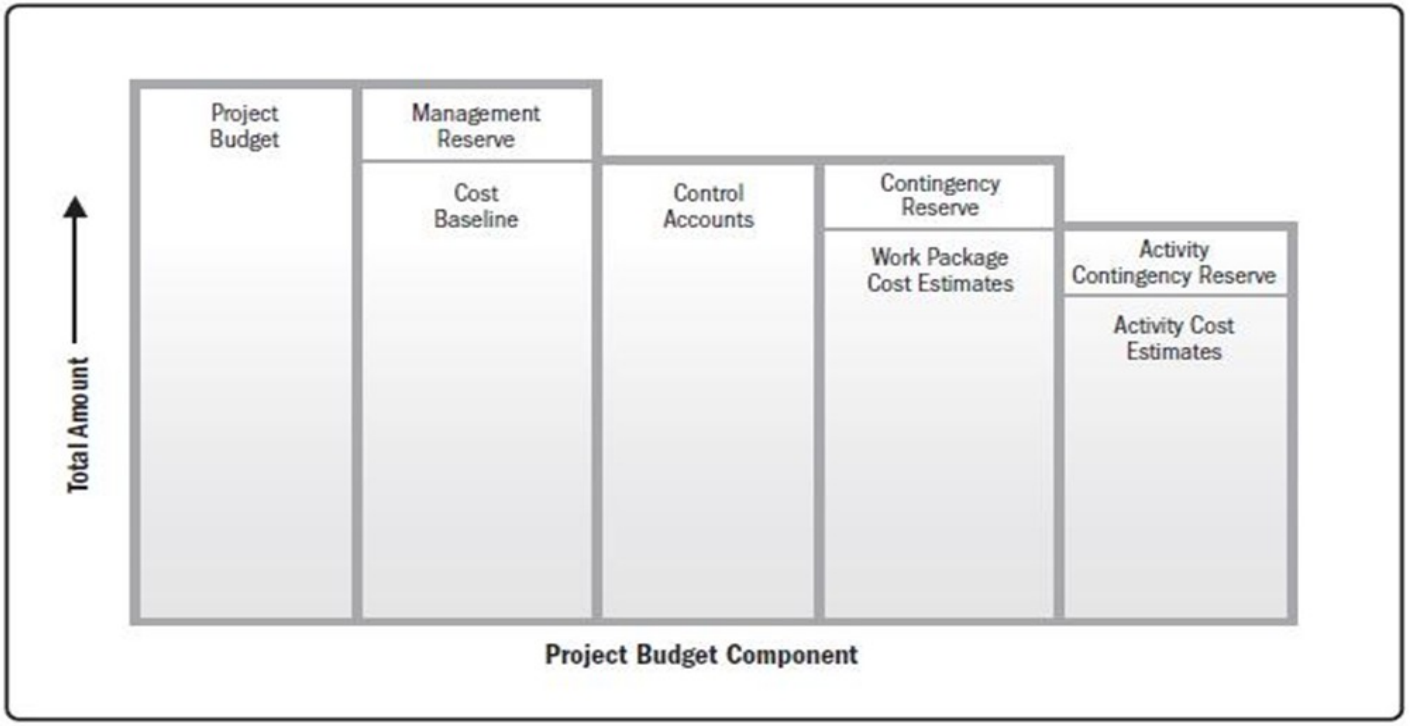
Project Budget Components

- **Earned value.** Earned value (EV) is a measure of work performed expressed in terms of the budget authorized for that work. It is the budget associated with the authorized work that has been completed. The EV being measured needs to be related to the PMB, and the EV measured cannot be greater than the authorized PV budget for a component. The EV is often used to calculate the percent complete of a project. Progress measurement criteria should be established for each WBS component to measure work in progress. Project managers monitor EV, both incrementally to determine current status and cumulatively to determine the long-term performance trends.

Project Budget Components

- **Actual cost.** Actual cost (AC) is the realized cost incurred for the work performed on an activity during a specific time period. It is the total cost incurred in accomplishing the work that the EV measured. The AC needs to correspond in definition to what was budgeted in the PV and measured in the EV (e.g., direct hours only, direct costs only, or all costs including indirect costs). The AC will have no upper limit; whatever is spent to achieve the EV will be measured.

Project Budget Components



Control Costs

Control Costs is the process of monitoring the status of the project to update the project budget and managing changes to the cost baseline.

Control Costs

Inputs	Tools and Techniques	Outputs
❑ Project management plan ❑ Project Funding Requirements ❑ Work Performance data ❑ Organizational Process Assets	❑ Earned value management ❑ Forecasting ❑ To-complete performance index TCPI ❑ Performance reviews ❑ Project management software ❑ Reserve analysis	❑ Work performance Information ❑ Cost forecasts ❑ Change Requests ❑ Project management plan Updates ❑ Project document updates ❑ Organizational Process assets updates

Control Costs - Inputs

Project Management Plan contains Cost management plan and Cost performance Baseline, these are helpful in controlling the cost.

Project Funding Requirements As explained earlier include projected expenditures plus anticipated liabilities

Work Performance Data provides information on the status and cost of project activities and ensures that the project goals and objectives are met as per the project management plan

Organizational Process Assets that can influence this process include cost budgeting policies, procedures, guidelines , cost budgeting tools and reporting methods

Control Costs – Tools and Techniques

Earned value management is used to measure the performance of the project, It integrates project scope, cost, and schedule measures to help the project management team assess and measure project performance and progress. The EVM technique involves evaluating the planned value , Earned Value and the Actual cost for each schedule activity , work package or control account.

1.Planned Value (PV) is the authorized budget assigned to the scheduled work to be accomplished for a schedule activity or work breakdown structure component. It is also known as Budgeted Cost of Work Scheduled (BCWS)

2.Earned Value (EV) is the value of work performed expressed in terms of the approved budget assigned to that work for a schedule activity or a WBS component. It is also known as Budgeted cost of Work Performed (BCWP)

3.Actual Cost (AC) is the total cost actually incurred and recorded in accomplishing work performed for an activity or work breakdown structure component.

Control Costs – Tools and Techniques

Variance Analysis helps in analyzing the performance of the project vis-à-vis cost and schedule baselines

Schedule Variance (SV) is a measure of schedule performance on a project.

It is equal to the earned value (EV) minus the planned value (PV). SV= EV-PV.

Cost variance (CV) is a measure of cost performance on a project. It is equal to the earned value (EV) minus the actual costs (AC). CV = EV-AC

Schedule performance Index (SPI) in addition to project status is used to predict the completion date. SPI = EV/PV

Cost Performance Index (CPI) measures the cost efficiency for the work completed, it is the ratio of EV to AC. CPI = EV/AC.

Control Costs – Tools and Techniques

Forecasting Techniques are used to calculate Estimate at completion (EAC) and Estimate to complete (ETC).

EAC = AC + ETC

EAC forecast for ETC work performed at the budgeted rate EAC=AC+BAC–EV

EAC forecast for ETC work performed at the present CPI EAC = BAC / cumulative CPI

EAC forecast for ETC work considering both SPI and CPI factors.

AC + [(BAC – EV) / (cumulative CPI x cumulative SPI)].

Control Costs – Tools and Techniques

To-complete performance index (TCPI) is the calculated projection of cost performance that must be achieved on the remaining work to meet a specified management goal, such as the BAC or the EAC.

TCPI based on the BAC: Work Remaining (BAC – EV) / Fund Remaining (BAC – AC).

TCPI based on EAC : (BAC-EV)/(EAC-AC)

Performance reviews involves following three types of analysis

1. **Variance Analysis** compares actual project performance to planned or expected performance. Cost and schedule variances are the most frequently analyzed.

2.Trend Analysis examines project performance over time to determine if performance is improving or deteriorating.

3.Earned value management compares the baseline plan to actual schedule and cost performance.

Control Costs – Tools and Techniques

Project management software is often used to monitor the three EVM dimensions (PV, EV, and AC), to display graphical trends, and to forecast a range of possible final project results.

Reserve Analysis – During cost control, reserve analysis is used to monitor the status of contingency and management reserves for the project to determine if these reserves are still needed or if additional reserves need to be requested.

Control Costs – Outputs

Work Performance Information includes the calculated CV, SV, CPI, and SPI values for WBS components, in particular the work packages.

Cost Forecasts includes a calculated EAC value or a bottom-up EAC value.

Organizational Process Assets Updates include Causes of variances, Corrective action chosen and the reasons, and Other types of lessons learned from project cost control.

Change Requests include change request to the cost performance baseline or other components of the project management plan. Change requests can include preventive or corrective actions and are processed for review and disposition through the Perform Integrated Change Control process

Control Costs – Outputs

Project Management Plan Updates Elements of the project management plan that may be updated include Cost performance baseline, Cost management plan.

Project Documents Updates include updates to cost estimates, basis of Estimates

Organizational Process Assets Updates

Organizational Process Assets that may be updated include, but are not limited to Cause of variances

- ✓ Corrective action choosen and the reasons
- ✓ Financial databases
- ✓ Other types of lesson learned from project cost control

Project Quality Management

By
Casmir, R. (Ph.D)

What is Quality?

Definitions:

- Quality is “the degree to which a set of inherent characteristics fulfil requirements” – ***PMBOK®Guide* (Glossary)**
- “Conformance of requirements of the customer”– Philip Crosby
- “Fitness for use” – Joseph Juran

What is Quality?

- It is an integrated framework of any organization’s total quality management process.
- ✓ The management of quality comprises processes required to deliver a project on time while ensuring that the demands of the stakeholders, including customers, are met.
- ✓ Moreover, it is the ability to manage a product and deliver output in conformity with the requirements of the users while maximizing profits for the company.

What is Quality?

- Project quality management combines two frameworks: quality management and project management.
- ✓ For instance, let’s take the example of a multimedia tech company that is working on a new high-quality software application.
- ✓ Now, to ensure that it meets customer expectations and delivers value, there are two aspects that the project management team needs to look into.
- ✓ First, it has to ensure that there are no functionality defects in the software (this is referred to as quality management).
- ✓ Next, it must ensure that the new software is equipped with the latest features that are in demand in the market so that the company can generate more revenue (this is referred to as project management)

What is Grade?

Grade is a category or rank given to entities having the same functional use but different technical characteristics. (as per *PMBOK® Guide*).

Importance of Quality?

Why is quality important?

- Quality => Customer Satisfaction => Business growth
- Quality => Pride in delivery => Higher Team Morale
- Quality => Better Decision Making – PQM provides relevant data to an organization and enables it to make strategic decisions by identifying quality requirements.
- Quality => Enhanced Management – PQM promotes a culture of continuous improvement

- Lack of quality => Increased Cost
- Lack of quality => Loss of customer confidence
- Lack of quality => Loss of Reputation => Difficulty in gaining new business
- Lack of quality =>could lead to loss of life

Project Quality Management Processes

- ***Quality Management Planning***

✓ It sets quality standards and objectives for the project and determines how these standards will be met. In addition, it includes identifying the processes, tools, and techniques to manage and monitor project quality.

- ***Perform Quality Assurance***

✓ PQM ensures that the project is executed by the quality plan. Additionally, this includes activities such as regular quality audits and inspections, as well as continuous monitoring of project performance.

- ***Control Quality***

✓ It monitors and controls the quality of project outputs, including products, services, and deliverables. Moreover, this is done by testing, inspections, and verifying the project requirements have been met.

Quality Principles

- Quality is concerned with **Prevention over Inspection**.

- Inspection is typically used to prove that the system works (because you've put good quality practices in place), rather than a way to reject the failures.

- A good quote to remember is, "**Quality should be planned in not inspected in**" because the cost of rework can be very high.

Project Quality Management Tools and Techniques

Project Quality Management (PQM) Tools

- **Cost of Quality (COQ) Analysis:** It calculates the cost of poor quality, including the cost of prevention, appraisal, and internal and external failure
- **Statistical Process Control (SPC):** This is a set of statistical methods used to monitor and control a process to ensure it is operating within defined limits
- **Control Charts:** This is a statistical tool used to monitor a process to determine if it is in control, and identifies any special cause variations
- **Pareto Charts:** These are graphical representations showing which factors are contributing the most to a problem or issue
- **Flowchart**: It represents a process or workflow, used for identifying potential bottlenecks or areas for improvement

Project Quality Management Tools and Techniques

Project Quality Management Techniques

- **Continuous Improvement:** This is a systematic approach that identifies and implements improvements in a process or product
- **Quality Audits**: This is an independent review of a process or product to assess its compliance with quality standards
- **Design of Experiments (DOE):** DOE is a statistical method used to determine the effect of different variables on a process or product
- **Lean Methodology:** This is a management approach that seeks to optimize the flow of work by eliminating waste and improving efficiency

Quality Management Processes

Process	Process Group
Plan Quality Management Perform Quality Assurance Control Quality	Planning Executing Monitoring and Controlling

Plan Quality Management

Identifying quality requirements and/or standards for the project and product, and documenting how the project will demonstrate compliance.

This process is important to ensure that the resulting product is of acceptable quality. This is performed in parallel with the development of the project management plan and other planning process

Plan Quality Management

Inputs	Tools and Techniques	Outputs
❑ Project Management Plan ❑ Stakeholder register ❑ Risk register ❑ Requirement documentation ❑ Enterprise Environmental Factors ❑ Organizational Process Assets	❑ Cost-benefit analysis ❑ Cost of quality (COQ) ❑ Seven basic quality tools ❑ Benchmarking ❑ Design of experiments ❑ Statistical sampling ❑ Additional quality planning tools ❑ Meetings	❑ Quality Management Plan ❑ Process improvement plan ❑ Quality Metrics ❑ Quality Checklists

Plan Quality Management - Inputs

Project Management Plan – The following contents of the project management plan are useful in creating Quality management plan

- Scope baseline include Project scope statement, Work breakdown structure and WBS dictionary.
- Schedule baseline contains the accepted schedule performance measures , including start and finish dates
- Cost Base line documents the accepted time interval being used to measure cost performance
- Other management plans . These plans contribute to the overall project quality and may highlight actionable areas of concern with regard to the project quality

Stakeholder register aids in identifying those stakeholders possessing a particular interest in, or having an impact on quality

Risk register will list all the risks that may have impact on the quality

Plan Quality Management - Inputs

Requirement Documentation . Requirement documentation captures the requirements that the project shall meet pertaining to stakeholder expectations. The component of the requirements document include but is not limited to project and quality requirement

Enterprise Environmental Factors that influence the plan quality management process include , Government agency regulation, standard & guidelines and rules specific to the application area, operating conditions that may impact the quality and cultural perception that may influence expectations about quality

Organizational Process Assets that influence the Plan Quality Management process include Organizational quality policies, procedures and guidelines. The performing organization's quality policy, as endorsed by senior management , sets the organization's intended direction on implementing its quality management approach; Historical databases and lesson learned from previous phases or projects

Plan Quality -Tools and Techniques

Cost-benefit analysis considers the cost benefit trade offs, the benefit of quality activities must outweigh the costs e.g. higher productivity, less rework and lower cost.

Cost of quality (COQ) considers the expense of all the activities within a project to ensure quality and is the sum of the following

- Appraisal Cost includes cost related to Audit, Inspection and Testing
- Prevention Cost includes cost related Training, documentation , quality planning and process reviews
- Internal Failure Cost includes cost related to rework , scrap etc
- External Failure Cost is related to the defects found by the customer. It includes liabilities, Warranty work and lost business

Plan Quality -Tools and Techniques

Seven Basic Quality Tools – The seven basic QC tools include

Control chart determines whether or not the process is stable or has predictable performance. It illustrates the performance of project over time and is typically used in projects , or operations , where there are repetitive activities (Help desk, manufacturing , a series of testing). Upper (UCL) and lower(LCL) control limits together form the acceptable range of variation in the process. The UCL is typically set at +3 or +6 sigma while the LCL is set at -3 0r -6 sigma. The table below shows the four sigma values representing normal distribution.

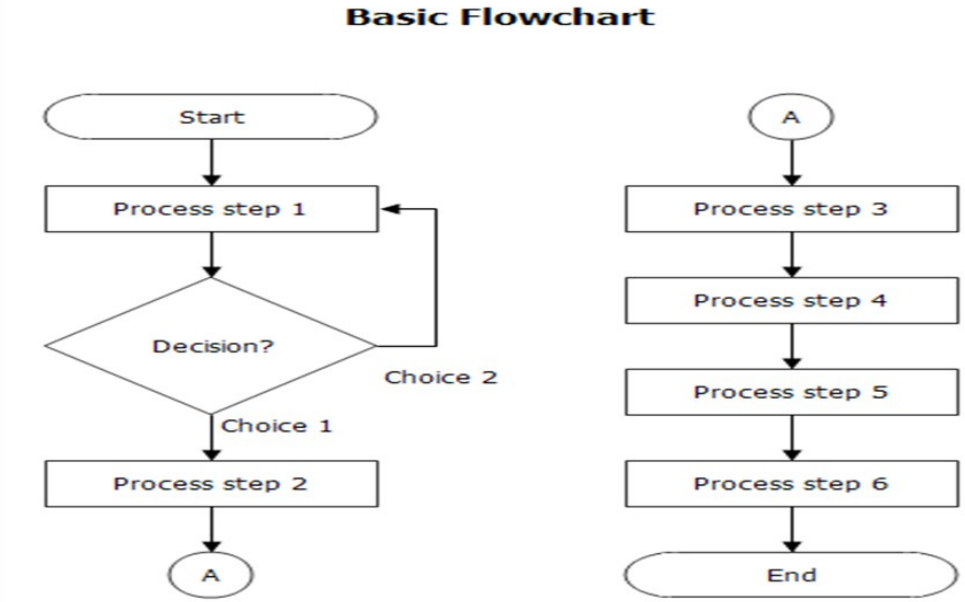
Tolerances means the result is acceptable if it falls within the range specified and, **Control Limits** means that the process is in control if the result falls within the control limits.

Plan Quality – Tools and Techniques

Cause and effect diagram (Ishikawa diagram or Fishbone diagram) is a graphical representation that depicts how different factors relate together and how these relations may contribute to inadequate quality.

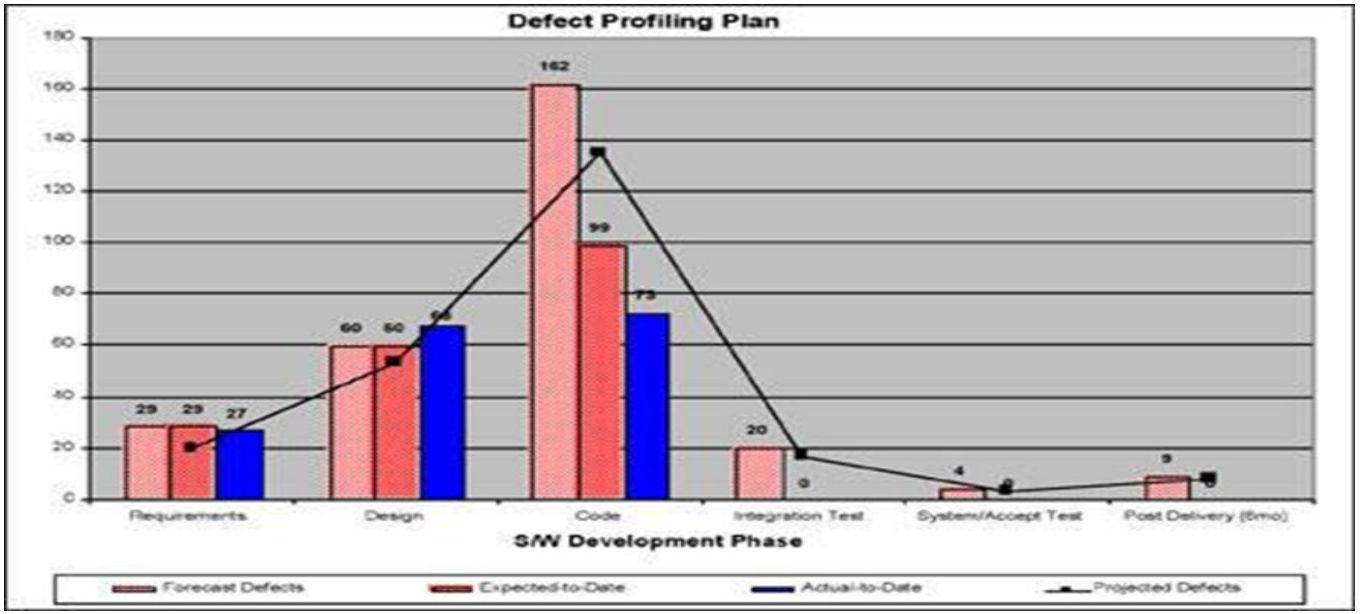
Plan Quality – Tools and Techniques

Flowcharting depicts the sequence of operation, inspection ,decisions etc in a system. This help in analyzing where quality problem may happen



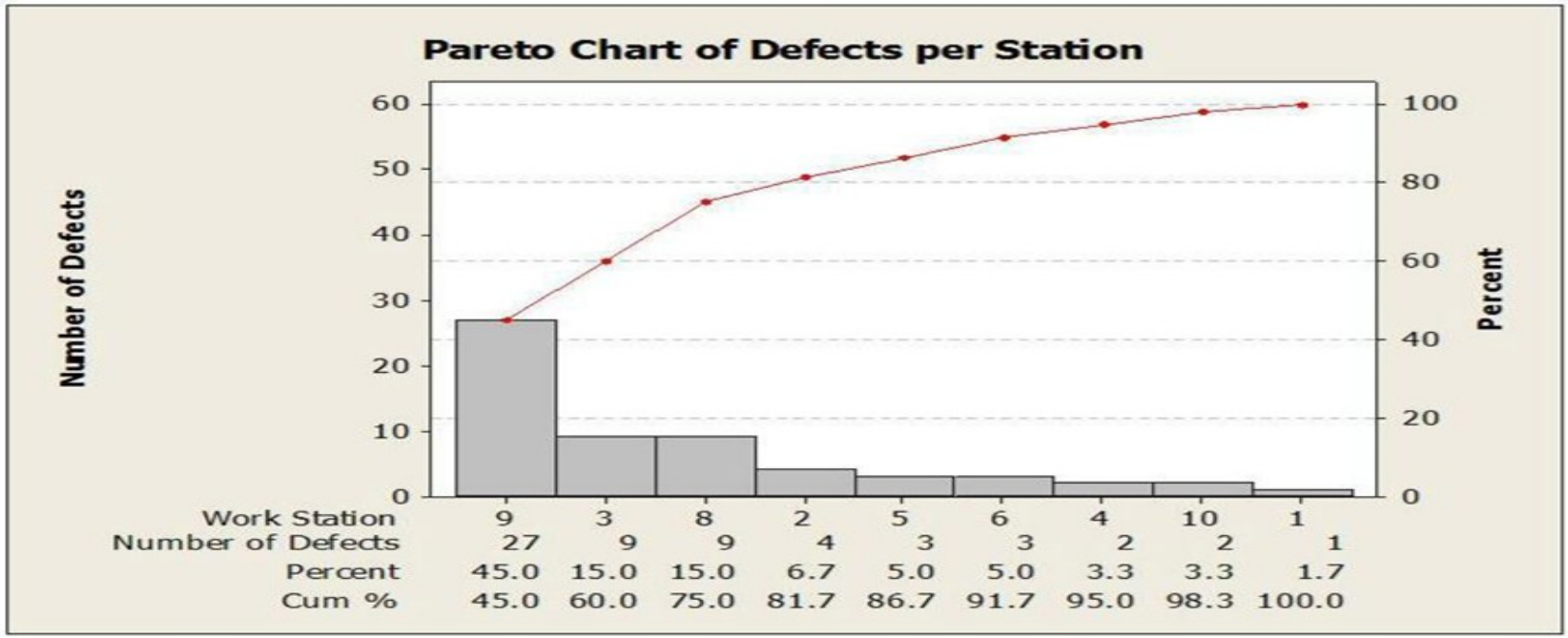
Plan Quality – Tools and Techniques

Histogram is a statistical tool used in the analysis and determination of possible solutions to quality control problem. It depicts how often the problem happens or its frequency



Plan Quality – Tools and Techniques

Pareto Chart A Pareto chart is a specific c type of histogram, ordered by frequency of occurrence. It shows how many defects were generated by type or category of identified cause . Rank ordering is used to focus corrective action. The project team should address the causes creating the



Plan Quality – Tools and Techniques

Check sheets, which are also known as tally sheets and may be used as a checklist when gathering data. Check sheets are used to organize facts in a manner that will facilitate the effective collection of useful data about a potential quality problem.

Plan Quality - Tools and Techniques

Benchmarking is a technique where project's quality standards are compared to best practices of other projects . Based on the difference corrective actions are made to the current project

Design of experiments approach relies on statistical what-if scenarios to determine what variables within project will result in the best outcome

Statistical sampling is used to test a random sample instead of testing the entire population. **Attribute Sampling** is the results conforms, or it does not where as, **Variable Sampling** is when the result is rated on a continuous scale that measures the degree of conformity.

Flowcharting depicts the relationship between the process steps graphically. It can be used to predict where quality problems may happen

Proprietary quality management methodologies includes methodologies such as Six Sigma, QFD, CMMI etc

Additional planning tool includes brainstorming, affinity diagrams and nominal group techniques

Plan Quality - Outputs

Quality Management Plan defines how the project team will implement the organization's quality policy

Process improvement plan explains how quality activities will be streamlined and improved. The areas to consider include process boundaries, process configuration, process metrics and targets for improved performance

Quality Metrics define how the quality will be measured. Quality metrics is also called as operational definitions

Quality Checklists ensures that a set of required steps have been performed and they were performed in the proper sequence

Project document updates that may be updated as plan quality is performed are Stakeholder register and responsibility assignment matrix.

Perform Quality Assurance

Auditing quality requirements and the results from quality control measurements to ensure appropriate quality standards and operational definitions are used. This is performed as an ongoing activity in the project life cycle

Perform Quality Assurance

Inputs	Tools and Techniques	Outputs
❑ Quality management Plan ❑ Process improvement plan ❑ Quality Metrics ❑ Quality control ❑ Measurements ❑ Project documents	❑ Quality management and control tools ❑ Quality Audits ❑ Process Analysis	❑ Change Requests ❑ Project ❑ Management Plan Updates ❑ Project Document Updates ❑ Organizational Process Asset Updates

Perform Quality Assurance - Inputs

Quality Management Plan describes the quality assurance and continuous process improvement approaches for the project.

Process Improvement Plan The project's quality assurance activities should be supportive of and consistent with the performing organization's process improvement plan.

Quality Metrics provide an objective means of measurement

Quality control measurements are used to analyze and evaluate the quality standards and processes of the performing organization

Project documents may influence quality assurance work and should be monitored within the context of a system for configuration management.

Perform Quality Assurance– Tools and Techniques

Quality Management and Control Tools

The perform Quality Assurance uses the tools and techniques of the Plan Quality Management and Control quality process. In addition, other tools that are available include

Affinity Diagram. Similar to mind-mapping techniques in that they are used to generate ideas that can be linked to form organized patterns of thought about a problem. The creation of WBS may be enhanced by using the affinity diagram to give structure to the decomposition of scope.

Process decision program charts (PDPC). Used to understand a goal in relation to the steps for getting to the goal. The PDPC is a useful method for contingency planning because it aids teams in anticipating intermediate steps that could derail achievement of the goal.

Interrelationship digraphs. Provide a process for creative problem solving in moderately complex scenarios that posses intertwined logical relationships for up to 50 relevant items.

Perform Quality Assurance– Tools and Techniques

Quality Management – Other tools

Tree Diagrams. Also known as systematic diagrams and may be used to represent decomposition hierarchies such as the WBS, RBS (Risk breakdown structure), and OBS (Organizational breakdown structure). As the tree diagrams permit the creation of nested branches that terminates into a single decision point , they are useful as decision trees for establishing an expected value for a limited number of dependent relationships that have been diagrammed systematically.

Prioritization matrices. Identify the key issues and the suitable alternatives to be prioritized as a set of decisions for implementation. Criteria are prioritized and weighted before being applied to all available alternatives to obtain a mathematical score that ranks the options.

Perform Quality Assurance– Tools and Techniques

Quality Management – Other tools

- **Activity networks diagrams .** Previously known as arrow diagrams. They include both the AOA (Activity on Arrow) and , most commonly used , AON (Activity on Node) formats of a network diagram.
- **Matrix diagrams.** A quality management and control tool used to perform data analysis within the organizational structure created in matrix. The matrix diagram seeks to show the strength of the relationships between factors , causes , and objectives that exist between the rows and columns that form the matrix.

Perform Quality Assurance– Tools and Techniques

Quality Audits - A quality audit is a structured , independent process to determine if project activities comply with organizational and project policies , processes , and procedures.

Process Analysis - follows the steps outlined in the process improvement plan to identify needed improvements . This analysis also examines problems experienced , constraints experienced , and non – value – added activities identified during process operations. Process analysis includes root cause analysis – a specific technique used to identify a problem , discover the underlying causes that lead to it, and develop preventive actions.

Perform Quality Assurance – Outputs

Change Requests are the outcome of updating quality standards and processes to improve organization's effectiveness and efficiency

Project Management Plan Updates includes updates in Quality management plan, Schedule management plan, Cost management plan as a result of Quality assurance activities

Project Document Updates include updates to training plans , process documentation and quality audit report

Organizational Process Asset Updates includes updating quality standards and processes to improve organization's effectiveness and efficiency

Control Quality

Monitoring and recording results of executing the quality activities to assess performance and recommend necessary changes. This process takes place throughout the project lifecycle , involves inspection , measurement and testing of deliverables and their comparison with the baseline

Control Quality

Inputs	Tools and Techniques	Outputs
❑ Project management plan ❑ Quality Metrics ❑ Quality Checklists ❑ Work Performance data ❑ Approved change requests ❑ Deliverables ❑ Project documents ❑ Organizational Process Assets	❑ Seven basic quality toold ❑ Statistical sampling ❑ Inspection ❑ Approved change requests review	❑ Quality Control measurements ❑ Validated changes ❑ Validated deliverables ❑ Work performance information ❑ Change requests ❑ Project management plan updates ❑ Project document updates ❑ Organizational Process Assets updates

Control Quality- Inputs

Project management plan contains the quality management plan, which describes how quality control will be performed within the project

Quality Metrics will be used to measure whether the work result meet quality specification

Quality Checklists will assist in assessing conformance of the deliverables to quality **Work Performance Data** evaluate actual progress against the planned progress.

Approved change requests include modification such as defect repairs, revised work methods and revised schedule

Deliverables of the project are primary input to this process

Control Quality - Inputs

Project Documents such as Agreements , Quality audit reports and change logs supported with corrective actions plans, Training plans and assessments of effectiveness.

Organizational Process Assets that influence the control quality process include, but are not limited to:

- The organization's quality standard and policies
- Standard work guidelines
- Issue and defect reporting procedures and communication policies

Control Quality – Tools and Techniques

Seven Basic Quality Tools

Statistical sampling is based on selecting and testing a random sample instead of the entire population.

Inspection is the examination (reviews, peer reviews, audits, or walkthroughs) of a work product to determine whether it conforms to documented standards.

Approve Change Request Review – All approved change requests should be reviewed to verify that they were implemented as approved

Control Quality – Outputs

Quality control measurements are the documented results of quality control activities in the format specified during quality planning.

Validated changes involve inspection of any changed or repaired items and they will be either accepted or rejected before notification of the decision is provided. Rejected items may require rework.

Verified deliverables - A goal of quality control is to determine the correctness of deliverables. The results of the execution quality control processes are validated deliverables.

Work Performance Information – is the performance data collected from various controlling processes, analyzed in context and integrated based on relationship across areas

Control Quality – Outputs

Change requests as a result of corrective or preventive action. These changes should be initiated in accordance with the Integrated change control Process

Project management plan updates includes update to quality management plan and the process improvement plan

Project documents that may be updated include, but are not limited to, quality standards, Agreements , Quality audit report and change log supported with corrective action plans.

Organization process assets updates includes updates to Completed checklists, lesson learned documentation etc.

Exercise 4: Quality Management

Revising Key Terms : Complete the following

1._____ is more important than inspection in Quality Management

2. When a process has data points above the upper limit or blow the lower limit , those data points are out of _____

3. _____ is the process of monitoring and recording results of executing the quality activities to assess performance and recommend necessary changes.

Project Human Resource Management

By

Casmir, R. (Ph.D)

Project Human Resource Management

Project Human Resource Management includes the processes that organize, manage, and lead the project team.

- The project team is comprised of the people with assigned roles and responsibilities for completing the project.
- Project team members may have varied skill sets, may be assigned full or part-time, and may be added or removed from the team as the project progresses.

Project Human Resource Management

- Project team members may also be referred to as the project's staff.
- Although specific roles and responsibilities for the project team members are assigned, the involvement of all team members in project planning and decision making is beneficial.
- Participation of team members during planning adds their expertise to the process and strengthens their commitment to the project.

Project Human Resource Management

- For you that are new to Project Management, how should you approach studying this Knowledge Area?
- It’s recommended to have quick review of what it is and how it interacts with other knowledge areas, and then an in-depth study of all its components and interactions.

Project HR Management Processess

This knowledge are is composed of the following processes:

- **Plan Human Resource Management**—The process of identifying and documenting project roles, responsibilities, required skills, reporting relationships, and creating a staffing management plan.
- **Acquire Project Team**—The process of confirming human resource availability and obtaining the team necessary to complete project activities.

Project HR Management Processess

- **Develop Project Team**—The process of improving competencies, team member interaction, and overall team environment to enhance project performance.

- **Manage Project Team**—The process of tracking team member performance, providing feedback, resolving issues, and managing changes to optimize project performance.

Project HR Management Processess

Some important questions you need to ask yourself when studying this area.

- What's the difference between Human Resource Management Plan and Project Management Plan?
- What's the difference between RBS and WBS?
- What's the difference between Project Team and Project Management Team?
- How does Human Resource Affects Project Risk Management?

Project HR Management Processess

- How does Human Resource Affects Project Time Management?
- How does Human Resource Affects Project Cost Management?
- Why does Human Resource Management has no processes in the Monitor and Control Process Group?
- What importance does the Activity Cost Estimates in Human Resources has in Time Management?
- Fast Tracking vs. Crashing which one impacts the quality most

RBS vs WBS

- In project management, the resource breakdown structure (RBS) is a tool to help with planning and controlling project work.
- Think of it as a work breakdown structure (WBS) for your resources, because that's what it is, and it's just as valuable.
- Work breakdown structure (WBS) is a graphical portrayal of the total scope of work to be completed, divided down into tiny chunks that each person in the group can comprehend.
- RBS, is a list of resources and its allocation done and required to complete a project

Schedule Compression

- Schedule compression refers to techniques used when a project manager wants to shorten the duration of the project without changing the scope of the project.
- It can be used when a project falls behind schedule and needs to catch up or to finish the project sooner than originally scheduled.
- Fast tracking involves the performance of activities in parallel, whereas crashing involves the addition of resources to a project.
- In fast tracking, there is increased risk, whereas in crashing, there is increased cost.

Schedule Compression

Fast Tracking

Advantages

- Reduces project duration without increasing costs (initially).
- Helps meet tight deadlines or recover lost time.
- Utilizes existing resources more efficiently.

Disadvantages

- Increases risk due to overlapping tasks.
- May lead to rework and errors if dependent activities are not adequately completed.
- Reduces flexibility in schedule management.

Schedule Compression

Crashing

Advantages

- Can significantly reduce project duration.
- Provides greater control over specific task durations.
- Effective when time is more critical than cost.

Disadvantages

- Increases project costs due to added resources (overtime, hiring, etc.).
- May lead to resource over-allocation or burnout.
- Diminishing returns if overused or applied ineffectively.

Project Communications Management

By
Casmir, R. (Ph.D)

Project Communications Management

Processes required to ensure timely and appropriate generation, collection, distribution, storage, retrieval and ultimate disposition of project information. This knowledge area is made up of five processes to determine what to communicate, to whom, how often, through which media and when to reevaluate the plan

An effective project manager spends about 90% of his time communicating and 50% of that time is spent communicating with the project team

Communication Model – Various Elements

Sender	The person who needs to initiate the communication
Receiver	The person who gets the message
Medium	The thing used to transmit the message
Feedback	A response to a message
Noise	Something that interferes with the message
Encode	Modify a message so that it can be sent
Decode	Modify a message that has been sent so that it can be understood

Kinds of Communication

- Formal vs Informal
- Verbal vs Non-verbal
- Official vs Non-official
- Written vs Oral
- Vertical vs Horizontal

Formal vs Informal Communication

Feature	**Formal Communication**	**Informal Communication**
Tone	Professional, structured	Casual, friendly
Channel	Official channels (e.g., memos, reports)	Personal chats, social media, phone calls
Usage	In organizations, business, and academics	Among friends, coworkers casually
Documentation	Usually documented	Often not recorded
Example	An email to your manager	Chatting with a colleague at lunch

Verbal vs Non-verbal Communication

Feature	**Verbal Communication**	**Non-verbal Communication**
Nature	Uses **spoken or written words**	Uses **body language, gestures, expressions**
Form	Oral or written	Facial expressions, posture, eye contact
Clarity	Often more explicit	Can be ambiguous, open to interpretation
Example	Giving a speech, sending an email	Nodding, smiling, frowning

Official vs Non-official Communication

Feature	**Official Communication**	**Non-official Communication**
Authority	Authorized by the organization	Personal and not sanctioned by the org
Purpose	Work-related, formal	Personal sharing or social interaction
Flow	Follows hierarchy or set structure	Informal and unstructured
Example	A company memo or meeting minutes	A WhatsApp message between coworkers

Written vs Oral Communication

Feature	**Written Communication**	**Oral Communication**
Medium	Emails, letters, reports	Meetings, phone calls, conversations
Permanence	Permanent record	Temporary unless recorded
Speed	Slower, more time to prepare	Fast, immediate
Feedback	Delayed	Instant
Example	Report to management	Giving instructions verbally in a meeting

Vertical vs Horizontal Communication

Feature	**Vertical Communication**	**Horizontal Communication**
Direction	Between different **levels** in an organization	Between people at the **same level**
Purpose	Instructions, reporting, feedback	Coordination, collaboration
Flow	Upward (to managers) or downward (to staff)	Lateral (peer to peer)
Example	A supervisor giving instructions to staff	Two department heads working on a project

Kinds of Communication

Communication Type	Example
Formal Written	RFP, Project Charter
Informal Written	E-mail, Memos
Formal Verbal	Speech
Informal Verbal	Conversations

Kinds of Communication

- **Nonverbal Communication**

Means gestures, facial expressions, and physical appearance while you are communicating your message

- **Para lingual Communication**

Is the tone and pitch of your voice when you are telling people what's going on with your project

⇒ Body language, gestures, facial expressions, tone and pitch of voice are all examples of paralinguistic features.

- **Feedback**

Is when you respond to communication.

- **Active listening**

When you give feedback to someone who is speaking

Communication Management Processes

Process	Process Group
❑ Plan Communications management ❑ Manage Communications ❑ Control Communications	❑ Planning ❑ Executing ❑ Monitoring and Controlling

Plan Communication Management

Determining the project stakeholder information needs and defining a communication approach. It entails:

- How often communication will be distributed and updated
- Mode of communication (email, teleconference etc)
- Information that will be included in the project communications
- Stakeholders that will receive this information

Plan Communications Management

Inputs	Tools and Techniques	Outputs
❑ Project management plan ❑ Stakeholder ❑ Register ❑ Enterprise Environmental Factors ❑ Organizational Process Assets	❑ Communications requirements analysis ❑ Communications technology ❑ Communication models ❑ Communication method ❑ Meetings	❑ Communications ❑ Management Plan ❑ Project document updates

Plan Communications Management - Inputs

Project Management Plan provides information on how the project will be executed , monitored , controlled , and closed.

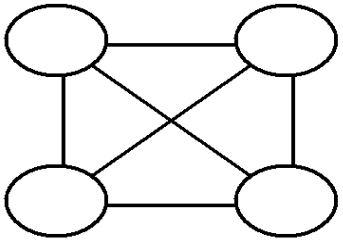
Stakeholder Register provides the information needed to plan the communication with project stakeholders.

Enterprise Environmental Factors , plan communication management process is tightly linked with enterprise environmental factors, since the structure of an organization will have a major effect on the project's communication requirements.

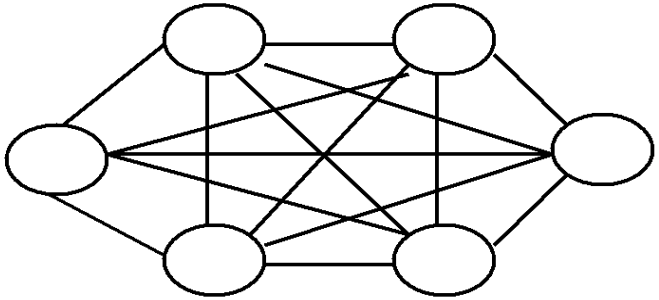
Organization Process Assets, Lesson learned and historical information are of particular importance because they can provide insights on both the decision taken regarding communication issues and the result of the decision in previous similar project

Plan Communications Management - Tools and Techniques

Communication Requirements Analysis , The analysis of the communication requirements determine the information needs of the project stakeholders. The project manager should also consider the number of potential communication channel or paths as an indicator of the complexity of a project's communications. For Example if there are n people involve in a project communication. The total number of channels would be n(n-1)/2



4 people – 6 channels



6 people – 15 channels

Plan Communications Management - Tools and Techniques

Communication Technology, The method used to transfer information among project stakeholder may vary significantly. For example, a project team may use technique from brief conversation to extended meetings, or from simple written documents to extensive materials which are accessible online as a method of communication .

Factors that can affect the choice of communication technology include :

- Urgency of the need for information
- Availability of technology
- Ease of use
- Project environment
- Sensitivity and confidentiality of the information

Plan Communications Management - Tools and Techniques

Communication Models, Basic model of communication includes following parameters

- Encode : Thoughts and ideas translated to a language which can be understood by other
- Message & Feedback Message : Output of encoding
- Medium : The method used to convey the message
- Decode : Converting feedback back to understandable form

Plan Communications Management - Tools and Techniques

Communication Methods are broadly classified into

- Interactive communication
- Push Communication
- Pull Communication

Meetings , The plan communication process requires discussion and dialogue with the project team to determine the most appropriate way to Update and communicate project information and to respond to request from various stakeholders for that information. These discussions and dialogue are commonly facilitated through meetings, which may be conducted face to face or online and in different locations, such as the project site or customer's site.

Plan Communications Management - Outputs

Communication Management Plan contains the following information

- Stakeholder communication requirements
- Information to be communicated including language , format , content and level of detail;
- Reason for distribution of that information
- Time frame and frequency for the distribution of required information and receipt of acknowledgement or response , if applicable
- Person responsible for communicating the information
- Person responsible for authorizing release of confidential information
- Person or groups who will receive the information
- Person or groups who will receive the information
- Methods or technologies used to convey the information, such as memos, e-mail, and /or press releases;

Plan Communications Management - Outputs

Communication Management Plan contains the following information

- Methods or technologies used to convey the information, such as memos, e-mail, and /or press releases;
- Glossary of common terminology
- Flow charts of the information flow in the project, workflows with possible sequence of authorization , list of report, and meeting plans etc and
- Communication constraints usually derived from a specific legislation or regulation , technology , and organization policies etc

The communications management plan can also include guidelines and templates for project status meetings , project team meetings , e-meetings and e-mail messages.

Communications Plan

Communication	Format	Frequency	Distribution
Team Briefing	Restricted Intranet	Daily at 9:00	Team and stakeholders with access to secure project info area
Weekly Web Bulletin	Internal Intranet	Weekly	Team, sponsor, senior management
Technical Incident Report	Email	Immediately after Incident	Webmaster, IT Department
Budget and Schedule Detail	Spreadsheets and Detailed Gantt Chart	Bi-Weekly	Sponsor, Senior Management
Accomplishments and Setbacks	Email and Intranet	Weekly	All internal stakeholders
Schedule Milestones	Email and Intranet	Weekly	All internal stakeholders
Cost-to-Date Milestones	Email and Intranet	Weekly	All internal stakeholders
Current Top 5 Risks	Email and Intranet	Weekly	All internal stakeholders

Manage Communications

This process updates stakeholders on the progress of the project as per the communications management plan. The information that gets distributed includes status reports, review meetings etc

Manage Communications

Inputs	Tools and Techniques	Outputs
❑ Communication management plan ❑ Work performance reports ❑ Enterprise environmental factors ❑ Organizational process assets	❑ Communication technology ❑ Communication models ❑ Communications method ❑ Information management system ❑ Performance reporting	❑ Project communications ❑ Project management plan updates ❑ Project document updates ❑ Organizational Process assets updates

Control Communications Process

This process tracks how the project is progressing against the plan and provide report to stakeholders on status, progress metrics (Schedule and Cost variance) and forecasts

Control Communications Process

Inputs	Tools and Techniques	Outputs
❑ Project management plan ❑ Project communications ❑ Issue log ❑ Work performance data ❑ Organizational Process assets	❑ Information management system ❑ Expert Judgment ❑ Meetings	❑ Work performance information ❑ Change requests ❑ Project management plan updates ❑ Project document updates ❑ Organizational Process assets updates

Project Communication Management

Communication Channels

- Channels = n(n-1)/2
  Where n is the number of people on the project

- If there are five people on the project, then we have 10 communication channels.

- This is calculated as follows: 5*(5-1) /2 = 10

Communication Filtering

When a portion of the message is lost in communication.

Common reasons of filtering:

- Ambiguous language
- Predefined mindset
- Hidden agenda in message content
- Senders reputation
- Senders status in the organization
- Dysfunctional emotional behavior
- Environmental Background

Exercise 1: Communication Model

Revising Key Terms : Communication Model

1. _____ **is used to transmit the message**

2.The person who gets the message is _____

3.Something that interferes with the message is_____

Exercise 2: Kinds of Communication

Revising Key Terms : Kinds of Communication

1.The Project manager writes a requirement specification for your project. This is an example of _____ kind of communication.

a) **Informal verbal**
b) **Informal written**
c) **Formal verbal**
d) **Formal written**

2. You sent an email to some of your team members to get more information about an issue that has been identified on your project. This is an example of _____ kind of communication

a) **Informal verbal**
b) **Informal written**
c) **Formal verbal**
d) **Formal written**

Exercise 3: Communication Management

Revising Key Terms : Complete the following

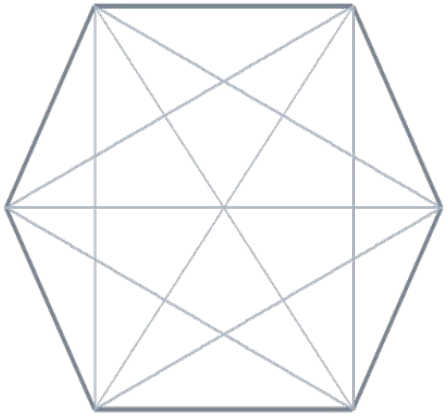
1. A contract is always _____ communication
2. A stakeholder with high interest and low power must be kept_____ of your project's progress
3. Information should be distributed to all _____

Exercise 4: Number of Lines of Communication

Practicing the concept

You are managing a project with five team members and one sponsor. Draw all the channels of communication on this picture.

Solution: Lines of Communication = 6 * (6 – 1)/2 = 15



Project Risk Management

Casmir, R. (PhD)

Goals and Objectives

- Project Risk
- Risk Management
- Risk Management Planning
- Identifying & Quantifying Risks
- Response Planning
- Risk Monitoring & Control

What is Risk?

- Project risk is an uncertain event that can have a positive or a negative impact on the result
- If a project risk materializes and has a negative impact on the project, it is called “Threat” to the project e.g.
  - Unproven technology impacting project schedule
  - Loosing critical resource from project
- If an uncertain event has a positive impact on the project, it is called “opportunities”. E.g.
  - Emergence of new tool to increase productivity
  - New resource replacing the lost resource is a better one!
- Not all risks are bad, however most of them are perceived as “Threat” to success

What is Project Risk?

- An event that, if it occurs, causes either a positive or negative impact on a project
- Keys attributes of Risk
  - Uncertainty
  - Positive and Negative
  - Cause and Consequence



Project Risks – Key characteristics

- Some of the risks are known risks e.g.
  - Working in a project with competitor (multi-vendor scenario)
  - Doing a project in a new geography (cultural differences)
  - Project attracting environmentalists attention
  - Accident during Travel
- To manage “known risks”, risk response plan is prepared
- Known risks are predicted / identified based on the past experience of similar projects

Risk Management

- Risk management is concerned with identifying risks and drawing up plans to minimise their effect on a project.
- A risk is a probability that some adverse (or positive) circumstance will occur
  - Project risks affect schedule or resources;
  - Product risks affect the quality or performance of the software being developed;
  - Business risks affect the organization developing or procuring the software.

Project Risks – Key characteristics

- All the risks can not be predicted and such risks are known as “unknown risks”. e.g.

  Natural disaster
- Unknown risks cannot be managed proactively
- Most of the organizations address such risks through “Business continuity” and “Disaster Recovery” plan
- What is the difference between “Risk” and “Issue”?

Risk Management Process

- PMBOK ® Definition
  - “The systematic process of identifying, analyzing, and responding to project risk”
- Steps
  - ✓ Risk Management Planning
  - ✓ Risk Identification
  - ✓ Qualitative/Quantitative Risk Analysis
  - ✓ Risk Response Planning
  - ✓ Risk Monitoring & Control

Value From Managing Risks

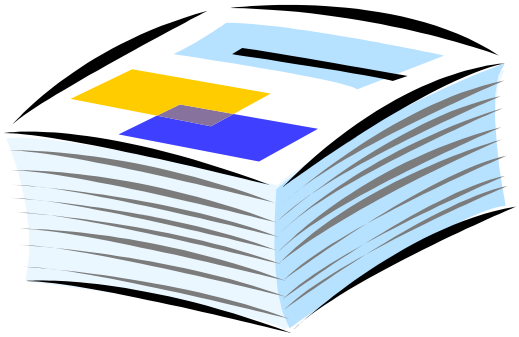
- Opportunity to move from “fire-fighting” to proactive decision making on the project.
- Better chance of project success.
- Improved project schedule and cost performance.
- Stakeholders and team members better understand the nature of the project.
- Helps define the strengths and weaknesses of the project.

Risk Management Planning

- Plan for the Planning
  - Risk planning should be appropriate for the project
  - Question you should ask:
    1. How risky is the project?
    2. Is it a new technology or something your organization is familiar with?
    3. Do you have past projects to reference?
    4. What is the visibility of the project?
    5. How big is the project?
    6. How important is the project?

The Risk Management Plan

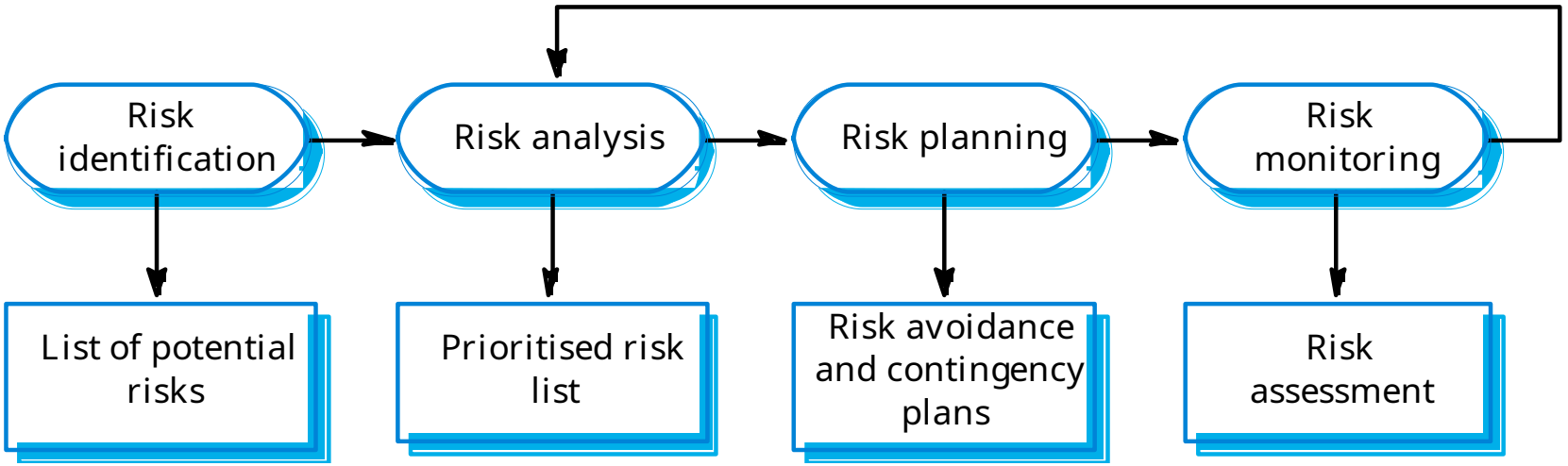
- What should it include?
  - How you will identify, quantify or qualify risk
    - Methods and tools
  - Budget…yes budget
  - Who is doing what
  - How often
  - Risk categories, levels, and thresholds for action.
  - Reporting requirements
  - Monitoring, tracking and documenting strategies



The Risk Management Process

- Risk identification
  - Identify project, product and business risks;
- Risk analysis
  - Assess the likelihood and consequences of these risks;
- Risk response planning
  - Draw up plans to avoid or minimize the effects of the risk;
- Risk monitoring
  - Monitor the risks throughout the project;

The Risk Management Process



Identifying Risk

- Continuous, Iterative Process
- What is it and what does it look like
- The sooner the better
- The more the merrier
- A fact is not a risk (it’s an issue).
- Be specific
- Don’t try to do everything at once



Identification Techniques

- Brainstorming
- Checklists
- Interviewing
- SWOT Analysis (strengths, weaknesses opportunities, threats)
- Delphi Technique (anonymous consensus building)
- Diagramming Techniques
  - Cause & effect
  - Flow Charts
  - Influence Diagrams

Identification Techniques

Delphi Technique (anonymous consensus building)

- ✓ The **Delphi Technique** is a method used to gather expert opinions and reach a group consensus through a series of structured questionnaires or surveys.

Key Features:

- **Panel of Experts**: A group of knowledgeable individuals in a specific field.
- **Multiple Rounds**: Experts answer questions anonymously over several rounds.
- **Anonymity**: Experts don’t know who the other participants are, which reduces the influence of dominant personalities.
- **Feedback Loop**: After each round, a summary of the group's responses is shared, and participants can revise their earlier answers.
- **Consensus Building**: The process continues until the group reaches a general agreement or the responses stabilize.

Risks and Risk Types

Risk type	Possible risks
Technology	The database used in the system cannot process as many transactions per second as expected. Software components that should be reused contain defects that limit their functionality.
People	It is impossible to recruit staff with the skills required. Key staff are ill and unavailable at critical times. Required training for staff is not available.
Organizational	The organization is restructured so that different management are responsible for the project. Organizational financial problems force reductions in the project budget.
Tools	The code generated by CASE tools is inefficient. CASE tools cannot be integrated.
Requirements	Changes to requirements that require major design rework are proposed. Customers fail to understand the impact of requirements changes.
Estimation	The time required to develop the software is underestimated. The rate of defect repair is underestimated. The size of the software is underestimated.

Software Risks

Risk	Affects	Description
Staff turnover	Project	Experienced staff will leave the project before it is finished.
Management change	Project	There will be a change of organizational management with different priorities.
Hardware unavailability	Project	Hardware that is essential for the project will not be delivered on schedule.
Requirements change	Project and product	There will be a larger number of changes to the requirements than anticipated.
Specification delays	Project and product	Specifications of essential interfaces are not available on schedule
Size underestimate	Project and product	The size of the system has been underestimated.
CASE tool under-performance	Product	CASE tools which support the project do not perform as anticipated
Technology change	Business	The underlying technology on which the system is built is superseded by new technology.
Product competition	Business	A competitive product is marketed before the system is completed.

Risk Analysis

- Assess probability, seriousness, and urgency of each risk.
- Probability may be very low, low, moderate, high or very high.
- Risk effects might be catastrophic, serious, tolerable or insignificant.
- Urgency might be immediate, short term, or long term.

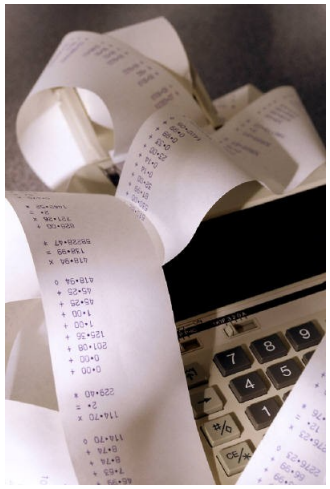
Analyzing Risk - Qualitative



- Subjective
- Educated Guess
- High, Medium, Low
- Red, Yellow, Green
- 1-10
- Prioritized/Ranked list of ALL identified risks
- First step in risk analysis!

Risk Analysis - Quantitative

- Numerical/Statistical Analysis
- Determines probability of occurrence and consequences of risks
- Should be focused to highest risks as determined by Qualitative Risk Analysis and Risk Threshold



Risk Analysis (i)

Risk	Probability	Effects
Organizational financial problems force reductions in the project budget.	Low	Catastrophic
It is impossible to recruit staff with the skills required for the project.	High	Catastrophic
Key staff are ill at critical times in the project.	Moderate	Serious
Software components that should be reused contain defects which limit their functionality.	Moderate	Serious
Changes to requirements that require major design rework are proposed.	Moderate	Serious
The organization is restructured so that different management are responsible for the project.	High	Serious

Risk Analysis (ii)

Risk	Probability	Effects
The database used in the system cannot process as many transactions per second as expected.	Moderate	Serious
The time required to develop the software is underestimated.	High	Serious
CASE tools cannot be integrated.	High	Tolerable
Customers fail to understand the impact of requirements changes.	Moderate	Tolerable
Required training for staff is not available.	Moderate	Tolerable
The rate of defect repair is underestimated.	Moderate	Tolerable
The size of the software is underestimated.	High	Tolerable
The code generated by CASE tools is inefficient.	Moderate	Insignificant

Probability & Impact Analysis

Risk	Probability	Impact	Expected Value
1	25%	$45,000	$11,250
2	50%	$2,000	$1,000
3	30%	$100,000	$30,000

Risk Response Planning

- “What are we going to do about it?”
- Techniques/Strategies:
  - Avoidance – Eliminate it
  - Transference – Pawn it off
  - Mitigation – Reduce probability or impact of it
  - Acceptance – Do nothing
- Strategy should be commensurate with risk
  - Hint: Don’t spend more money preventing the risk than the impact of the risk would be if it occurs ☺
- The Risk Response Plan/Risk Response Register

Risk Management Strategies (i)

Risk	Strategy
Organizational financial problems	Prepare a briefing document for senior management showing how the project is making a very important contribution to the goals of the business.
Recruitment problems	Alert customer of potential difficulties and the possibility of delays, investigate outsourcing work.
Staff illness	Reorganize team so that there is more overlap of work and people therefore understand each other's jobs.

Risk Management Strategies (ii)

Risk	Strategy
Requirements changes	Derive traceability information to assess requirements change impact, and maximise information hiding in the design.
Organizational restructuring	Prepare a briefing document for senior management showing how the project is making a very important contribution to the goals of the business.
Database performance	Investigate the possibility of buying a higher-performance database.
Underestimated development time	Investigate outsourcing components, investigate use of a program generator

Risk Monitoring

- Assess each identified risk regularly to decide whether or not it is becoming less or more probable.
- Also assess whether the effects of the risk have changed.
- Each key risk should be discussed at management progress meetings.

Risk Monitoring & Control

- Continuous, Iterative Process
- Done right the risk impact will be minimized:
  - Someone IS responsible
  - Watch for risk triggers
  - Communicate…Communicate…Communicate
  - Take corrective action - Execute
  - Re-evaluate and look for new risk constantly
- Tools:
  - Risk Reviews
  - Risk Audits

Risk indicators

Risk type	Potential indicators
Technology	Late delivery of hardware or support software, many reported technology problems
People	Poor staff morale, poor relationships amongst team member, job availability
Organizational	Organizational gossip, lack of action by senior management
Tools	Reluctance by team members to use tools, complaints about CASE tools, demands for higher-powered workstations
Requirements	Many requirements change requests, customer complaints
Estimation	Failure to meet agreed schedule, failure to fix reported defects

PROJECT PROCUREMENT MANAGEMENT

CASMIR, R.O. (PhD)

Procurement Definition

- Procurement means buying, purchasing, renting, leasing or otherwise acquiring any goods, works or services by a procuring entity
- Purchasing is often used to refer to contracts and big products, *whereas*
- Buying is inferred for small products

Defining Procurement

- It is the process of acquiring goods, works and services, covering both acquisition from third parties and from in-house providers.
- The process spans the **whole life cycle** from identification of needs, through to the end of services contract or the end of the useful life of an asset. It involves options appraisal and the critical make or buy decision.

- Processes necessary to purchase or acquire products, services or results needed from outside the project team.
- Purchasing focuses on the cost of the order, while procurement focuses on value creation and Total Cost of Ownership. While purchasing aims to minimize the cost of an order, procurement aims at other objectives like risk mitigation, contract compliance, cost savings, ongoing supplier relationships

Types of Procurement

- The procurement procedure choice and subsequently the type of contract depend on the following categories of procurement;

(a) Works

- Design, construction, rehabilitation, maintenance and repair of buildings, factories, roads, plant, bridges and others temporary or permanent constructions.

(b) Supplies

- Supply of various requirements such as stationary, building materials, finished goods, spare parts .

(c) Non-Consultancy Services

- Maintenance and repair of equipment, security services, cleaning services, operating staff canteen.

(d) Consultancy Services

- Studies, designs, provision of technical assistance, and training by a consultant.

Fundamental Procurement Principles

- **Procurement Principles**
  - Professionalism
  - Transparency
  - Value for Money
  - Accountability
  - Competitiveness
  - Fairness
  - Ethical Approach

Principles of Public Procurement

What is Professionalism?

- It is the discipline whereby educated, experienced and responsible procurement officers make informed decisions regarding purchase operations.
- Enables more informed judgments about what is available in national and international markets.
- Gain knowledge of good domestic and internationally acceptable procurement practices.
- Enables individuals and organisations to implement global and national business practices within the governmental procurement laws.

Principles of Public Procurement

What is Transparency?

- Transparency means that the same rules apply to all suppliers of goods, works and services and that these rules are publicised as the basis of procurement decisions prior to their use.

- Enables the creation of open, fair and transparent procedures.

- Creates goodwill between organisations and their stakeholders

Principles of Public Procurement

What is Value for Money?

- It is the optimum combination of whole-life cost and quality to meet the customers requirements.
- Enables the cost effective use of scarce funds
- Reduces operational costs, waste and losses.

What is Accountability?

- It is the process of holding an individual(s) or an organisation fully responsible for all aspects of the procurement process over which they exert authority.
- Strengthens the perception of transparency and fairness.
- Reduces the incidence of corruption.

Principles of Public Procurement

What is Competitiveness?

- Actively encouraging greater supplier participation in the bidding process through advertising, sourcing reviews, prequalification etc.
- Potential for cost savings.
- Increase the potential supplier base.
- Greater awareness of new developments.

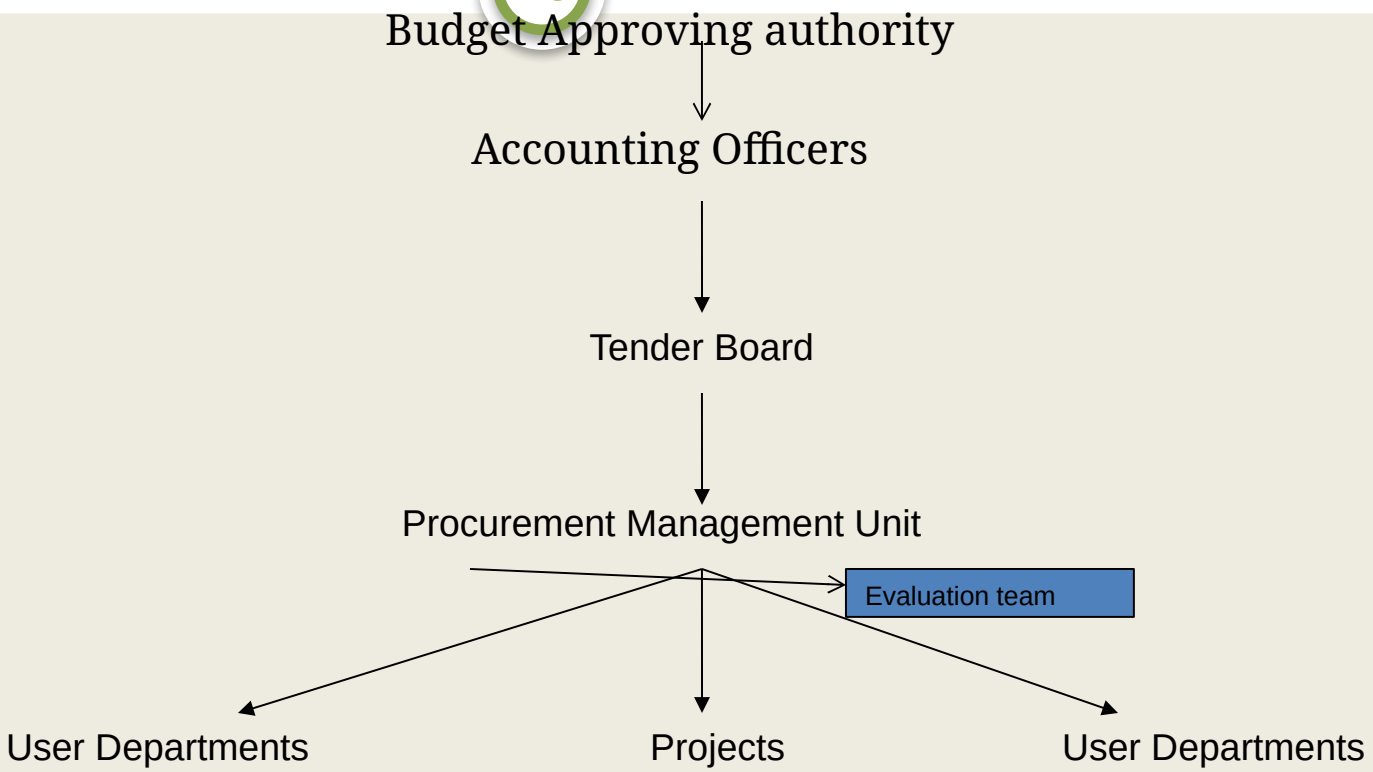
What is Fairness?

- Showing consideration and impartiality in all stages of the procurement process, especially demonstrating equality in bid evaluations.
- Development of mutual trust.
- Increases the potential supplier base.
- Greater awareness of new developments.

What is Ethical Approach?

- Exemplary approach to all procurement processes that cannot be questioned or criticized.
- Reduction in corruption.
- Embodies all procurement principles.
- Enhanced profile for procurement.

Procuring Entities Structure



Legal Establishments and Institutions

- Tender Board
- Accounting Officer
- Procurement Management Unit (PMU)
- Evaluation Committee
- User Department
- [Budget Approving Authority]

Choice of Procurement Method

- Choice of Procurement Method may determined by:
  - Value
  - Circumstances such as:
    - Urgency
    - Limited number of supply sources
    - Nature of requirement
    - Currency of payment

METHODS OF SELECTION AND LIMIT OF APPLICATION PERCONTRACT
FOR GOODS, WORKS AND NON-CONSULTANCY SERVICES

Method of tendering	**Goods**	**Works**	**Non Consultancy Services**	**Disposal of Public Assets**
International competitive tendering	No limit	No limit	No limit	No limit
National competitive tendering	Up to Tshs 1,000,000,000	Up to Tshs 5,000,000,000	Up to Tshs 1,000,000,000	Up to Tshs 5,000,000,000
Restricted tendering	No limit but must be justified	No limit but must be justified	No limit but must be justified	No limit but must be justified
Competitive quotations (Shopping)	Up to Tshs 120,000,000	Up to Tshs 200,000,000	Up to Tshs 100,000,000	Not applicable
Single source procurement	No limit, but must be justified.	No limit, but must be justified.	No limit, but must be justified.	Not applicable
Minor value procurement	Up to 10,000,000	Up to 20,000,000	Up to 10,000,000	Not applicable
Micro value procurement	5,000,000	Not applicable	Not applicable	Not applicable

Project Procurement Management

Negotiations Techniques

- **Deadline**: imposing a deadline to reach an agreement
- **Stalling**: one party may claim that an agreement cannot be finalized because of his limited authority
- **Fair and Responsible**: a negotiator may claim that the price for a product is equitable because that's what another company is paying.
- **Delay**: are useful when tempers are beginning to flare, to divert from a subject, etc (Ex: arrival of refreshments, request for recess etc)
- **Withdrawal**: one party may attack an issue, then retreat

Project Procurement Management

Negotiations Techniques

- **Arbitration**: a third party may be brought in when agreement cannot be reached. (out of court settlement)
- **Fait Accompli**: a party may claim that what is being asked for has already been accomplished and cannot be changed (*a thing that has already happened or been decided before those affected hear about it, leaving them with no option but to accept it.*)

Close Procurements



- Process of completing each project procurement.
- It involves administrative activities such as updating the project records with results and archiving the information for future use and finalizing open claims in accordance with the terms and conditions stated in the contract document.
- Close Procurement process closes the procurement applicable to that phase of the project.

Project Stakeholder Management

By
Casmir, R. (Ph.D)

What is Stakeholder Management?

Stakeholder Management includes the processes required to identify people, groups or organizations that could impact or be impacted by the project, to analyze stakeholder expectations and their impact on the project and to develop appropriate management strategies for effectively engaging stakeholder in project decisions and execution.

What is Stakeholder Management?

Stakeholder management also focuses on:

- continuous communication with stakeholders to understand their needs and expectations,
- addressing issues as they occur, managing conflicting interests, and
- fostering appropriate stakeholder engagement in project decisions and activities.
- Stakeholder satisfaction should be managed as a key project objective.

Project Stakeholder Management Key processes?

This knowledge are is composed of the following processes:

- *Identify Stakeholders*—The process of identifying the people, groups, or organizations that could impact or be impacted by a decision, activity, or outcome of the project; and analyzing and documenting relevant information regarding their interests, involvement, interdependencies, influence, and potential impact on project success.

Identify Stakeholders

The key benefit of Identify stakeholder process is that it allows the project manager to identify the appropriate focus for each stakeholder or group of stakeholders

What is Stakeholder Management?

This knowledge are is composed of the following processes:

- *Plan Stakeholder Management*—The process of developing appropriate management strategies to effectively engage stakeholders throughout the project life cycle, based on the analysis of their needs, interests, and potential impact on project success

What is Stakeholder Management?

This knowledge are is composed of the following processes:

- *Manage Stakeholder Engagement*—The process of communicating and working with stakeholders to meet their needs/expectations, address issues as they occur, and foster appropriate stakeholder engagement in project activities throughout the project life cycle.

What is Stakeholder Management?

- Every project will have stakeholders who are impacted by or can impact the project in a positive or negative way.
- While some stakeholders may have a limited ability to influence the project, others may have significant influence on the project and its expected outcomes.
- The ability of the project manager to correctly identify and manage these stakeholders in an appropriate manner can mean the difference between success and failure.

Identify Stakeholders

Inputs	Tools & Techniques	Outputs
❑ Project Charter ❑ Procurement documents ❑ Enterprise ❑ Environmental Factors ❑ Organizational process assets	❑ Stakeholder analysis ❑ Expert judgment ❑ Meetings	❑ Stakeholder Register

Identify Stakeholders - Inputs

Project Charter - can provide information about internal and external parties related with the project and affected by the result or the execution of the project, such as project sponsor(s), customers, team members , groups and departments participating in the project, and other people or organizations affected by the project.

Procurement Documents – If the project is the result of a procurement activity or is based on an established contract , the parties in that contract are key project stakeholders. Other relevant parties such as suppliers should also be considered as part of the project stakeholder list.

Identify Stakeholders - Inputs

Enterprise Environmental Factors – relevant to Identify stakeholders are:

- Organizational culture and structures
- Governmental or industry standards (e.g. Regulations, product standards); and
- Global , regional or local trends and practices or habits.

Organizational Process Assets – relevant to Identify stakeholder are :

- Stakeholder register templates
- Lessons learned from previous projects or phases and
- Stakeholder register from previous projects.

Identify Stakeholders -Tools and Techniques

1. **Stakeholder Analysis -** Identifies the interests, expectations and influence of the stakeholders and relates them to the purpose of the project.

There are multiple classification models used for stakeholders analysis , such as :

- *Power /Interest grid*, grouping the stakeholders based on their level of authority
- ("power") and their level or concern ("interest") regarding the project outcomes
- *Power /Influence grid*, grouping the stakeholders based on their level of authority
- ("power") and their active involvement ("influence") in the project.
- *Influence/ Impact grid ,* grouping the stakeholders based on their active involvement (" Influence") in the project and their ability to effect changes to the project's planning or execution (" impact ").
- *Salience model,* describing classes of stakeholders based on their power (ability to impose their will), urgency (need for immediate attention), and legitimacy (their involvement is appropriate).

Identify Stakeholders -Tools and Techniques

2. Expert Judgement include suggestions and recommendations of Subject Matter Experts such as :

- Senior Management
- Other units within the organization
- Identified key stakeholders
- Project managers who have worked on projects in the same area
- Subject Matter Experts in the business project area
- Industry groups and consultants
- Professional and technical associations , regulatory bodies and non governmental organizations (NGOs)

3. Meetings – Project meetings designed to develop an understanding of major project stakeholders and they an be used to exchange and analyze information about roles , interest , knowledge and the overall position of each stakeholder facing the project.

Power Interest Grid

- The **Power/Influence Grid**, which is also known as the **Power/Influence** Matrix in stakeholder management, is a simple tool helps you categorize project stakeholders by the **power** and **influence** they have on the project.
- By definition, **power** is the level of authority a stakeholder has in the project.

How do you use a power-interest grid?

- The **power/interest grid** has an x axis for **interest** and y axis for **power**.
- The **grid** is split into four quadrants, the bottom left is keep informed; the top left is actively consult; top right is actively engage; and bottom right is maintain **interest**.

Power Grid Matrix

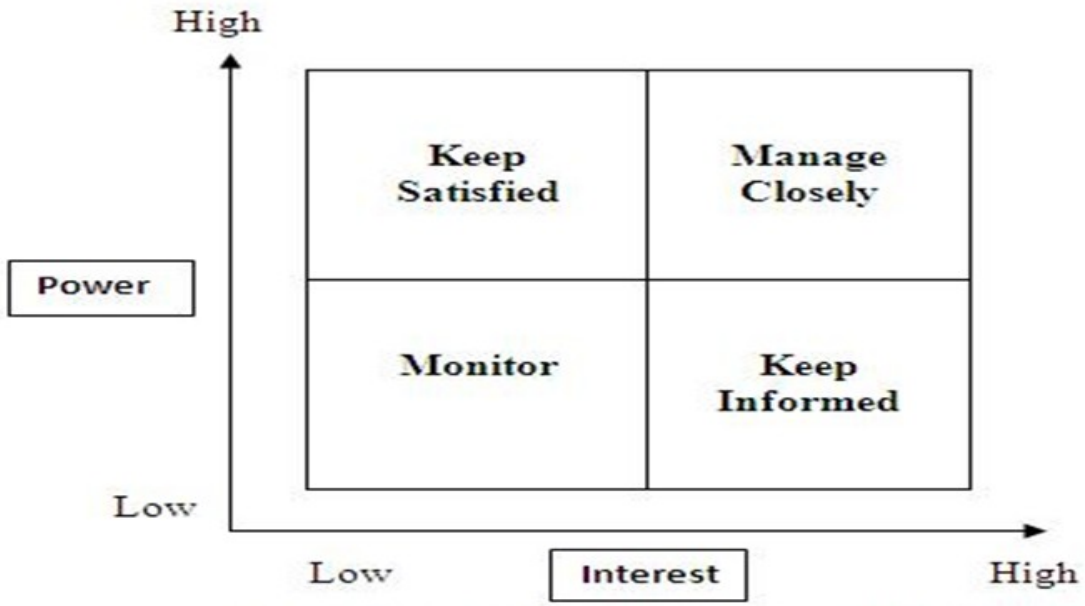


Figure: Power / Interest Grid with Stakeholders

Identify Stakeholders - Output

The main output of Identify Stakeholder process is stakeholder register. It contains following details.

Identification Information :

Name, Organizational Position, Location, Role in the project, Contact info

Assessment Information:

Major Requirement, Main expectation, Potential Influence

Stakeholders Category:

External/Internal, Supporter, Neutral, Resister

Plan Stakeholder Management

Plan Stakeholder Management is the process of developing appropriate management strategies to effectively engage stakeholders throughout the project life cycle, based on the analysis of their needs, interests and potential impact on the project success.

Plan Stakeholder Management

Inputs	Tools & Techniques	Outputs
❑ Project Management ❑ Plan Stakeholder Register ❑ Enterprise Environmental Factors ❑ Organizational process assets	❑ Expert judgment ❑ Meetings ❑ Analytical Techniques	❑ Stakeholder ❑ Management Plan ❑ Procurement document updates

Plan Stakeholder Management –Inputs

Project Management Plan - The information that is taken from Project Management Plan to create stakeholder plan includes

1. Life cycle selected for the project and the processes that will be applied to each phase;

2.Description of how work will be executed to accomplish the project objectives;

3.Description of how human resources requirements will be met and how roles and responsibilities, reporting relationships, and staffing management will be addressed and structured for the project;

4.Change management plan that documents how changes will be monitored and controlled

5.Need and techniques for communicating among stakeholders

Plan Stakeholder Management –Inputs

Stakeholder Register – Provides the information needed to plan appropriate ways to engage project stakeholders.

Enterprise Environmental Factors - such as organizational culture, structure, and the political climate. These factors help in determining the best options to support a better adaptive process for managing stakeholders

Organizational Process Assets – such as lessons learned database and historical information are of particular importance as they provide insight into previous stakeholder management plans and their effectiveness. These can be used to plan the stakeholder management activities for the current project.

Plan Stakeholder Management – Tools and Techniques

Expert Judgment – Suggestions and guidance should be sought from groups or individuals with specialized training or subject matter expertise or insight into the relationships within the organization, such as

1. Senior Management
2. Project Team Members
3. Other units or individuals within the organization
4. Identify key stakeholders
5. Project managers who have worked on the projects in the same area
6. Industry groups and consultants
7. Professional and technical associations , regulatory bodies and non governmental organizations.

Plan Stakeholder Management – Tools and Techniques

Meetings Should be held with experts and the project team to define the required engagement levels of all stakeholders. This information can be used to prepare the stakeholder management plan

Plan Stakeholder Management – Tools and Techniques

Analytical Techniques includes comparison of the current engagement level of stakeholder to the planned engagement levels required for successful project completion. Stakeholder engagement throughout the lifecycle of the project is critical to project success.

The engagement level of the stakeholders can be classified as follows:

- **Unaware –** Unaware of project and potential impacts
- **Resistant** – Aware of project and potential impacts and resistant to change
- **Neutral –** Aware of project yet neither supportive nor resistant
- **Supportive** – Aware of project and potential impacts and supportive to change
- **Leading** – Aware of project and potential impacts and actively engaged in ensuring the project is a success.

Stakeholders Engagement Assessment Matrix

Stakeholder	Unaware	Resistant	Neutral	Supportive	Leading
Stakeholder 1	C			D	
Stakeholder 2			C	D	
Stakeholder 3				D C	

Figure 13-7, where C indicates the current engagement, and D indicates the desired engagement

The example in Figure 13-7 shows that stakeholder 3 is at the desired engagement level, while stakeholders
1 and 2 require further communications and additional actions to move them to the desired level of engagement.

Plan Stakeholder Management - Outputs

Stakeholder Management Plan is a component of the project management plan that explains the management strategies required to effectively engage stakeholders. In addition to the data gathered in the stakeholder register it provides.

- Desired and current engagement levels of key stakeholders
- Scope and impact of change to stakeholders
- Identified interrelationships and potential overlap between stakeholders
- Stakeholder communication requirements for the current project phase;
- Information to be distributed to stakeholders, including language, format, content , and level of detail;
- Reason for the distribution of that information and the expected impact to stakeholder engagement;
- Time frame and frequency for the distribution of required information to stakeholders.
- Method for updating and refining the stakeholder management plan as the project progresses and develops.

Plan Stakeholder Management - Outputs

Project Document Updates includes project schedules and stakeholder register

Manage Stakeholder Engagement

Manage Stakeholder Engagement is the process of communicating and working with stakeholders to meet their need/expectations, address issues as they occur, and foster appropriate stakeholder engagement in project activities throughout the project life cycle.

Manage Stakeholder Engagement

Inputs	Tools & Techniques	Outputs
❑ Stakeholder Management Plan ❑ Communication Management Plan ❑ Change Log ❑ Organizational process assets	❑ Communication Methods ❑ Interpersonal skills ❑ Management Skills	❑ Issue Log ❑ Change Requests ❑ Project Management Plan updates ❑ Project documents updates ❑ Organizational process assets updates

Manage Stakeholder Engagement - Inputs

Stakeholder Management Plan – This plan explains the management strategies required to effectively engage stakeholders.

Communication Management Plan – This plan explains what to be communicated o whom and at what frequency. What to be communicated to external stakeholders or not

Change log – Document changes that occur during a project. It is important to communicate . The impact of these changes on cost, time and quality should be communicated to the relevant stakeholders.

Organizational Process Assets - The organizational process assets that can influence the Manage Stakeholder Engagement process include Issue management procedures, Change management procedure and Organizations communication requirement.

Manage Stakeholder Engagement – Tools and Techniques

Communication Methods – The methods of communication identified for each stakeholder in the communications management plan are utilized during stakeholder engagement management. Based on the stakeholders' communication requirements,the project manager decides how, when, and which of these communication methods are to be used in the project

Interpersonal skills - This include conflict management skills,communication skills, ability to overcome resistance to the change.

Management Skills - The project manager applies management skills such as negotiation, facilitation and ability to influence people to coordinate and harmonize the group toward accomplishing the project objectives

Manage Stakeholder Engagement – Outputs

Issue Log – It is good to maintain an issue log for managing stakeholder issues. This log is updated as new issues are identified and current issues are resolved.

Change Requests - Managing stakeholder engagement may result in a change request to the product or the project. It may also include corrective or preventive actions to the project itself or to the interaction with the impacted stakeholders, as Appropriate

Project Management Plan updates - Stakeholder management plan or communication management plan might be updated when new stakeholder is identified Project documents updates Organizational process assets updates

Manage Stakeholder Engagement – Outputs

Project documents updates - Project documents that may be updated include, but are not limited to, the stakeholder register. This is updated as information on stakeholders change, when new stakeholders are identified, or if registered stakeholders are no longer involved in or impacted by the project, or other updates for specific stakeholders are required.

Organizational process assets updates - The organizational process assets that may be updated include,

Project records. Project records include correspondence, memos, meeting minutes, and other documents describing the project.

Lessons learned documentation. Documentation includes the root cause analysis of issues faced, reasoning behind the corrective action chosen, and other types of lessons learned about stakeholder management. Lessons learned are documented and distributed, and become part of the historical database for both the project and the performing organization.

Control Stakeholder Engagement

Control Stakeholder Engagement is the process of monitoring overall project stakeholder relationship and adjusting strategies and plans for engaging stakeholders.

Control Stakeholder Engagement

Inputs	Tools & Techniques	Outputs
❑ Project Management Plan ❑ Issue Log ❑ Work Performance Data ❑ Project Documents	❑ Information Management Systems ❑ Expert Judgment ❑ Meetings	❑ Work Performance Information ❑ Change Requests ❑ Project Management Plan updates ❑ Project documents updates ❑ Organizational process assets updates

Inputs - Control Stakeholder Engagement

Project Management Plan - The information related to Project Management that will be utilized include

- The lifecycle selected for the project and the processes that will be applied to each phase.
- How work will be executed to accomplish the project objectives
- How human resources requirements will be met, how roles and responsibilities , reporting relationships, and staffing management will be addressed and structured for the project;
- A change management plan that documents how change will be monitored and controlled
- Needs and techniques for communication among stakeholders

Issue Log – The issue log is updated as new issues are identified and current issues are resolved

Work Performance Data – include reported percentage of work completed , technical performance measures, start and finish dates of schedule activities, number of change requests, number of defects , actual costs, actual durations etc.

Project Documents – Multiple project documents originating from initiation , planning, execution or control processes may be used as supporting inputs for controlling stakeholder engagement. These include

- Project schedule
- Stakeholder register
- Issue Log
- Change Log
- Project Communications

Tools and Techniques - Control Stakeholder Engagement

Information Management Systems – An information management system provides a standard tool for the project manager to capture , store, and distribute information to stakeholders about the project cost, schedule progress, and performance.

Expert Judgement - Suggestions should be sought from subject matter expert such as :

- Senior Management
- Other units or individuals within the organization
- Identified key stakeholders
- Project Managers who have worked on projects in the same area
- Subject matter experts in the business or project area
- Industry groups and consultants
- Professional and technical associations, regulatory bodies and non governmental organization

Meetings – Status review meetings are used to exchange and analyze information about stakeholder engagement

Outputs- Control Stakeholder Engagement

Work Performance Information is the performance data collected from various controlling processes, analyzed in context , and integrated based on relationships across areas. Thus work performance data have been transformed into work performance information.

Change Requests - Analysis of project performance and interactions with stakeholders often generates change requests. These change requests include

- Recommended corrective actions include changes that bring the expected future performance of the project in line with the project management plan and ;
- Recommended preventive actions can reduce the probability of incurring future negative project performance.

Outputs- Control Stakeholder Engagement

Project Management Plan Updates – Elements of the project management plan that may be updated include :

- Change Management Plan
- Communication management plan
- Cost Management Plan
- Human resource management plan
- Procure management plan
- Quality management plan
- Requirements management plan
- Risk management plan
- Schedule management plan
- Scope management plan
- Stakeholder management plan

Outputs- Control Stakeholder Engagement

Project Documents Updates – Following documents may be updated.

Stakeholder register - This is updated as information on the stakeholders change, when new stakeholders are identified, or if registered stakeholders are no longer involved in or impacted by the project, or other updates for specific stakeholders are required.

Issue log – This is updated as new issues are identified and current issues are resolved.

Outputs- Control Stakeholder Engagement

Organizational Process Assets Updates – Following are the organization process assets that may be updated.

- **Stakeholder notifications -** Information may be provided to stakeholders about resolved issues, approved changes and general project status.
- **Project reports -** Formal and informal project reports describe project status and include lessons learned, issue logs, project closure reports , and outputs from other knowledge areas
- **Project presentations -** Information formally or informally provided by the project team to any or all project stakeholders
- **Project records –** Project records include correspondence, memos, meeting minutes, and other documents describing the project.
- **Feedback from stakeholders –** Information received from stakeholders concerning project operations can be distributed and used to modify or improve future performance of the project.
- **Lesson learned documentation -** Documentation includes the root cause analysis of issues faced, reasoning behind the corrective actions chosen, and other types of lessons learned about stakeholder management. Lesson learned are documented and distributed so that they become part of the historical database for both the project and performing organization.